Multi-Agent Architecture with Generative AI: Structural Plasticity

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Abstract

This work presents a **comprehensive and generalizable framework for the design and construction of multi-agent systems** based on Large Language Models (LLMs). The work addresses theoretical foundations, architectural patterns, and essential capabilities that these systems must implement across any application domain. To validate and demonstrate the proposed concepts, we use **Lumen**—a conversational Business Intelligence system—as an **implementation case study**.

Problem: Isolated Large Language Models (LLMs) are insufficient for complex enterprise applications due to hallucinations, lack of access to real data, absence of persistent memory, and context limitations. Architectures combining multiple specialized agents with structured access to data sources are required. This problem is transversal across multiple domains: administration, operations, process automation, data intelligence, and others.

Main Novelty: This work introduces **PEMA (Plastic Evolving Multi-Agent)**, a theoretical extension that provides **structural plasticity** to multi-agent systems. While LLM parameters θ remain fixed post-training, PEMA allows the *coordination topology between agents* to evolve dynamically through Hebbian learning—solving what we term the *adaptation barrier*.

Approach: We propose a **domain-independent** reference framework for multi-agent systems that includes: (1) theoretical foundations of functional agency, (2) taxonomy of multi-agent architectures, (3) PEMA theoretical framework adding structural plasticity to the system, (4) ten essential capabilities, and (5) reusable design patterns. Each capability is presented with **general theory** applicable to any domain, illustrated with Lumen as a case study.

Contributions (hierarchized by novelty):

🔬 **Main Contribution (Theoretical)**:

- **PEMA (Plastic Evolving Multi-Agent)**: Extension of the functional agency framework with **structural plasticity** inspired by neuroscientific principles. First formalization of Hebbian learning for inter-agent trust weights, with theoretical guarantees of bounded predictability (Theorem 10.1) and benchmark protocol for plastic systems. *This contribution opens a new research line in adaptive multi-agent systems.*

📐 **Secondary Contributions (Methodological)**:

1. **Functional Agency Theoretical Framework**: Rigorous formalization of necessary conditions (Definitions 1.1-1.5) for genuine agentic behavior—universally applicable
2. **Novel Design Patterns**, highlighting:
    - *Two-Layer Routing*: Keywords for 85% of traffic, LLM only for ambiguous cases—reduces costs 5x
    - *Speculative Gap RAG (SG-RAG)*: RAG that detects documentation gaps and persists them for analysis—first RAG that learns from its limitations
    - *Experience-Weighted Routing*: Routing where weights evolve by accumulated collaborative success
    - *Adaptive Memory with Decay*: Memory with controlled forgetting (inspired by EWC) balancing retention vs relevance

📋 **Tertiary Contributions (Practical)**: 3. **Architecture Taxonomy**: Classification of five main architectures with transferable decision framework 4. **Ten Essential Capabilities**: Universal checklist for agentic systems 5. **Lumen Case Study**: Complete validation in BI domain demonstrating +20pp accuracy vs baseline

Results: The proposed framework provides practical guidance for building multi-agent systems in any enterprise domain. Validation through Lumen demonstrates improvements of +20 percentage points in accuracy (91.5% vs 71.4% for monolithic architecture) with controlled latency increase (+58%), confirming the viability and benefits of the proposed approach. The PEMA extension provides theoretical foundations for adaptive systems with stability guarantees.

Keywords: Multi-Agent Systems, **Adaptation Barrier**, **Structural Plasticity**, Hebbian Learning, LLM, Functional Agency, Generative AI, PEMA

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PART I: THEORETICAL FOUNDATIONS

Chapter 1: Theory of Agentic Systems

1.1 The Era of Agentic AI

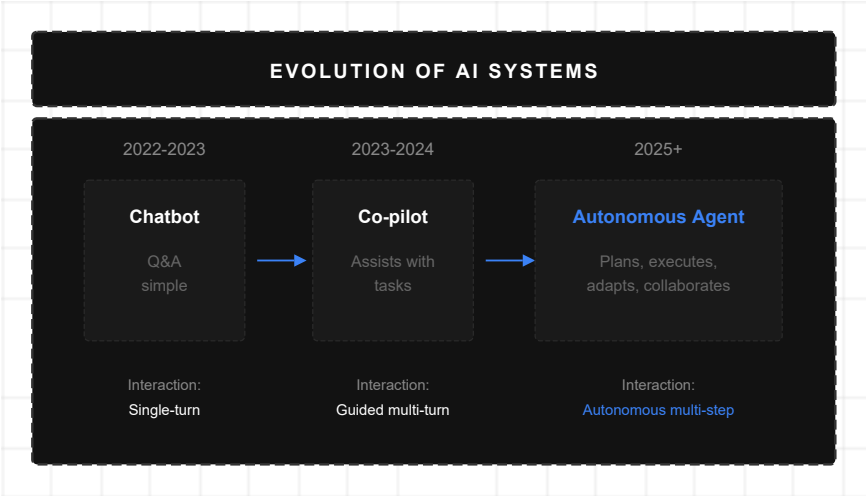
The artificial intelligence landscape has undergone a fundamental transformation. While individual Large Language Models (LLMs) like GPT, Gemini, and Claude dominate in 2025, the future belongs to **agentic systems**: architectures where multiple specialized AI agents collaborate to solve complex problems with increasing autonomy.

According to recent research, the global market for agentic AI tools is experiencing explosive growth, with a compound annual growth rate (CAGR) of approximately 56.1%, reaching $10.41 billion in 2025. Deloitte predicts that in 2025, 25% of enterprises using generative AI will launch agentic AI pilots, growing to 50% by 2027.

Definition: What is Agentic AI

Agentic AI refers to artificial intelligence systems capable of completing complex tasks and achieving objectives with little or no human supervision. It differs from current chatbots and co-pilots in that:

- **Generates outcomes, not just responses**: Instead of single-turn responses, generates complete outcomes
- **Plans and executes**: Decomposes objectives into subtasks and executes them autonomously
- **Interacts with the environment**: Uses tools, navigates systems, and adapts in real-time
- **Self-corrects**: Evaluates its own results and iterates until satisfying criteria



The evolution shows four distinct eras: from rule-based systems (1980s-2000s) through statistical ML (2000s-2017), to transformer-based LLMs (2017-2023), and finally to multi-agent agentic systems (2023+).

Figure 1: Evolution of AI Systems

1.2 Functional Agency: A Theoretical Framework

Recent research proposes the concept of **functional agency** as a theoretical framework for agentic systems. A system exhibits functional agency when it satisfies three conditions:

Mathematical Formalization

Let an agentic system be defined as a tuple:

Definition 1.1 (Agentic System): An agentic system is a tuple $A = (S, O, G, \pi, M, \alpha)$ where:

- S: Set of environment states
- O: Set of observations the agent can perceive
- G: Set of objectives
- π: $S \times G \rightarrow A$, policy function mapping states and objectives to actions
- M: $A \times S \rightarrow S$, transition model (actions and states to new states)
- α: adaptation function that modifies π based on feedback

Definition 1.2 (Functional Agency): A system A exhibits functional agency if and only if it satisfies the following three conditions:

Condition 1: Action Generation

The system must be capable of **generating actions based on environmental information toward an objective**. It is not sufficient to passively process information; the agent must make decisions that modify its environment.

Formally: $\forall s \in S, \forall g \in G: \pi(s, g)$ produces an action $a \in A$ that modifies the environment state.

Condition 2: Outcome Model

The system must **represent relationships between actions and their consequences**. This implies a causal understanding (or strong correlation) of how actions affect the world.

Formally: The system maintains a model M: A × S → S such that M(a, s) predicts the resulting state s' from executing action a in state s.

Condition 3: Adaptation

The system must **modify its behavior when its outcome model changes**. If it discovers that an action no longer produces the expected effect, it must adapt its strategy.

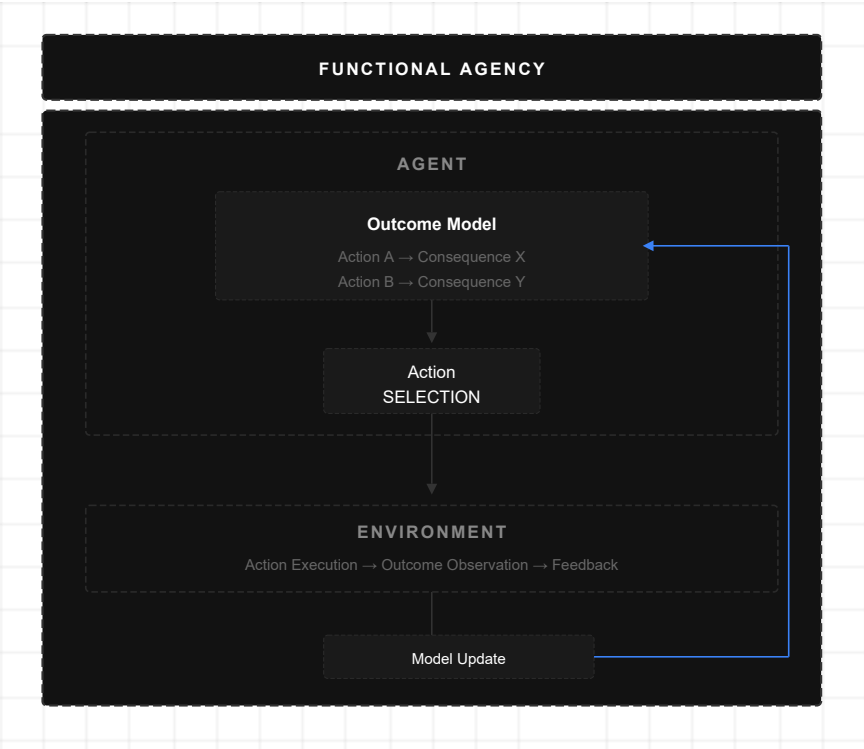
Formally: If M(a, s) ≠ s' (incorrect prediction), then α updates π such that π'(s, g) ≠ π(s, g) for similar cases.

Proposition 1.1: An isolated LLM does not satisfy Condition 3 of functional agency, since its parameters are static post-training and it cannot adapt its policy based on environment feedback at runtime.

Proof sketch: Let L be an LLM with fixed parameters θ post-training. L's policy function π is determined by θ: $\pi_L(s, g) = f_\theta(s, g)$. Condition 3 requires the existence of an adaptation function α that modifies π when M(a, s) ≠ s'. For L, modifying π would require modifying θ, but θ is immutable during inference (the model is "frozen"). Therefore, α cannot exist for L at runtime, and L does not satisfy Condition 3. ∎

Corollary 1.1: To build systems with genuine functional agency, it is necessary to complement LLMs with external mechanisms for memory, tools, and feedback loops.

Justification: From the result of Proposition 1.1, if we want an LLM-based system to satisfy Condition 3, we must externalize the adaptation function α. External mechanisms (persistent memory to remember errors, tools to execute actions, feedback loops to evaluate results) allow implementing α outside the LLM, complementing its static capabilities with dynamic adaptation. ∎



The three conditions for functional agency: (1) Action Generation; (2) Outcome Model—causal understanding; (3) Adaptation—modify behavior when predictions fail. An isolated LLM fails Condition 3.

Figure 2: Functional Agency

1.3 Components of an Agentic System

Modern agentic systems are built on fundamental components that work together:

1.3.1 Reasoning Engine (LLM)

The core of the agent is a Large Language Model that provides:

- **Natural language understanding**: Interprets instructions and context
- **Reasoning**: Decomposes problems and plans solutions
- **Generation**: Produces text, code, decisions

1.3.2 Memory

Agents require multiple types of memory:

Type	Function	Persistence
Working Memory	Context of current task	Session
Episodic Memory	Indexed past experiences	Long-term
Semantic Memory	Domain knowledge	Permanent
Procedural Memory	Learned execution patterns	Permanent
Structural Memory	Inter-agent trust weights (PEMA)	Adaptive

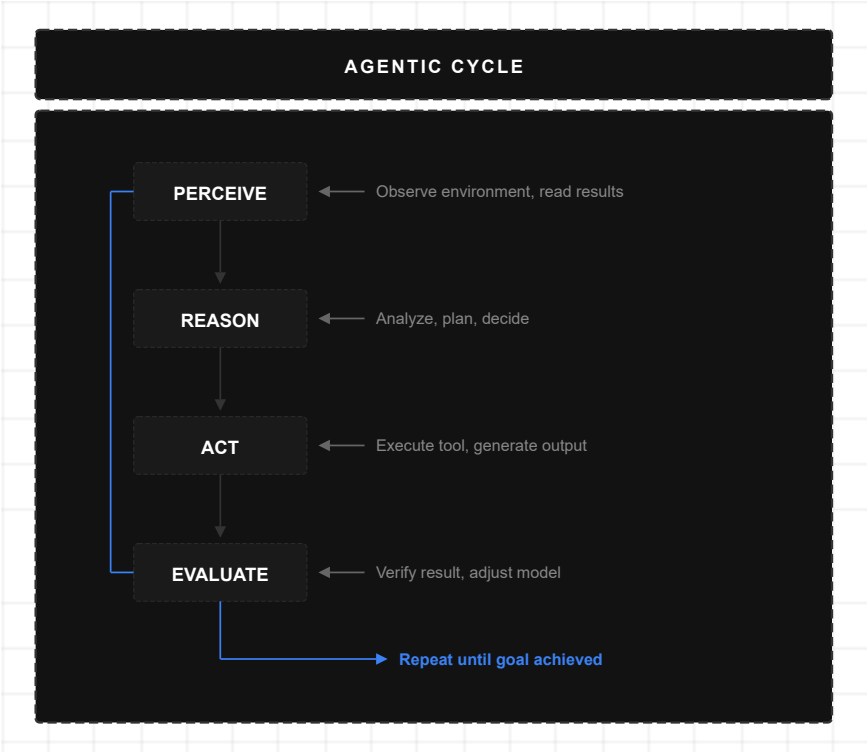
1.3.3 Tools

Tools extend the agent's capabilities beyond text generation:

- **Code execution**: Compilers, interpreters, debuggers
- **Data access**: APIs, databases, file systems
- **Web interaction**: Navigation, search, scraping
- **Communication**: Email, messaging, notifications

1.3.4 Feedback Loop

The agentic system operates through integrated feedback loops:



The continuous agentic loop: Perceive → Think → Act → Observe → Adapt. The agent receives input, reasons, executes with tools, observes results and updates its internal state.

Figure 3: Agentic Cycle

1.4 Agentic Reasoning Patterns

Research has identified fundamental patterns for agent reasoning:

1.4.1 Chain-of-Thought (CoT)

The agent explicates its reasoning process step by step before acting:

```
Question: "What were the total sales for Q3?"

CoT Reasoning:
1. I need to query sales data
2. Q3 corresponds to July-September
3. I must sum all transactions in that period
4. I will use the execute_dax tool with the appropriate query

Action: execute_dax("CALCULATE(SUM(Sales[Amount]), QUARTER(Sales[Date])=3)")
```

1.4.2 ReAct (Reasoning and Acting)

Alternates between reasoning and action, using observations to guide next steps:

```
Thought: I need to find information about the Acme client
Action: search_documents("Acme client contracts")
Observation: Found 3 relevant documents...
Thought: The main document is the 2024 contract
```

```
Action: read_document("acme_contract_2024.pdf")
Observation: The contract specifies...
Thought: Now I can answer the question
Answer: According to the 2024 contract, the Acme client...
```

1.4.3 Reflection

The agent critically evaluates its own output and improves it iteratively:

```
Attempt 1: [Generates response]
Evaluation: "This response doesn't cite specific sources"
Attempt 2: [Generates improved response with citations]
Evaluation: "Now it's precise and verifiable"
Final Output: [Refined response]
```

1.4.4 Planning

For complex tasks, the agent generates a plan before executing:

```
Objective: "Create a sales report by region"

Plan:
1. Identify the correct data model
2. Query sales by region
3. Query previous period sales for comparison
4. Calculate percentage variations
5. Generate visualization
6. Write main insights

Execution: [Executes each step of the plan]
```

1.5 Current Capabilities (State of the Art 2025)

1.5.1 What Agents Can Do Today

Current agentic systems demonstrate impressive capabilities:

Agentic Programming:

- Generate code from natural language specifications
- Interact with compilers, debuggers, and version control systems
- Iterate on errors and refine solutions

Workflow Automation:

- Complete multi-step tasks in enterprise applications
- Coordinate between multiple systems (CRM, ERP, BI)

- Handle exceptions and edge cases

Research and Analysis:

- Search and synthesize information from multiple sources
- Generate structured reports
- Identify patterns and anomalies in data

Tool Interaction:

- Navigate web interfaces
- Execute database queries
- Generate and modify documents

1.5.2 Autonomy Levels

Agentic system maturity is measured in levels:

Level	Description	2025 Status
1	Basic assistance with limited tools	Mature
2	Multi-step execution with supervision	Mature
3	Autonomy in specific domains (<30 tools)	Emerging
4	Broad autonomy with self-correction	Research
5	Complete autonomy with continuous learning	Future

According to industry analysis, as of Q1 2025 most agentic applications remain at Levels 1 and 2, with some exploring Level 3 within specific domains.

1.6 Limitations and Challenges

1.6.1 Limitations of Current LLMs

Current language models present critical deficiencies for agency:

Limited Causal Reasoning:

- Confuse correlation with causation
- Difficulty predicting consequences of novel actions
- Bias toward patterns observed in training

Lack of Metacognition:

- Unawareness of their own limitations
- Systematic overconfidence in incorrect answers
- Difficulty distinguishing sufficient from insufficient information

Hallucinations:

- Generation of plausible but false information

- Inventing citations, data, or facts
- Filling knowledge gaps with fabrications

1.6.2 System Challenges

Memory Scalability:

- Limited context windows (8K-200K tokens)
- Lack of structured persistence between sessions
- Performance degradation with long contexts

Security and Reliability:

- Risk of executing unsafe actions without supervision
- Unanticipated emergent behaviors
- Difficulty auditing decisions

Tool Incompatibility:

- Tools designed for humans, not agents
- Inconsistent APIs and incomplete documentation
- Need for wrappers and adapters

Inadequate Evaluation:

- Existing benchmarks don't capture real workflows
- Precision metrics don't reflect practical utility
- Difficulty measuring autonomy and reliability

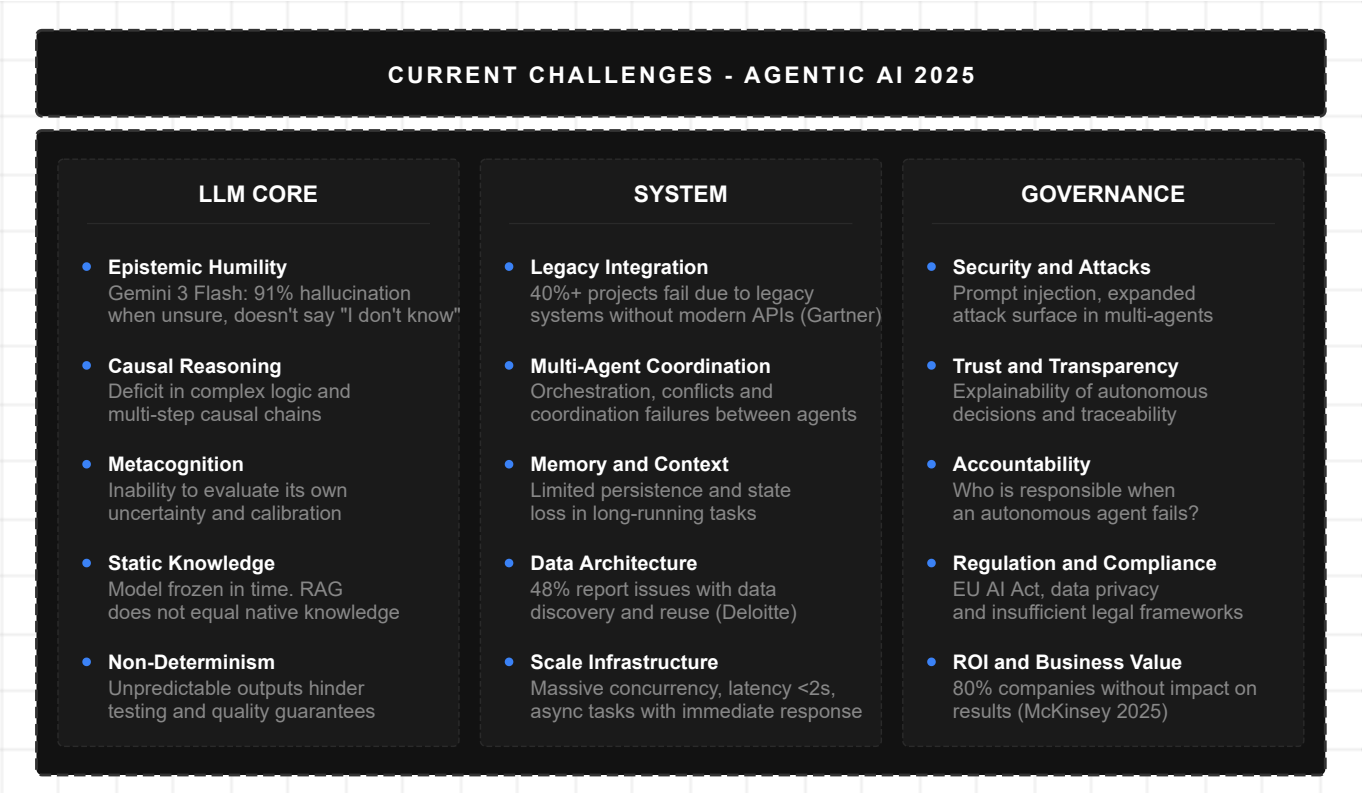


Figure 4: Current Challenges

1.7 Horizon: Vision 2025-2030

1.7.1 Emerging Trends (2025)

Multi-Agent as the Norm:

- Systems where specialized agents collaborate
- Decentralization and specialization as dominant architecture
- Pattern similar to microservices in software development

Small Language Models for Agents:

- SLMs powerful enough for specialized repetitive tasks
- More economical and efficient for massive invocations
- Ideal for components of complex agentic systems

Fusion with IoT and Robotics:

- Warehouse robots managed by decision agents
- Autonomous vehicles coordinated with supply chain agents
- Foundations for smart factories and automated logistics networks

1.7.2 Industry Predictions

Horizon	Prediction
2025	25% of enterprises with gen AI launch agentic pilots
2027	50% of enterprises with gen AI use agentic systems
2029	Agents resolve 80% of customer service issues (Gartner)
2030	Multi-agent ecosystems across industries

1.7.3 Vision: Autonomous Enterprises

The horizon points toward **enterprises with incremental autonomy**:

Phase 1 (Current): Co-pilots assist in specific tasks **Phase 2 (2025-2027)**: Agents execute complete workflows with supervision **Phase 3 (2027-2029)**: Multi-agent ecosystems coordinate across departments **Phase 4 (2030+)**: Autonomy in most operational processes

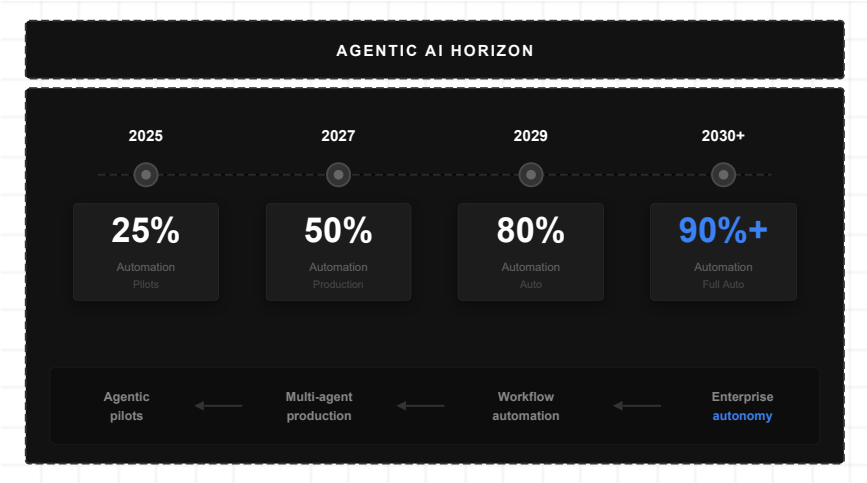


Figure 5: Agentic AI Horizon

The 2025-2030 trajectory: Phase 1 (current) co-pilots; Phase 2 (2025-27) agents with supervision; Phase 3 (2027-29) multi-agent ecosystems; Phase 4 (2030+) operational autonomy.

Chapter 2: Multi-Agent Architectures

2.1 From Monolithic to Multi-Agent

The transition from individual LLMs to multi-agent systems represents a paradigmatic shift similar to the evolution from monolithic applications to microservices:

Aspect	Monolithic LLM	Multi-Agent System
Responsibility	One model does everything	Specialized agents
Scalability	Vertical (larger model)	Horizontal (more agents)
Reliability	Single point of failure	Redundancy and fallbacks
Maintenance	Change affects everything	Isolated changes
Specialization	Generalist	Domain experts

Advantages of the Multi-Agent Approach

Specialization:

- Each agent optimized for its domain
- Specific prompts and tools
- Lower cognitive load per agent

Composition:

- Reusable agents across applications
- New capabilities by adding agents
- Architectural flexibility

Resilience:

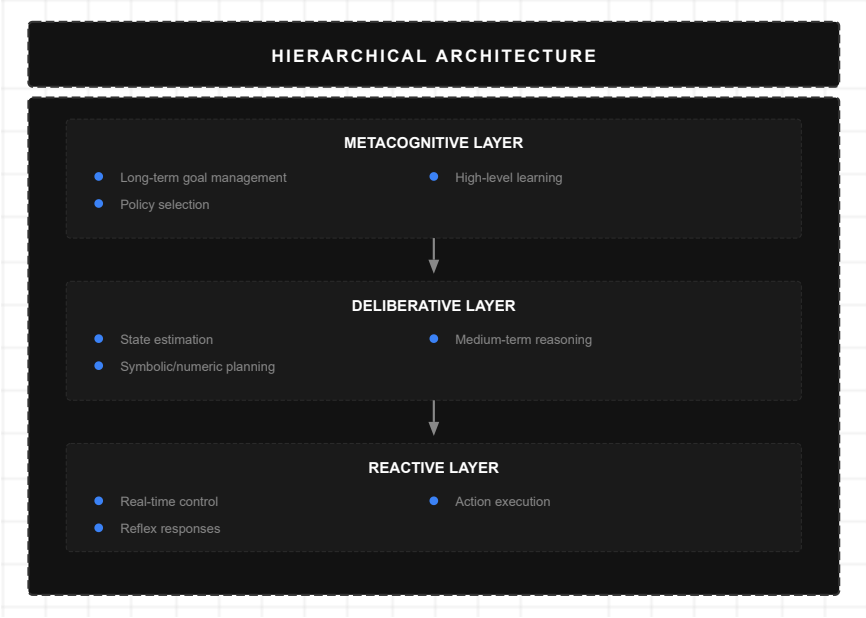
- Failure of one agent doesn't collapse the system
- Retries and fallbacks possible
- Graceful degradation

2.2 Main Architectures (2025)

Research and industrial practice have converged on five main architectures:

2.2.1 Cognitive Hierarchical Architecture

Divides intelligence into layers with different temporal scales and abstraction levels:

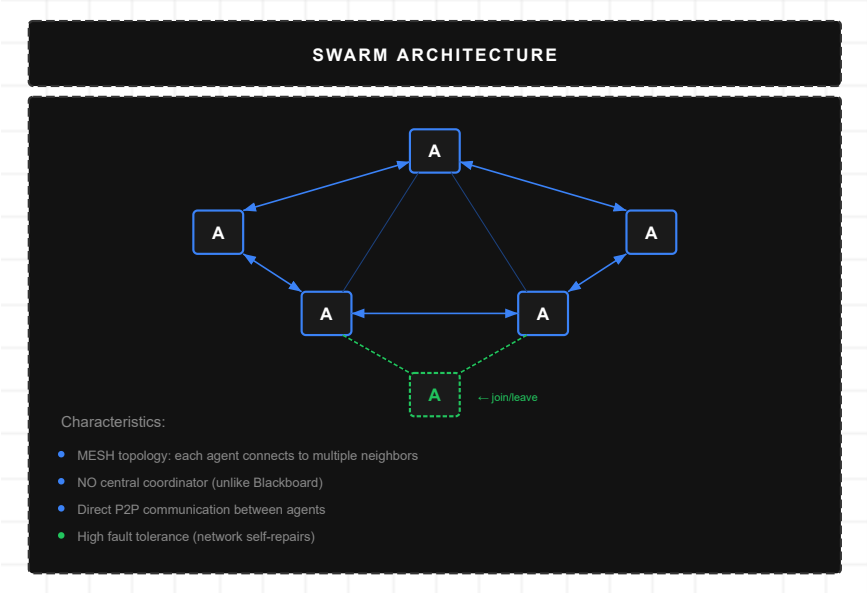


Cognitive layers with different temporal scales: Strategic layer (long-term planning), Tactical layer (intermediate decisions), Operational layer (real-time execution). **Use**: Robotics, trading.

Figure 6: Hierarchical Architecture

2.2.2 Swarm Architecture

Minimalist approach with simple agents that emerge complex behavior:



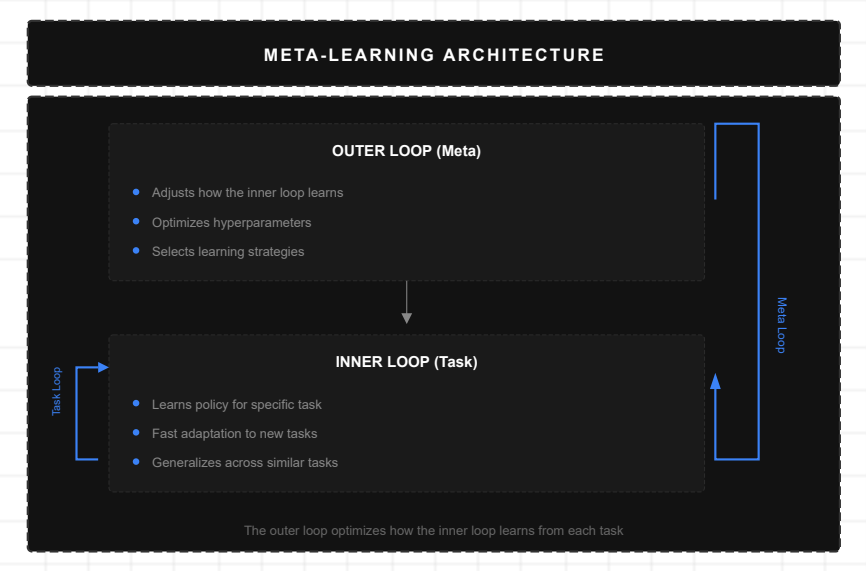
Simple homogeneous agents with emergent behavior: Peer-to-peer communication, no central coordinator. **Use**: Rapid prototyping, distributed search.

Figure 7: Swarm Architecture

> **Note: Swarm vs Blackboard vs PEMA**
>
> It's important to distinguish three related but distinct concepts:
>
> - **Swarm** (this section): *Architecture* where homogeneous agents communicate peer-to-peer and behavior emerges from local interactions. No central state or coordinator.
> - **Blackboard** (Section 2.3.4): *Coordination pattern* where heterogeneous agents read/write to a shared central state. Compatible with multiple architectures.
> - **PEMA** (Chapter 11): *Plasticity mechanism* that can be applied to any architecture, allowing trust weights between agents to evolve through Hebbian learning.
>
> In practice, a system could combine Swarm architecture with Blackboard coordination and PEMA plasticity.

2.2.3 Meta-Learning Architecture

Separates task learning from meta-learning (learning to learn). **Magentic-One** from Microsoft Agent Framework exemplifies this pattern:



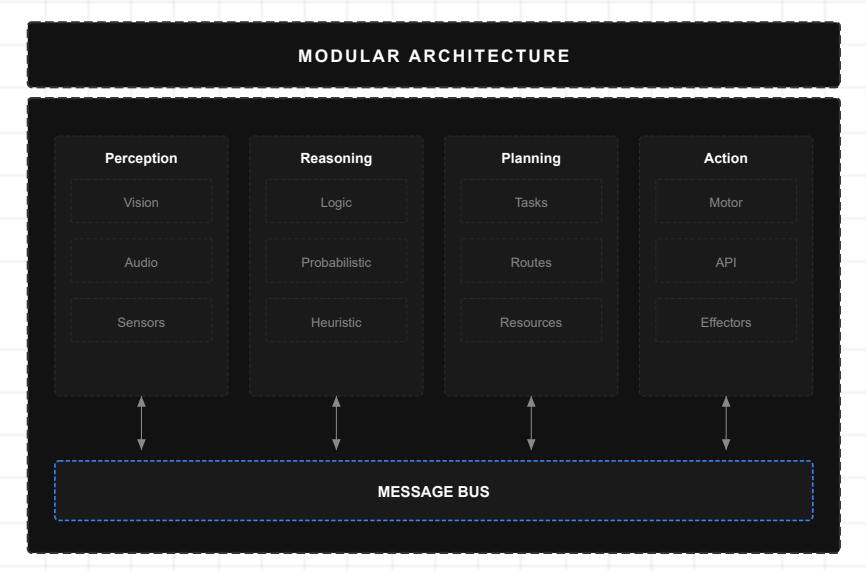
Learning to learn: Separates task execution from meta-learning. Magentic-One exemplifies this pattern. **Use**: Systems requiring rapid adaptation.

Figure 8: Meta-Learning

> **Connection with PEMA**: Meta-learning is the theoretical precursor to the structural plasticity proposed in PEMA (Chapter 11). While traditional meta-learning operates on *model parameters*, PEMA extends the concept to *inter-agent trust weights*, allowing the coordination topology to evolve adaptively. This represents a higher level of adaptation: not just "learning to learn," but "learning to collaborate."

2.2.4 Modular Architecture

Interchangeable components with well-defined interfaces:

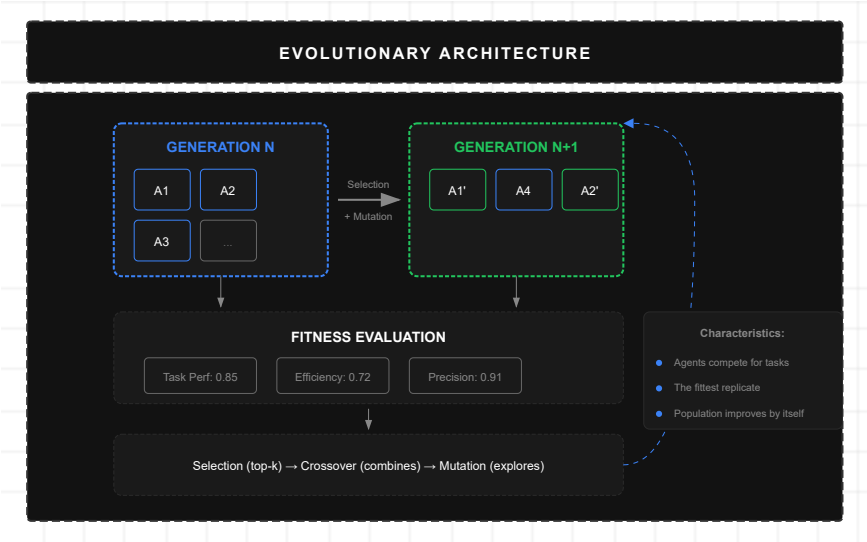


Interchangeable components with defined interfaces: Plug-and-play agents, hot-swappable modules. **Use**: Enterprise systems requiring flexibility.

Figure 9: Modular Architecture

2.2.5 Evolutionary Architecture

Agents that evolve and adapt through selection and mutation:

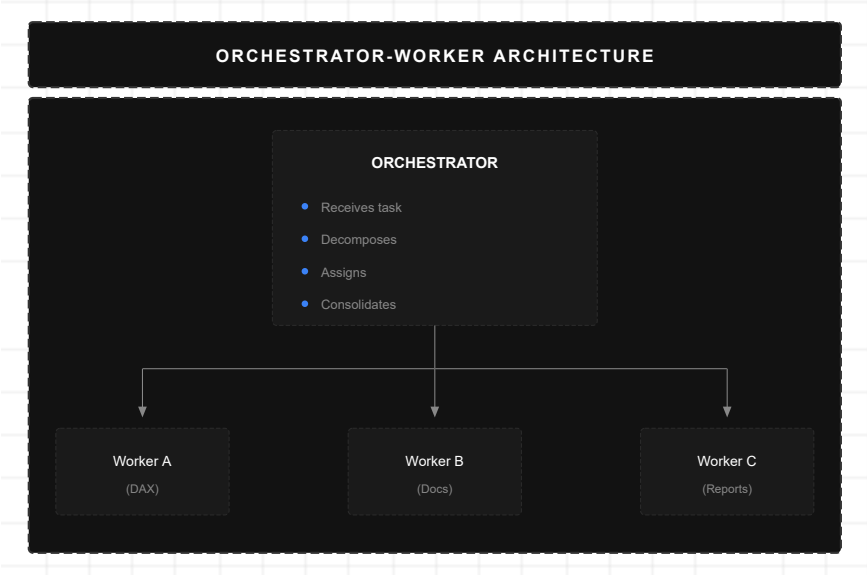


Agents that evolve through selection and mutation: Population of variants compete; successful strategies propagate. **Use**: Long-term optimization, self-improving systems.

Figure 10: Evolutionary Architecture

2.2.6 Orchestrator-Worker Architecture

A central agent coordinates specialized workers:

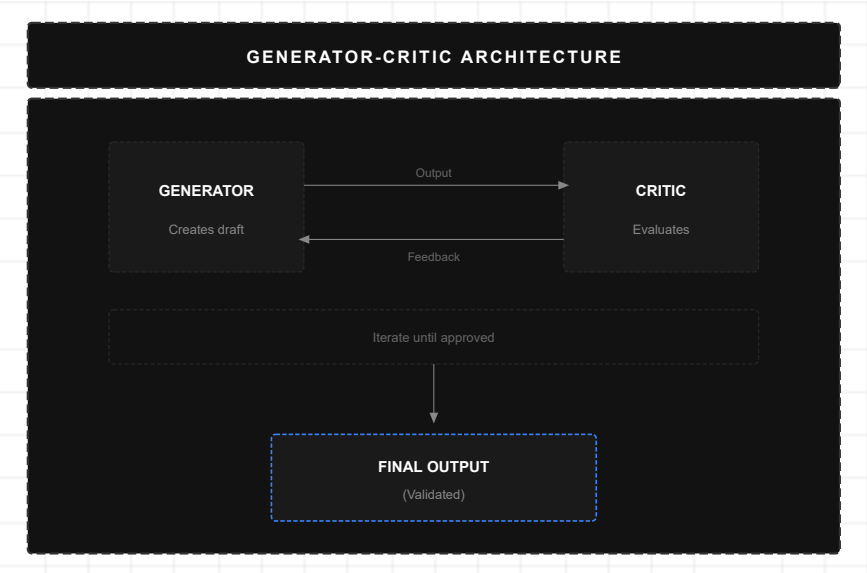


Central coordinator with specialized workers: Orchestrator decomposes tasks, assigns to workers, aggregates results. **Use**: Most enterprise multi-agent systems.

Figure 11: Orchestrator-Worker Architecture

2.2.7 Generator-Critic Architecture

Separates content creation from its validation:

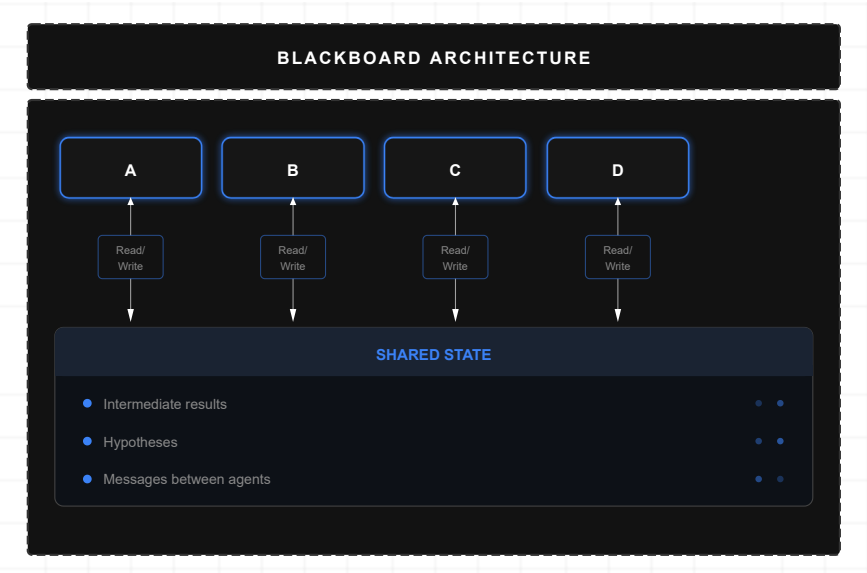


Separates creation from validation: Generator produces content; Critic evaluates and provides feedback. Iterates until quality threshold. **Use**: Code generation, high-quality content.

Figure 12: Generator-Critic Architecture

2.2.8 Blackboard Architecture

Agents share state via centralized storage:

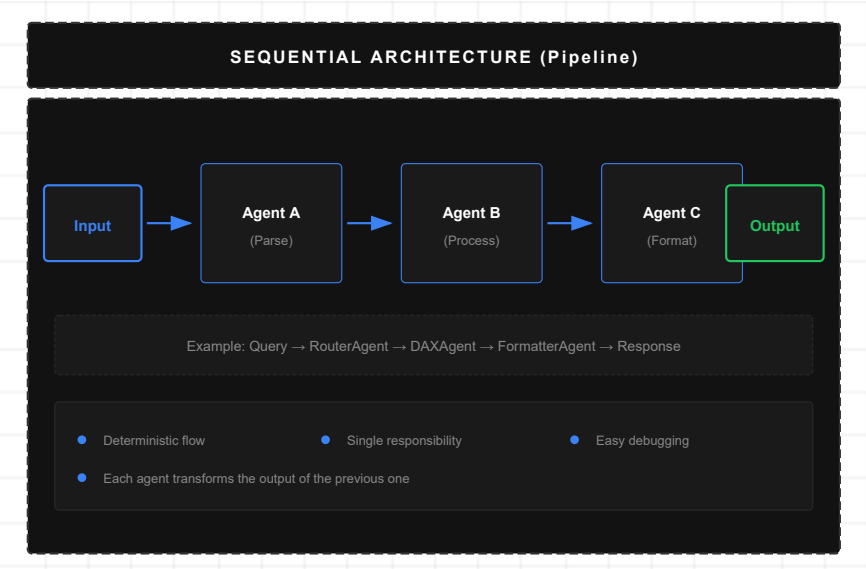


Centralized shared state: Agents read/write to common storage. No direct inter-agent communication. **Use**: Heterogeneous agent collaboration.

Figure 13: Blackboard Architecture

2.2.9 Sequential Architecture (Pipeline)

Agents process in chain, each transforming the previous one's output:

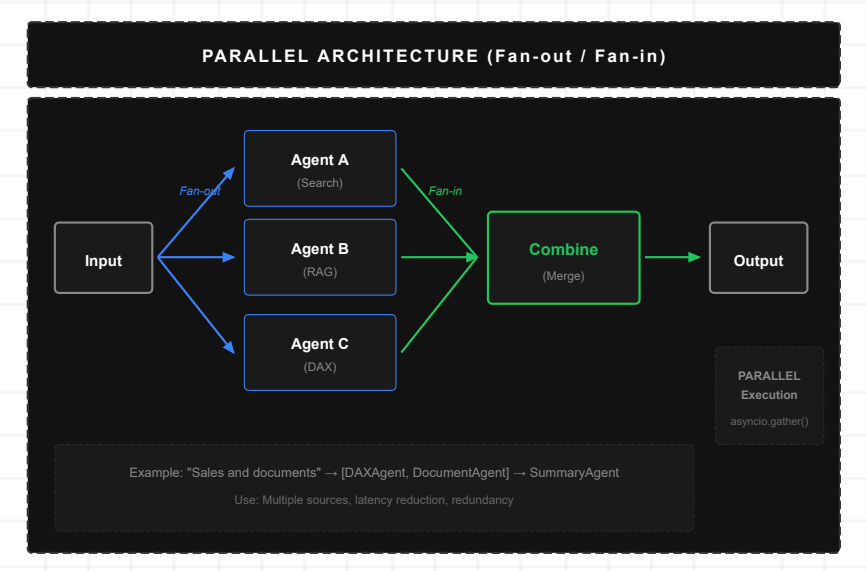


Chain processing: Each agent transforms output of previous one. Deterministic flow, easy to debug. **Use**: Structured processing, ETL, pipelines.

Figure 14: Sequential Architecture

2.2.10 Parallel Architecture (Fan-out/Fan-in)

Multiple agents process simultaneously and combine results:

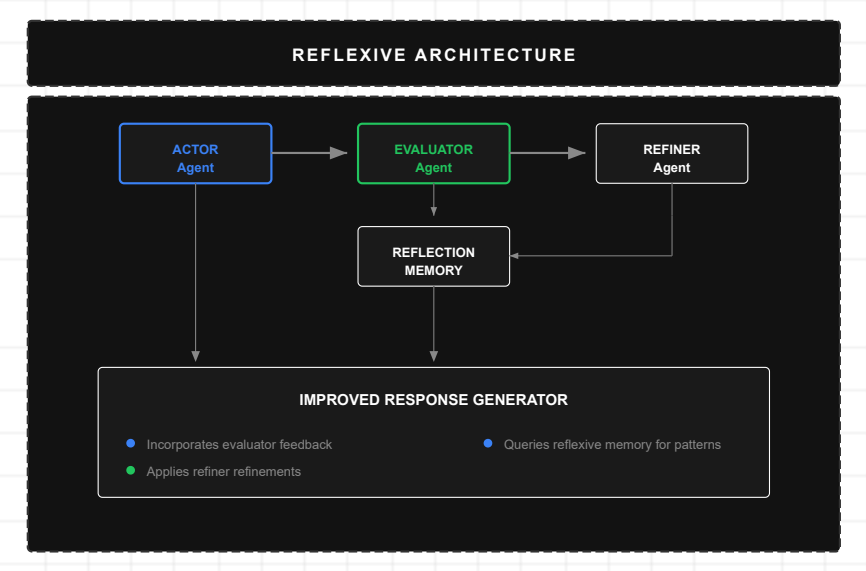


Simultaneous processing with aggregation: Fan-out distributes work; Fan-in combines results. Minimizes latency. **Use**: Multi-source queries, fault tolerance.

Figure 15: Parallel Architecture

2.2.11 Reflexive Architecture (Loop)

Agent evaluates its output and refines iteratively until satisfying criteria:



Self-evaluation and iterative refinement: Agent produces output, evaluates against criteria, refines until satisfactory. **Use**: Code generation, high-quality content.

Figure 16: Reflexive Architecture

2.3 Frameworks and Tools (2025 Status)

Framework	Architecture	Main Use
LangGraph	State graphs	Production agents with persistence and cycles
AutoGen	Async conversational	Event-driven multi-agent cooperation
CrewAI	Role-playing	Agent teams with defined roles
Semantic Kernel	Enterprise orchestration	Enterprise integration (.NET/Python/Java)
Swarm	Lightweight handoffs	Educational, maximum simplicity
LlamaIndex	RAG-first	Document-aware agents
DSPy	Prompt compiler	Automatic prompt optimization
MS Agent Framework	Unified	SK + AutoGen combined (**Lumen's base**)

2.4 Performance Metrics

2.4.1 Effectiveness Metrics

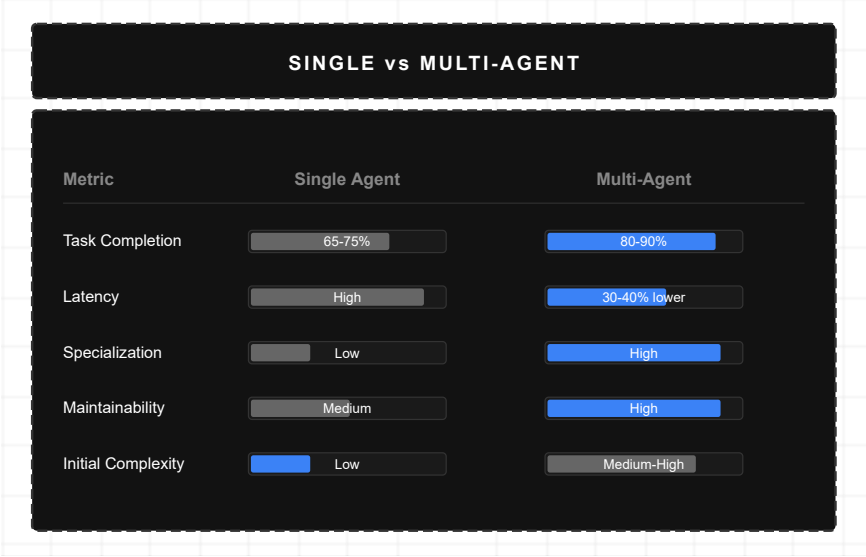
Metric	Description	Benchmark
Task Completion Rate	% of tasks completed successfully	80-90% (routed)
First-Pass Success	Success without retries	60-75%
Error Recovery Rate	Recovery from failures	70-85%

2.4.2 Efficiency Metrics

Metric	Description	Impact

Metric	Description	Impact
Latency	Total response time	Router reduces 30-40%
Token Usage	LLM token consumption	Specialization reduces 20-30%
Tool Invocations	Tool calls	Fewer with better routing

2.4.3 Comparison: Single vs Multi-Agent



Performance comparison: Multi-agent achieves +20pp accuracy vs monolithic LLMs. Trade-off: latency (+58%) but better specialization, error recovery, and maintainability.

Figure 17: Single vs Multi-Agent

Chapter 3: Related Work

This chapter positions the presented work in relation to existing literature, critically analyzing previous approaches and identifying the gaps this work addresses.

3.1 Theoretical Foundations

3.1.1 Intelligent Agents

The study of intelligent agents has roots in classical artificial intelligence. Wooldridge and Jennings (1995) defined an agent as a computational system situated in an environment, capable of autonomous action to achieve objectives. This definition has evolved with the arrival of LLMs, but the fundamental principles remain.

Russell and Norvig (2020) classify agents according to their architecture:

- **Simple reflex agents**: Respond directly to perceptions
- **Model-based agents**: Maintain internal state of the world
- **Goal-based agents**: Plan to achieve goals
- **Utility-based agents**: Optimize a utility function

Modern agentic systems with LLMs combine elements from all these categories, adding natural language reasoning capabilities.

3.1.2 Multi-Agent Systems

Research in Multi-Agent Systems (MAS) established principles of coordination and communication between agents (Ferber, 1999). Key concepts include:

- **Coordination**: Mechanisms for agents to work together without conflicts
- **Negotiation**: Protocols for resolving goal differences
- **Organization**: Structures defining roles and relationships

These principles inform the design of multi-agent systems with LLMs, although communication via natural language introduces new possibilities and challenges.

3.1.3 Business Intelligence

The Business Intelligence field was formalized with Kimball's (1996) work on dimensional modeling and Inmon's (2005) work on data warehouse architectures. Fundamental concepts include:

- **Dimensional model**: Facts and dimensions as base structure
- **OLAP**: Multidimensional analysis operations
- **ETL**: Extraction, transformation, and loading processes

Lumen operates on Power BI semantic models that implement these principles, adding a natural language layer to democratize access.

3.2 Related Work in Multi-Agent Systems with LLMs

3.2.1 Orchestration Frameworks

LangChain/LangGraph (Harrison Chase, 2022-2024): Pioneer framework that introduced the concept of "chains" for LLM composition. LangGraph extends this with state graphs for complex workflows.

Strengths: Fine control over execution flow, persistent state, active community. *Limitations*: Steep learning curve, sometimes opaque abstraction, not optimized for BI.

Microsoft Agent Framework (2024): Unifies Semantic Kernel and AutoGen. Provides `ChatCompletionAgent`, `@ai_function`, Agent-as-Tool, Guardrails (groundedness), and Magentic-One. **Lumen's implementation base**.

Strengths: Azure integration, strict typing, native MCP, Guardrails, telemetry. *Limitations*: Younger ecosystem than LangChain, evolving documentation.

CrewAI (Moura, 2024): Simplified framework with "crew" (team) metaphor of agents with defined roles.

Strengths: Simplicity, rapid prototyping, clear roles. *Limitations*: Less control than LangGraph, limited for complex workflows.

OpenAI Swarm (OpenAI, 2024): Minimalist experimental framework for "handoffs" between agents.

Strengths: Minimalism, ease of use, clear transfer concept. *Limitations*: Experimental, no production support, no persistent state.

3.2.2 Conversational BI Systems

Power BI Q&A (Microsoft): Allows natural language questions on Power BI models.

Strengths: Native integration, no additional configuration. *Limitations*: Simple queries only, no agents, no documents, limited responses.

ThoughtSpot (ThoughtSpot Inc.): Analytics platform with natural language search.

Strengths: Search-centered design, automatic visualizations. *Limitations*: Proprietary solution, not extensible, no agentic architecture.

Tableau Ask Data (Salesforce): Natural language interface for Tableau.

Strengths: Integration with Tableau ecosystem. *Limitations*: Limited capabilities, not multi-agent, basic responses.

3.2.3 Adaptive Systems and Multi-Agent Learning

A critical area that previous works don't address is **structural adaptation** of multi-agent systems. Existing approaches present static topologies:

DyLAN (Liu et al., 2024): Dynamic LLM-Agent Network proposes agent networks that can change dynamically, but operates at the level of agent selection per task, not cumulative trust weights.

Strengths: First work considering dynamism in LLM agent networks. *Limitations*: Doesn't formalize continuous learning; no theoretical stability guarantees.

GTD (Graph-based Task Decomposition) (Chen et al., 2024): Decomposes tasks into graphs where nodes are agents.

Strengths: Structured representation of dependencies between agents. *Limitations*: Graphs are static per task; no memory of past successes between tasks.

MetaGPT (Hong et al., 2024): Multi-agent framework simulating software development teams.

Strengths: Well-defined roles, realistic workflows, intermediate artifacts. *Limitations*: Fixed topology (PM → Architect → Engineer → QA); doesn't learn from experiences.

Connection with Multi-Agent Reinforcement Learning (MARL): The MARL field has developed techniques for joint policy learning in multi-agent systems:

- **COMA** (Foerster et al., 2018): Counterfactual Multi-Agent Policy Gradients introduces counterfactual baselines for credit assignment between agents.
- **MADDPG** (Lowe et al., 2017): Multi-Agent Deep Deterministic Policy Gradient allows centralized policies during training with decentralized execution.
- **QMIX** (Rashid et al., 2018): Factorizes the joint Q-function allowing implicit coordination.

Difference with PEMA: MARL operates on policies learned end-to-end via gradients, requiring millions of training interactions. PEMA operates on pre-trained agents (LLMs), adjusting only the *inter-agent trust weights* via local Hebbian rules—allowing adaptation with orders of magnitude less data and without base model retraining.

Critical analysis: None of these works addresses the **adaptation barrier**—the inability of multi-agent systems to modify their coordination patterns based on observed results. All assume that the optimal topology can be designed a priori, ignoring that:

1. Performance of specific collaborations varies with context
2. New effective collaboration patterns can emerge with use
3. The "institutional memory" of which combinations work is lost between sessions

This gap motivates this work's main contribution: **PEMA (Plastic Evolving Multi-Agent)**, which formalizes how inter-agent trust weights can evolve through Hebbian learning, with theoretical stability guarantees (Theorem 10.1).

3.3 Positioning and Comparative Analysis

3.3.1 Comparative Table

Dimension	LangGraph	AutoGen	CrewAI	Swarm	Power BI Q&A	Lumen
Architecture	Graphs	Conversational	Role-based	Handoff	Monolithic	Multi-Agent
Persistent State	☑	Partial	✗	✗	✗	☑
Intelligent Routing	Manual	Implicit	By role	Handoff	Rule-based	Two-Layer
BI Integration	Manual	Manual	Manual	Manual	Native	Native (Fabric)
RAG	Plugin	Plugin	Plugin	✗	✗	Integrated
Cross-Agent Memory	Via state	Via chat	✗	✗	✗	Persisted Memory
Enterprise OAuth	Manual	Manual	Manual	✗	Integrated	Integrated
Visualizations	Manual	Manual	Manual	✗	Native	Generated

3.3.2 Identified Gaps

Analysis of previous works reveals the following gaps that this work addresses:

1. **BI Integration Gap**: Agent frameworks aren't optimized for BI; BI solutions don't have agentic architecture.
2. **Memory Gap**: No system combines structured persistent memory (not just text) between specialized agents.

3. **Routing Gap**: Approaches are either too simple (keywords) or too expensive (always LLM). Cost/accuracy optimization is missing.
4. **Specialization Gap**: Generic frameworks don't have pre-built agents for DAX, reports, enterprise documents.

3.3.3 Contribution of This Work

This work fills these gaps through:

1. **Native BI Integration**: Specialized agents for Power BI/Fabric via MCP and OAuth.
2. **Persisted Memory Pattern**: Structured memory enabling data sharing (not just text) between agents.
3. **Two-Layer Routing**: Optimization using keywords for clear cases and LLM only for ambiguous ones.
4. **Specialized Agents**: Pre-built and tested DAXAgent, DocumentAgent, ReportAgent.
5. **Structural Plasticity (PEMA)**: First formalization of Hebbian learning for inter-agent trust weights, with theoretical stability guarantees.

From Theory to Practice

Part I established the theoretical foundations: what is functional agency (Section 1.2), available multi-agent architectures (Section 2.2), and the structural plasticity gap this work addresses (Section 1.3).

Part II demonstrates how these concepts materialize in Lumen:

Theoretical Concept	Section	Implementation in Lumen	Capability
Functional Agency (Def. 1.1)	1.2	Agents with outcome model, action selection, feedback	Cap. 4
Outcome Model	1.2	Typed tools with @ai_function	Cap. 4
Hierarchical Architecture	2.2.1	BIWorkflow as supervisor	Cap. 2
Parallel Architecture	2.2.10	Embedding parallel, concurrent queries	Cap. 8
Reflexive Architecture	2.2.11	Self-critique in DAXAgent	Cap. 3
Memory (agency requirement)	1.2	Persisted Memory in SQL Server	Cap. 1
Structural Plasticity (PEMA)	3.2	Adaptive trust weights	Cap. 10

Lumen is not just an implementation—it's a laboratory where each capability validates a theoretical principle. The following 10 capabilities demonstrate the practical applicability of the framework.

PART II: INTRODUCTION TO LUMEN

Introduction

1.1 Problem Context

The adoption of Artificial Intelligence in Business Intelligence presents a fundamental challenge: business users need answers, not technical interfaces. A user asking "What were Q3 sales compared to Q2?" expects an immediate response with numbers, context, and perhaps a visualization. However, traditionally they must navigate reports, configure filters, and interpret charts manually.

This challenge is not exclusive to the BI domain. Industries such as administration, operations, process automation, scientific and legal research face similar problems: the need for systems combining natural language reasoning with structured access to domain-specific data.

1.2 Motivation

Isolated Large Language Models (LLMs) don't solve this problem. They suffer from multiple critical limitations:

- **Hallucinations**: They invent data that seems plausible but is incorrect
- **Lack of access to real data**: They cannot query enterprise databases
- **Absence of persistent memory**: Each conversation starts from scratch
- **Limited capabilities**: They only generate text, without executing actions in external systems

These limitations motivate the need for more sophisticated architectures that extend LLM capabilities while maintaining their natural language reasoning strengths.

1.3 Research Gap: The Adaptation Barrier

Autonomous agents combine reasoning, tool access, memory, and controlled autonomy. A **multi-agent** architecture distributes responsibilities among specialized agents, where each masters a specific area (DAX, documents, reports) while collaborating to solve complex problems.

However, current frameworks (AutoGen, LangGraph, CrewAI) present a fundamental limitation we call the **adaptation barrier**: while LLM parameters θ remain fixed post-training, the coordination topology between agents also remains static. This prevents the system from learning from accumulated experiences to improve its performance.

Identified gap: There is no theoretical framework formalizing how multi-agent systems can exhibit *structural plasticity*—the ability for connections and trust weights between agents to evolve dynamically based on observed results.

1.4 Contributions

This work presents **Lumen**, a conversational BI platform built on **Microsoft Agent Framework** with multi-agent architecture, whose purpose is to democratize access to business insights through conversational interface combining LLMs with transactional systems.

Central Thesis

> **This work demonstrates that a multi-agent architecture based on functional agency—with specialized agents, dynamic workflows, contextual handoffs, and intent detection—outperforms monolithic approaches in Business Intelligence.** Results show +20pp in accuracy (91.5% vs 71.4%) with acceptable latency (+58%). Additionally, **PEMA** (Plastic Evolving Multi-Agent) is introduced, a theoretical framework for multi-agent systems with structural plasticity inspired by Hebbian principles, allowing relationships between agents to evolve with experience.

This thesis is defended through:

1. **Theoretical framework** (Part I): Formalization of functional agency and architecture taxonomy
2. **Implementation** (Parts II-IV): Design and construction of Lumen with 10 essential capabilities
3. **Empirical validation** (Part V): Evaluation with real users and quantitative metrics

The specific contributions, hierarchized by novelty, are detailed in the Abstract (see page 1).

1.5 Document Organization

This whitepaper is organized by **capabilities**. Each section presents:

1. **Theory**: Fundamentals and industry techniques
2. **Lumen Implementation**: How Lumen implements this capability
3. **Future Versions**: Planned improvements or how it would be implemented if not existing

Figure 18: LUMEN Architecture

Component Summary

Component	Type	Function
BIWorkflow	Workflow	Orchestrates complete BI queries
GeneralWorkflow	Workflow	Handles general queries
QueryWorkflowSimple	Workflow	Optimized DAX queries
DocumentOnlyWorkflow	Workflow	Documents-only mode
ReportWorkflow	Workflow	Report embedding
GeneralAgent	Agent	Main orchestrator, composes tools
DAXAgent	Agent	DAX queries via MCP
DocumentAgent	Agent	RAG over Weaviate
ReportAgent	Agent	OAuth + Embed API
FabricAgent	Agent	Fabric REST API

Component	Type	Function
PowerBIAgent	Agent	PowerBI MCP
PDFAgent	Agent	Extraction with Docling
PlanningAgent	Agent	Task planning
SearchAgent	Agent	Semantic search
SummaryAgent	Agent	Context summaries
FormatterAgent	Agent	Output formatting

Transition to Technical Capabilities

*With the general architecture and specialized agents defined, the following sections detail each of the **ten technical capabilities** that constitute the complete implementation of Lumen. Each capability represents an empirically validated design pattern, from memory and context management (Capability 1) to structural plasticity via PEMA (Capability 10). These capabilities are independent but complementary—a system can implement a subset according to its specific requirements.*

Capability 1: Memory and Context

Theory

Types of Memory in Agentic Systems

Memory is fundamental for agents to maintain coherence and learn from previous interactions. Research in agentic systems identifies four types of memory:

Definition 3.1 (Agentic Memory System): An agentic memory system is a tuple $M = (W, E, S, P, \gamma)$ where:

- $W \subseteq Token^*$: Working Memory (token sequence of current context)
- $E: T \rightarrow Experience$: Episodic Memory (function from timestamps to experiences)
- $S: Concept \rightarrow Fact^*$: Semantic Memory (mapping from concepts to facts)
- $P: Task \rightarrow Procedure$: Procedural Memory (mapping from tasks to procedures)
- $\gamma: M \rightarrow M$: Consolidation function that moves information between memory types

Type	Description	Example	Formalization
Working Memory	Immediate context of current conversation	Recent chat messages	$W \subseteq Token^*$,
Episodic Memory	Specific past experiences with timestamps	"User asked about Q3 sales yesterday"	$E: T \rightarrow Experience$
Semantic Memory	Factual domain knowledge	"Company has 5 sales regions"	$S: Concept \rightarrow Fact^*$

Type	Description	Example	Formalization
Procedural Memory	How to execute learned tasks	"To calculate YTD, use TOTALYTD"	P: Task → Procedure

Proposition 3.1: The effective capacity of an agentic system is limited by |W|, but can be extended indefinitely through E, S, and P if an efficient retrieval mechanism exists.

Proof sketch: Let C_eff be the system's effective capacity. Without external memory, C_eff ≤ |W| (limited by context window). With external memory M_ext = E ∪ S ∪ P and a retrieval function r: Query → M_ext with O(log n) or O(1) complexity via indices, the system can access arbitrarily sized information. At each step, W contains the relevant retrieved information, allowing reasoning over knowledge much larger than |W|. Therefore, C_eff → ∞ if |M_ext| → ∞ and r is efficient. ∎

> **Connection with PEMA**: Episodic memory E provides the history H that PEMA uses for Hebbian learning (Chapter 11). Specifically, E records which agents participated in each interaction and their results, allowing calculation of reinforcement δ(o) for updating trust weights W_ij. Without persistent memory, structural plasticity would be impossible—the system couldn't "remember" which collaborations were successful.

The Limited Context Problem

LLMs have finite context windows (8K-128K tokens). As conversations grow, decisions must be made about what information to retain. Main techniques are:

Sliding Window: Keeps only the N most recent messages. Simple but loses important historical context.

Summarization: Summarizes old messages to compress information. Preserves topics but loses details.

Observation Masking: Hides tool outputs that have already been processed, keeping only relevant results.

Hybrid Approach: Combines multiple techniques according to content type.

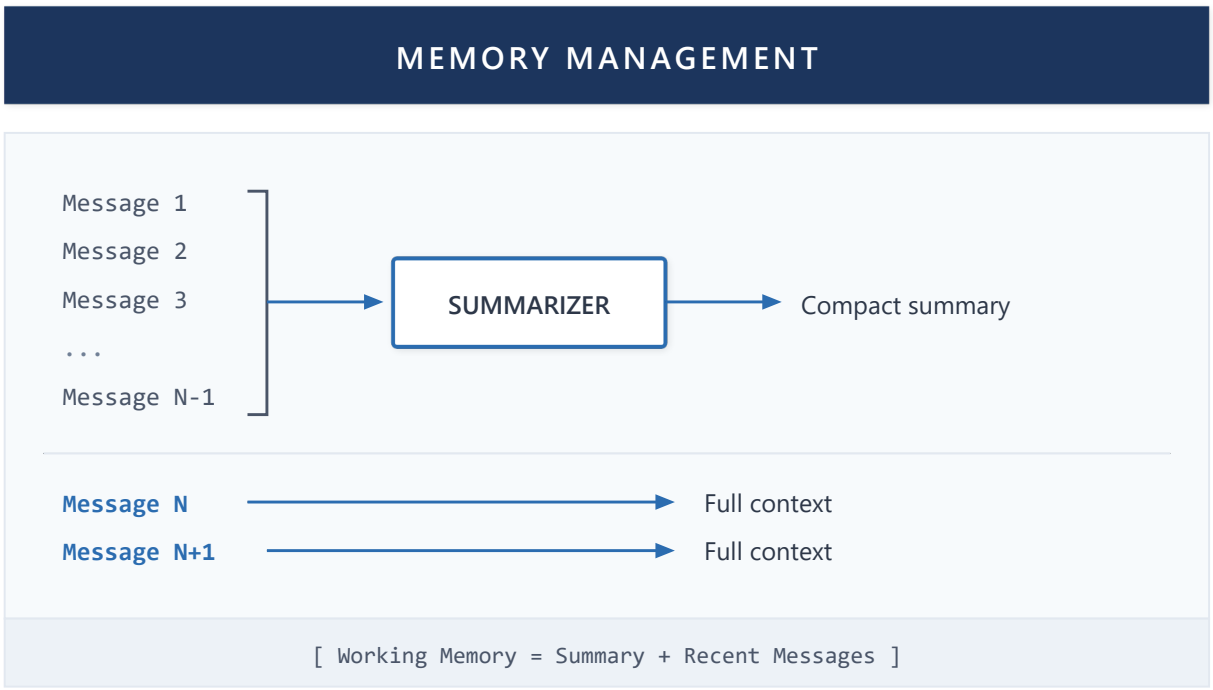


Figure 19: Memory Management

Persisted Memory Pattern

An emerging pattern in multi-agent systems is **Persisted Memory**: agents store structured data (not just text) that other agents can reuse without re-querying sources.

For example, if an agent executes a DAX query and obtains results, these are persisted in a structured format. A subsequent visualization agent can read this data directly without executing the query again.

Implementation in Lumen

Persisted Memory Pattern

Lumen implements the Persisted Memory pattern through the `persisted_memory` attribute in BaseAgent. When DAXAgent executes a query:

1. Results are stored in `persisted_memory` as structured JSON
2. FormatterAgent reads this JSON to generate visualizations
3. Message history doesn't inflate with raw data

This pattern significantly reduces token usage because large tabular data is stored outside the main conversational flow.

MessageStore for Persistence

Lumen uses Microsoft Agent Framework's MessageStore connected to SQL Server. Each session maintains its message history, allowing conversations to resume even after server restarts.

Each agent's `AgentThread` is linked to the user's `session_id`, guaranteeing isolation between different users' conversations.

Working Memory via Conversation History

Working context is maintained via the `messages` table in SQL Server. Each message includes:

- Role (user/assistant)
- Content
- Timestamp
- Metadata (model used, tokens, etc.)

Capability 2: Routing and Intent

Theory

The Routing Problem

In multi-agent systems, determining which agent should handle each message is critical. Incorrect routing leads to suboptimal responses or errors. Main strategies are:

Rule-Based Routing

Uses explicit rules based on keywords or patterns:

```
IF message contains "DAX" OR "measure" → DAXAgent
IF message contains "report" OR "dashboard" → ReportAgent
ELSE → GeneralAgent
```

Advantages: Fast, predictable, no LLM cost **Disadvantages**: Fragile to language variations, doesn't scale

LLM-Based Routing

The LLM itself classifies the message intent and decides the agent:

```
System: Classify this message into one of the categories:
- QUERY_DAX: Data and metrics queries
- REPORT: Report requests
- DOCUMENT: Document questions
- GENERAL: General conversation

User: "How many sales were there in January?"
LLM: QUERY_DAX
```

Advantages: Flexible, handles natural variations **Disadvantages**: Additional latency, token cost

Semantic Routing

Uses embeddings to compare the message with descriptions of each agent:

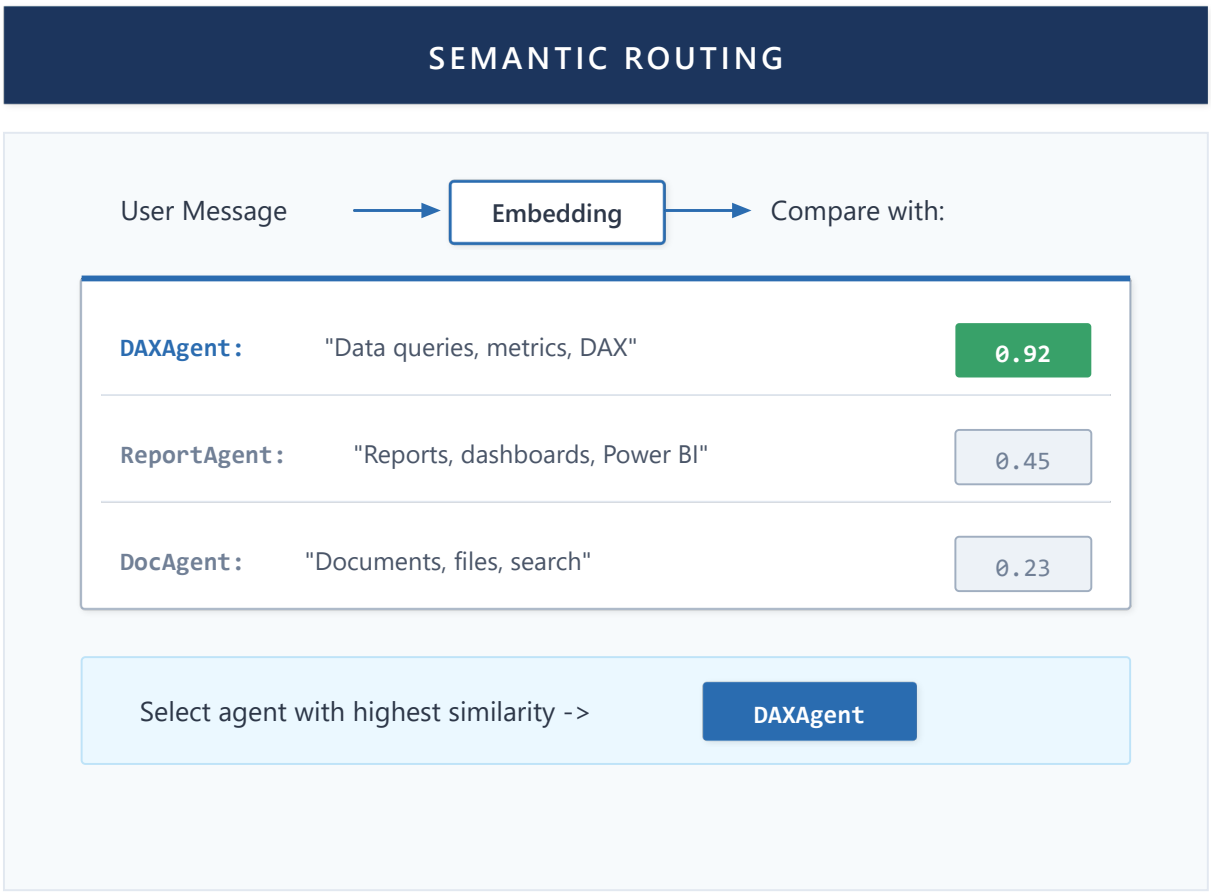


Figure 20: Semantic Routing

Advantages: Very fast (embeddings only), semantically robust **Disadvantages**: Requires description tuning, can confuse similar intents

Two-Layer Routing

Combines rule-based and LLM to optimize cost/accuracy:

1. **Layer 1 (Keywords)**: Fast rules for obvious cases
2. **Layer 2 (LLM)**: Classification for ambiguous cases

> **Connection with PEMA**: The routing strategies described use **static weights**. PEMA (Chapter 11) extends these approaches by allowing weights to evolve through Hebbian learning. For example, if DAXAgent consistently resolves type X queries better than those initially routed to GeneralAgent, the weight W_{DAX,X} would automatically increase. This transforms routing from manual configuration to a **self-optimizing** system.

Implementation in Lumen

Two-Layer Routing in BIWorkflow

Lumen implements two-layer routing in BIWorkflow:

Layer 1 - Keyword Detection: Analyzes the message looking for keywords that clearly indicate intent:

- "show the report", "dashboard" → ReportWorkflow
- "search in documents", "according to the file" → DocumentWorkflow
- Numbers, "how much", "sales", "DAX" → QueryWorkflow

Layer 2 - LLM Classification: If there's no keyword match, the LLM classifies between:

- `query`: Data queries
- `report`: Power BI reports
- `general`: Conversation/other

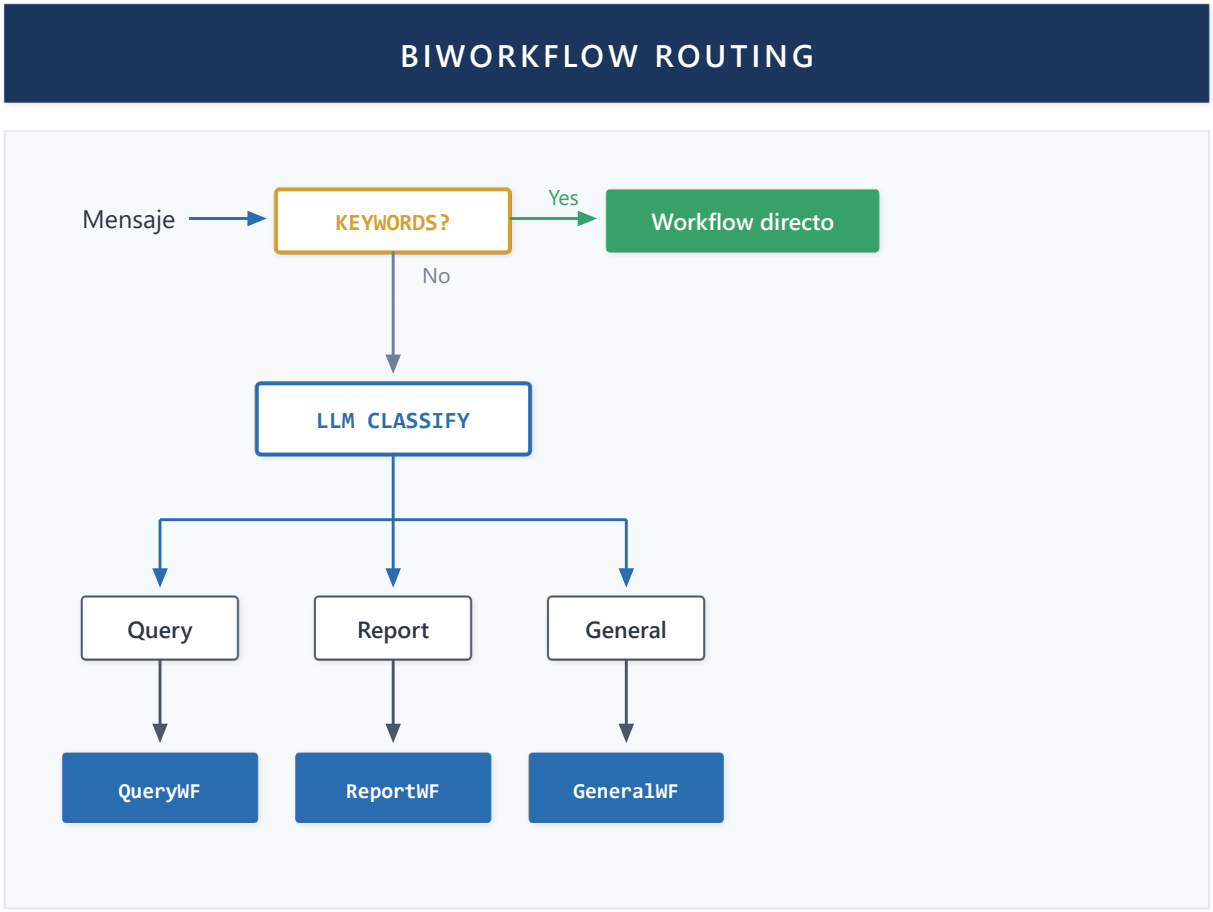


Figure 21: BIWorkflow Routing

Advantages of the Approach

- Keywords handle ~60% of cases without LLM cost
- LLM intervenes only in ambiguous cases
- Easy to add new keywords without changing logic
- Safe fallback to GeneralWorkflow

Capability 3: RAG - Knowledge Retrieval

Theory

RAG Evolution

Retrieval-Augmented Generation (RAG) has evolved significantly since its introduction:

Generation 1 - Basic Vector Search: Query embedding, similarity search, chunk injection into prompt.

Generation 2 - Hybrid Search: Combines vector search with keyword search (BM25) for better precision.

Generation 3 - Late Chunking: Instead of pre-embedding chunking, embeds the complete document and extracts relevant context post-search.

Generation 4 - GraphRAG: Builds knowledge graphs from documents, enabling reasoning about entity relationships.

Generation 5 - Agentic RAG: The agent dynamically decides which documents to search, when it needs more context, and can make multiple iterative queries.

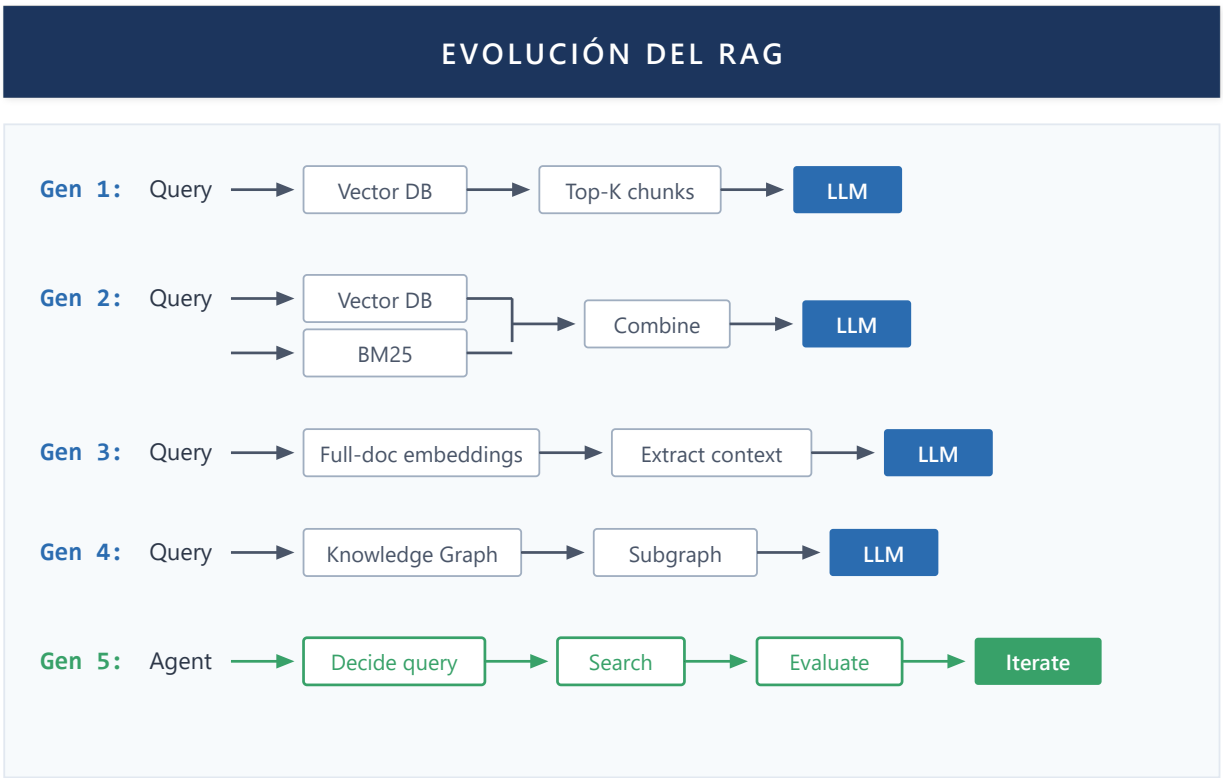


Figure 22: RAG Evolution

Chunking Strategies

How documents are divided into chunks directly affects RAG quality:

Strategy	Description	Best for
Fixed Size	N-token chunks	Homogeneous documents
Semantic	By paragraphs/sections	Structured documents
Sentence Window	Sentence + surrounding context	High precision

Strategy	Description	Best for
Hierarchical	Nested chunks (document > section > paragraph)	Long documents

Implementation in Lumen

Hybrid Search in Weaviate

Lumen uses Weaviate as vector database with hybrid search:

1. **Vector Search**: Embeddings generated with Azure OpenAI (text-embedding-3-small)
2. **Keyword Search**: Native BM25 from Weaviate
3. **Fusion**: Fusion algorithm weights both results

Speculative Gap RAG (SG-RAG)

Design contribution: Traditional RAG only retrieves what exists. SG-RAG detects what *should exist but doesn't*.

Definition 3.1 (Gap Detection): Let Q be a query and C = {c_1, c_2, ..., c_n} the chunk corpus. The system generates an "ideal chunk" c* that would perfectly answer Q. A gap is detected when:

$$\max_{c_i \in C} \text{similarity}(c^*, c_i) < \theta$$

Mechanism:

Component	Function
Ideal Chunk Generation	LLM generates description of perfect chunk for query
Gap Detection	If no real chunk exceeds threshold θ similarity → gap
Gap Memory	Persists: (query, timestamp, user, suggested_section)
Gap Analytics	Prioritizes gaps by frequency × recency × importance
Gap Resolution	When indexing new document, checks if it covers existing gaps

Priority formula:

```
gap_priority(G) = frequency(G) × recency(G) × user_importance(G)
```

Connection with PEMA: Extends plasticity to knowledge. The system not only learns which agents work better (inter-agent trust weights), but what knowledge it lacks (gap weights). Bidirectional plasticity: structure + content.

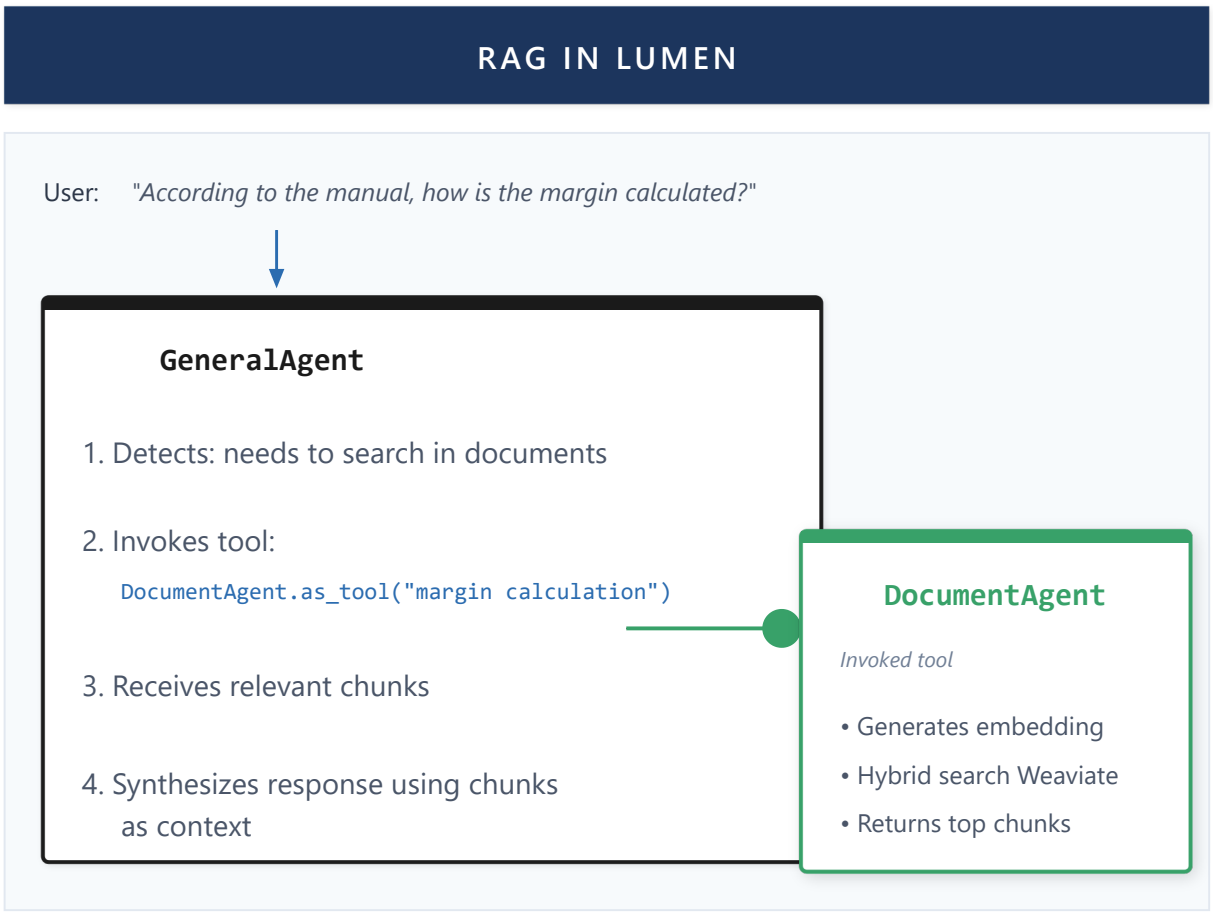


Figure 23: RAG in LUMEN

Capability 4: Orchestration, Workflows and Agent Composition

Theory

Workflow patterns (Sequential, Parallel, Hierarchical, Router, Reflexive) are described in **Section 2.2**. Here we detail how Lumen implements them.

Agents as Building Blocks

In multi-agent systems, agents must be composable. A well-designed agent can:

- Execute tasks autonomously
- Be invoked as a tool by other agents
- Share memory/context when necessary
- Scale independently

Tool System

Tools are the extensibility mechanism provided by Microsoft Agent Framework. A tool is a function decorated with `@ai_function` (from `semantic_kernel.functions`) that extends agent capabilities to execute actions in the real world.

Definition 4.2 (Tool): A tool T is a function f: Args → String decorated with `@ai_function` that:

- Receives typed parameters with semantic descriptions
- Returns a string interpretable by the LLM
- Includes a docstring describing when and how to use it

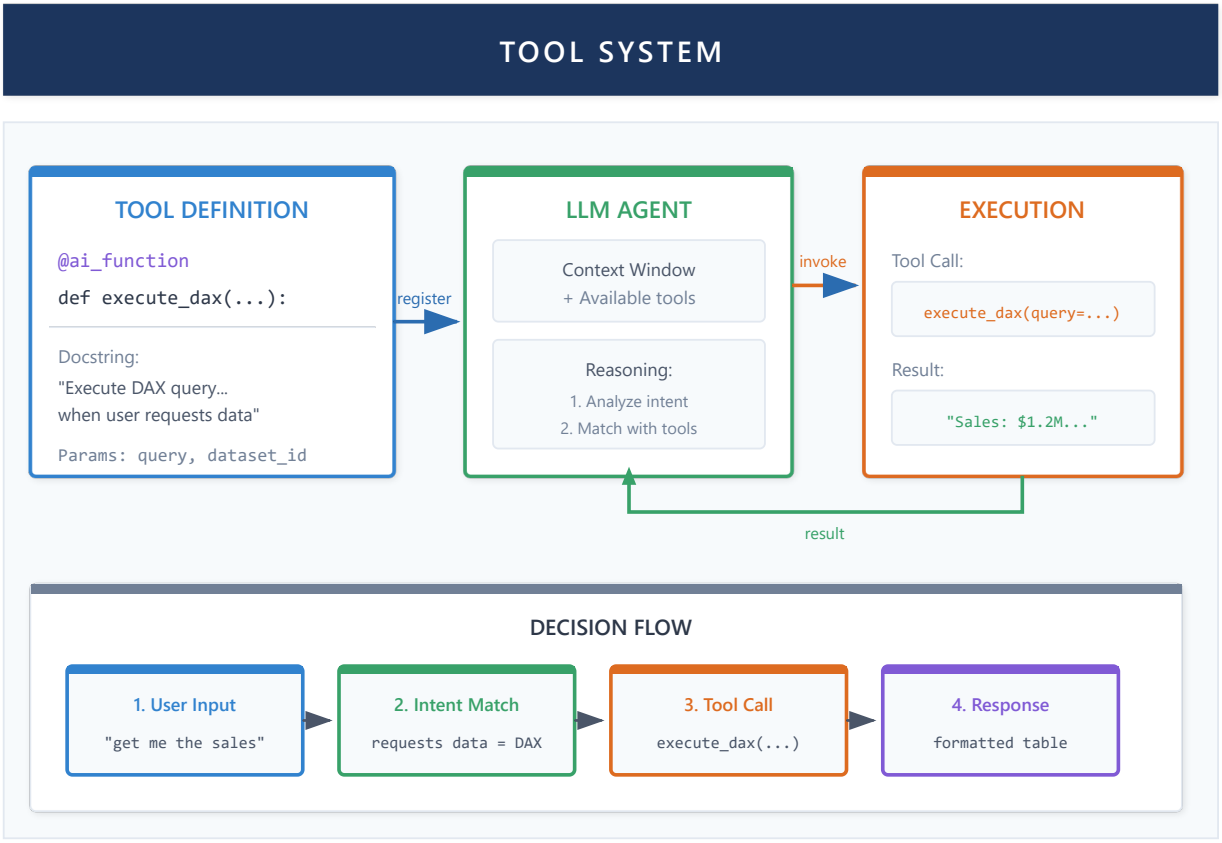


Figure 23b: Tool System - Definition, registration, and decision flow

Agent-as-Tool Pattern

Native pattern from Microsoft Agent Framework. **Agent-as-Tool** allows an agent to expose its capabilities as a tool that other agents can invoke.

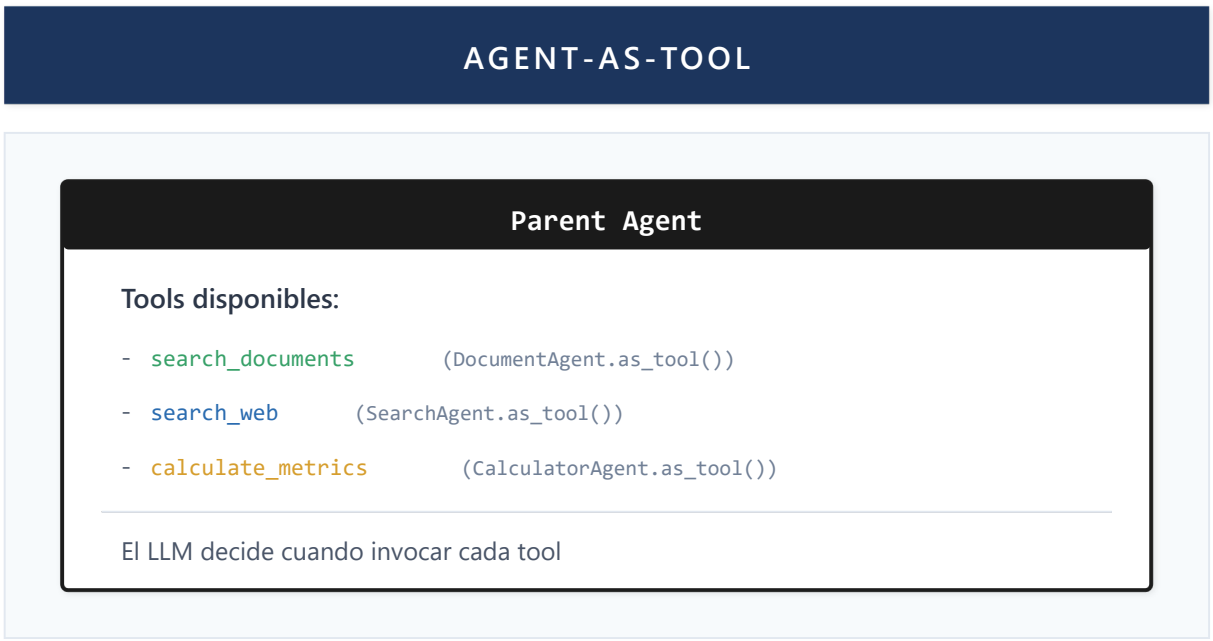


Figure 24: Agent as Tool Pattern

Advantages:

- LLM reasons about when to use each agent
- Flexible composition without hardcoding dependencies
- Same agent can serve multiple contexts

Handoff Pattern

The Handoff pattern allows an agent to **completely** transfer control to another agent when it detects that the task exceeds its expertise or requires specialized capabilities.

Definition 4.3 (Handoff): A handoff H from agent A_i to agent A_j is a tuple H = (trigger, target, context, query) where:

- **trigger**: Condition that triggers the transfer (e.g., "requires DAX execution")
- **target**: Target agent with required expertise
- **context**: State and metadata transferred (session_id, thread, history)
- **query**: Original user input preserved

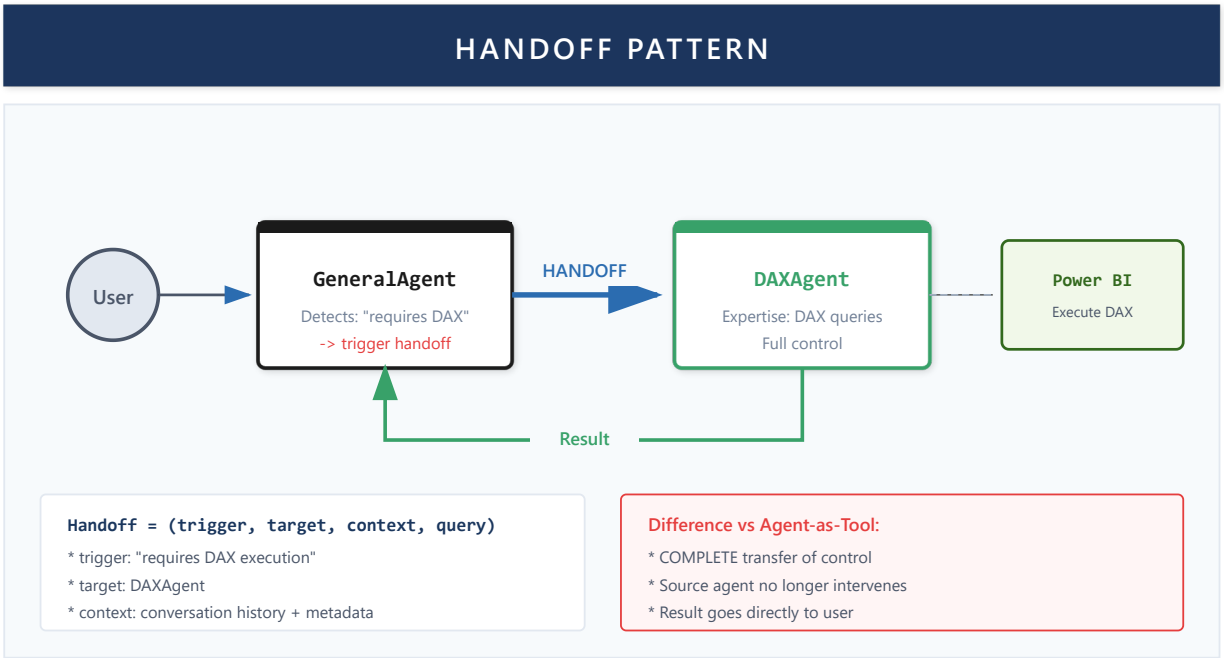


Figure 25: Handoff Pattern - Complete Transfer of Control

Types of Handoff

Type	Description	Example
Expertise-based	Current agent lacks specialized knowledge	GeneralAgent -> DAXAgent for data queries
Capability-based	Required tools not available	PowerBIAgent -> FabricAgent for workspace APIs
Context-based	Context indicates another domain	DocumentAgent -> SearchAgent when no documents
Planning-based	Multi-step query requires orchestration	FabricAgent -> PlanningAgent for "execute sales on Contoso"

Detection Mechanism

Handoff is triggered through two complementary mechanisms:

1. **Proactive detection by workflow**: The workflow evaluates if the current message requires an agent change based on domain keywords. If the user was with DocumentAgent but asks about "dollar price", a handoff to SearchAgent is detected.

2. **Explicit request by agent**: When an agent internally detects it cannot complete the task (e.g., FabricAgent receives "execute Q3 sales" but has no dataset_id), it requests handoff to PlanningAgent which can orchestrate the complete sequence.

Context Preservation

During handoff the following are preserved:

- **Conversation thread**: Target agent continues in the same multi-turn thread
- **Session metadata**: user_id, session_id, available tokens
- **Original query**: User input unmodified
- **Handoff reason**: For logging and debugging

Flow Example

```
User: "Execute 2024 sales on the Contoso model"
         |
FabricAgent: Detects execution verb + model name (no ID)
         |
         v
     [HANDOFF to PlanningAgent]
         |
PlanningAgent: Creates step plan
         |
     [Execute each step via tools]
         |
         v
     [Result to user]
```

Agent Hierarchy

Agents can be organized hierarchically:

- **Base Agent**: Common functionality (logging, streaming, tools)
- **Specialized Agents**: Inherit from base, add specific capabilities
- **Composite Agents**: Compose multiple specialized agents

Implementation in Lumen

Router Pattern: BIWorkflow

Lumen's main entry point is BIWorkflow, implementing the Router pattern:

1. Receives user message
2. Classifies intent (keywords + LLM)
3. Delegates to specialized sub-workflow
4. Streams response to frontend

Available sub-workflows:

- **QueryWorkflowSimple**: DAX queries
- **ReportWorkflow**: Report embedding
- **GeneralWorkflow**: General conversation with RAG
- **DocumentOnlyWorkflow**: Strict documents-only mode

Sequential Pattern: QueryWorkflowSimple

For DAX queries, Lumen uses a sequential pipeline:

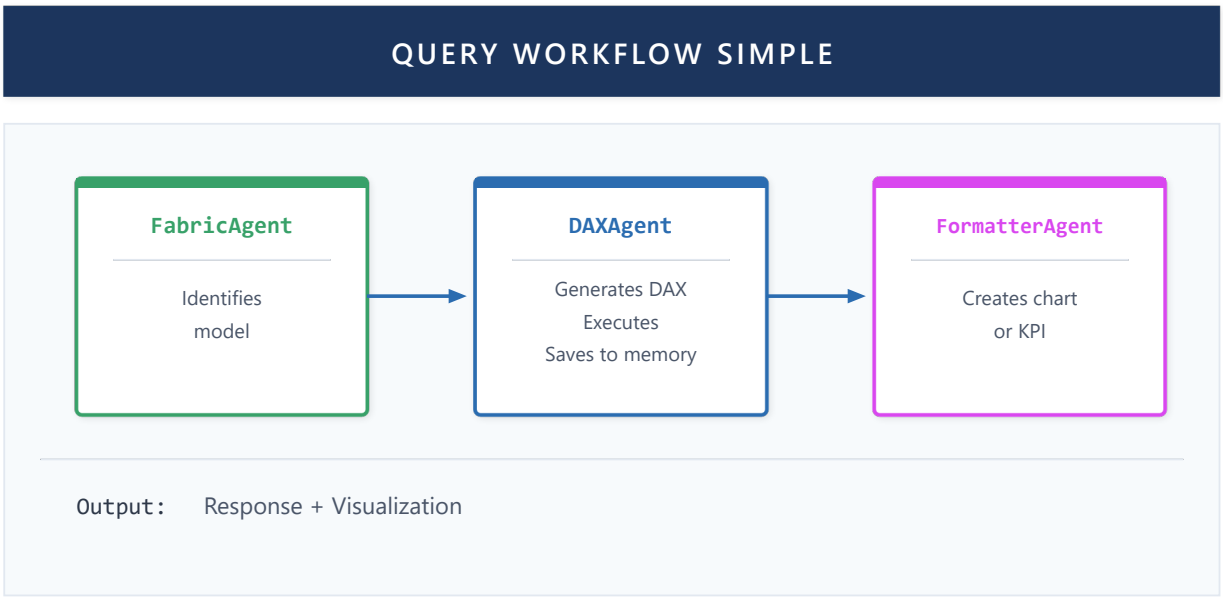


Figure 25: Query Workflow Simple

Each agent in the pipeline:

1. **FabricAgent**: Identifies appropriate semantic model
2. **DAXAgent**: Generates and executes DAX query
3. **FormatterAgent**: Converts results to visualization

Agent Hierarchy

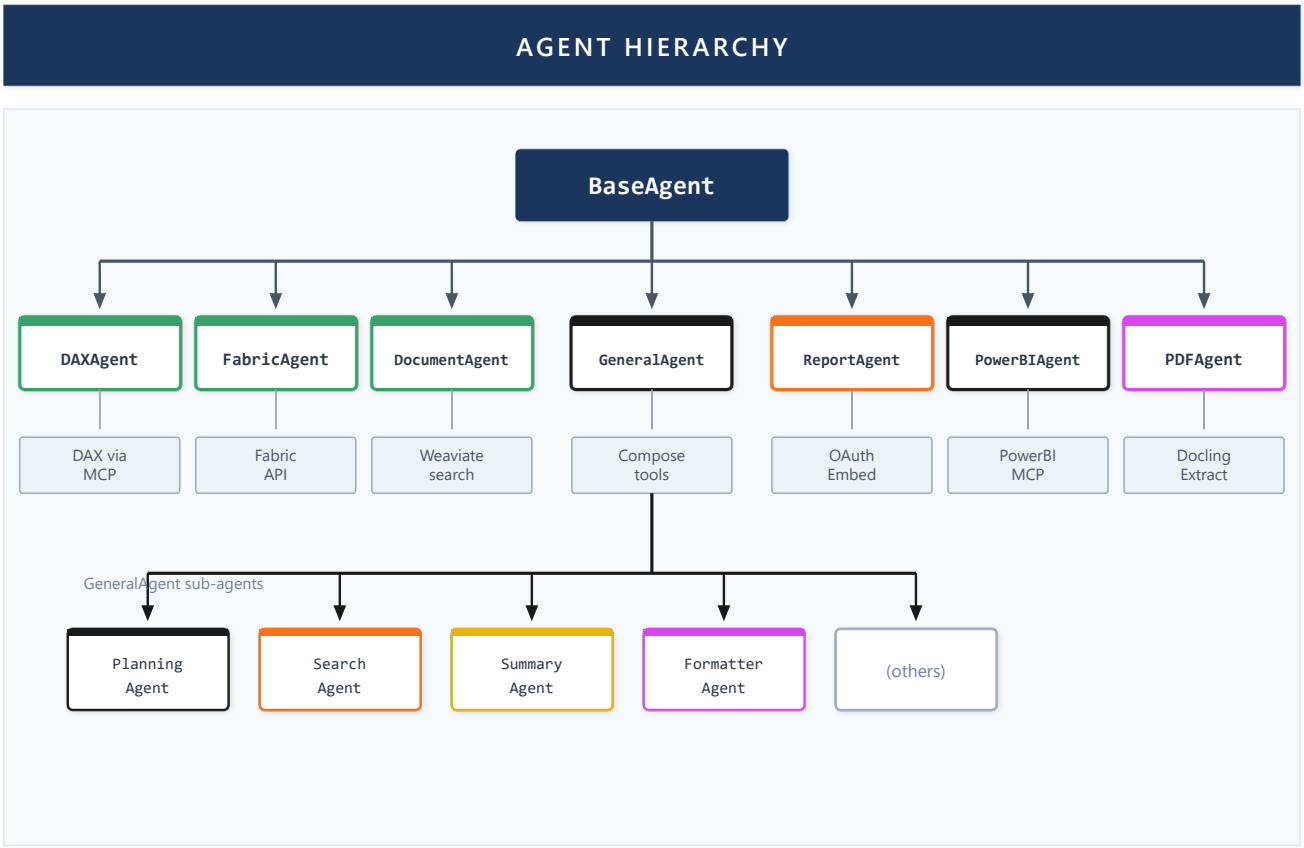


Figure 26: Agent Hierarchy

Specialized Agents

Agent	Specialization	Own Tools
DAXAgent	DAX queries	execute_dax (MCP), validate_dax
FabricAgent	Fabric API	list_workspaces, get_model_info
DocumentAgent	RAG	search_chunks (Weaviate)
ReportAgent	Power BI Embed	get_embed_token
FormatterAgent	Visualization	(generates JSON blocks)
SearchAgent	Web search	serper_search
SummaryAgent	Summarization	(internal map-reduce)
PowerBIAgent	DAX Modeling	(best practices expert)

Agent Development Guide

This section provides a conceptual guide for developing new agents in LLM-based multi-agent systems.

Agent Anatomy

All agents inherit from `ChatCompletionAgent` (Microsoft Agent Framework). BaseAgent extends this class providing:

- Azure OpenAI integration (Fast and Response pools)
- Tool management
- Response streaming (SSE)
- MCP integration (Model Context Protocol)
- Memory persistence

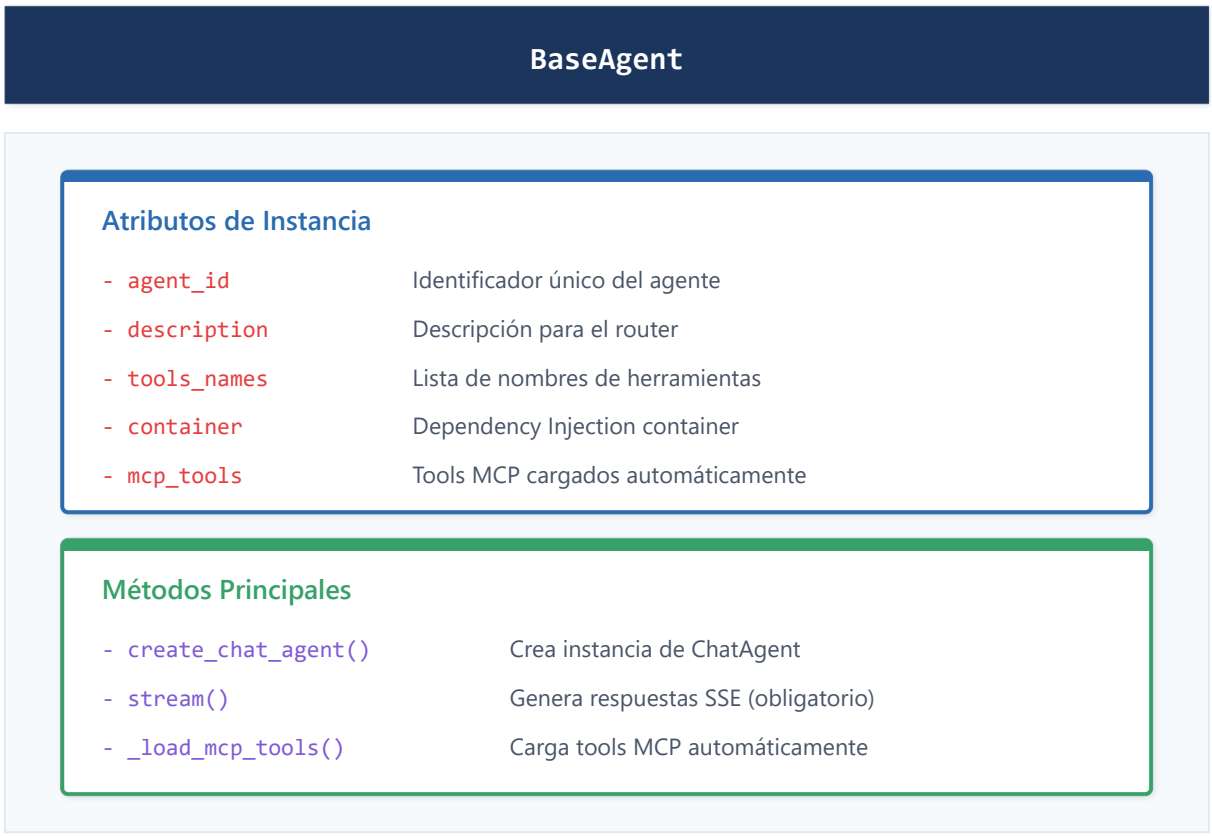


Figure 27: BaseAgent Class Structure

AgentContext

The context received by each agent contains all session information:

Field	Description
session_id	Unique session identifier
user_id	Authenticated user identifier
conversation_history	History of previous messages
metadata	Additional data (has_documents, reasoning_enabled, image_data)
thread	Agent Framework thread for handoffs between agents

Agent Creation Process

Developing a new agent follows four fundamental steps:

Step 1: Define the Class - Create a file extending BaseAgent. Configure: agent_id, description, tools, instructions.

Step 2: Implement stream() - The `stream()` method is the agent's core: receives context, configures thread_id, creates ChatAgent, iterates over LLM response, handles errors.

Step 3: Register in Container - The agent must be registered in the Container for Dependency Injection.

Step 4: Configure Routing - If the agent should be selected automatically, add keywords in router_service.

Tool System

Tools extend agent capabilities by allowing them to execute concrete actions.

Aspect	Description
Decorator	Uses `@ai_function` from Agent Framework
Typed parameters	Uses `Annotated` with `Field` for descriptions
String return	Always returns text the LLM can interpret
Docstring	Describes the function so the LLM knows when to use it

Tools can access system services (Database, Embedding, Weaviate, OAuth) through the container. They must be synchronous; for async operations use `asyncio.run()`.

Advanced Agent Patterns

Dual-Model Pattern

Pool	Model	Usage
Fast	gpt-4-turbo	Quick responses, routing, internal calls
Response	gpt-5	Final responses, extended reasoning, long context

The agent dynamically decides which pool to use based on context size (>50K chars → Response), task type, and `reasoning_enabled` flag.

Persisted Memory Pattern

Type	Storage	Duration
Working Memory	`AgentThread` (MS Agent Framework)	Session
Persisted Memory	`persisted_memory` field → Database	Permanent
Semantic Memory	Weaviate (embeddings)	Permanent
Structural Memory	W_ij trust matrix (PEMA)	Adaptive

MCP Integration

Model Context Protocol enables connecting external tool servers:

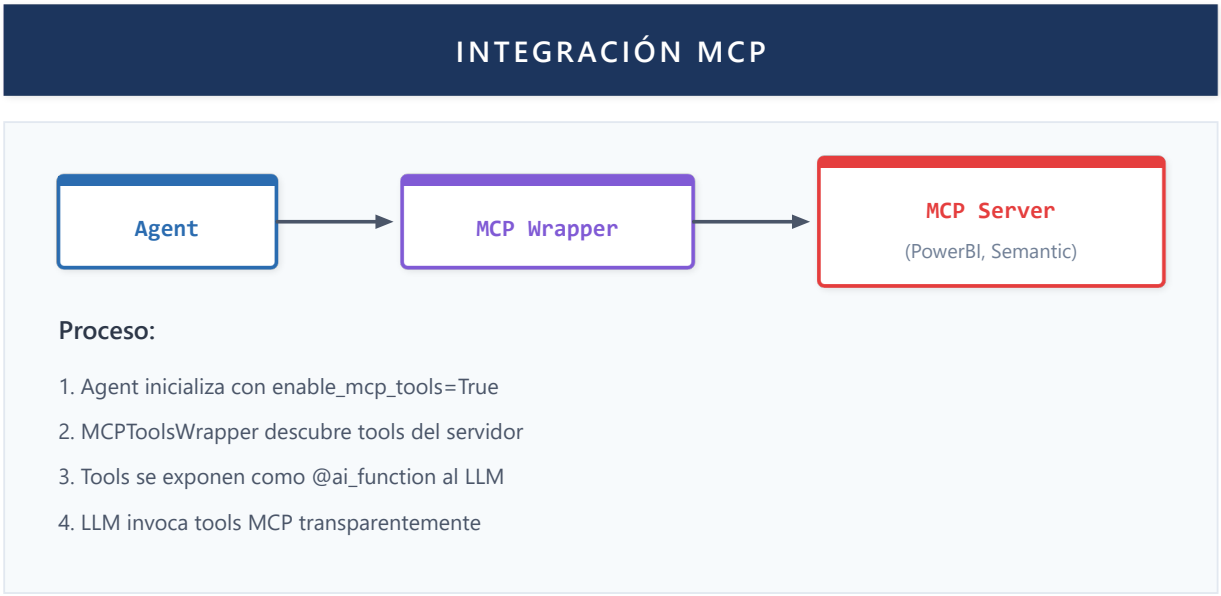


Figure 28: MCP Integration

Agent Categories

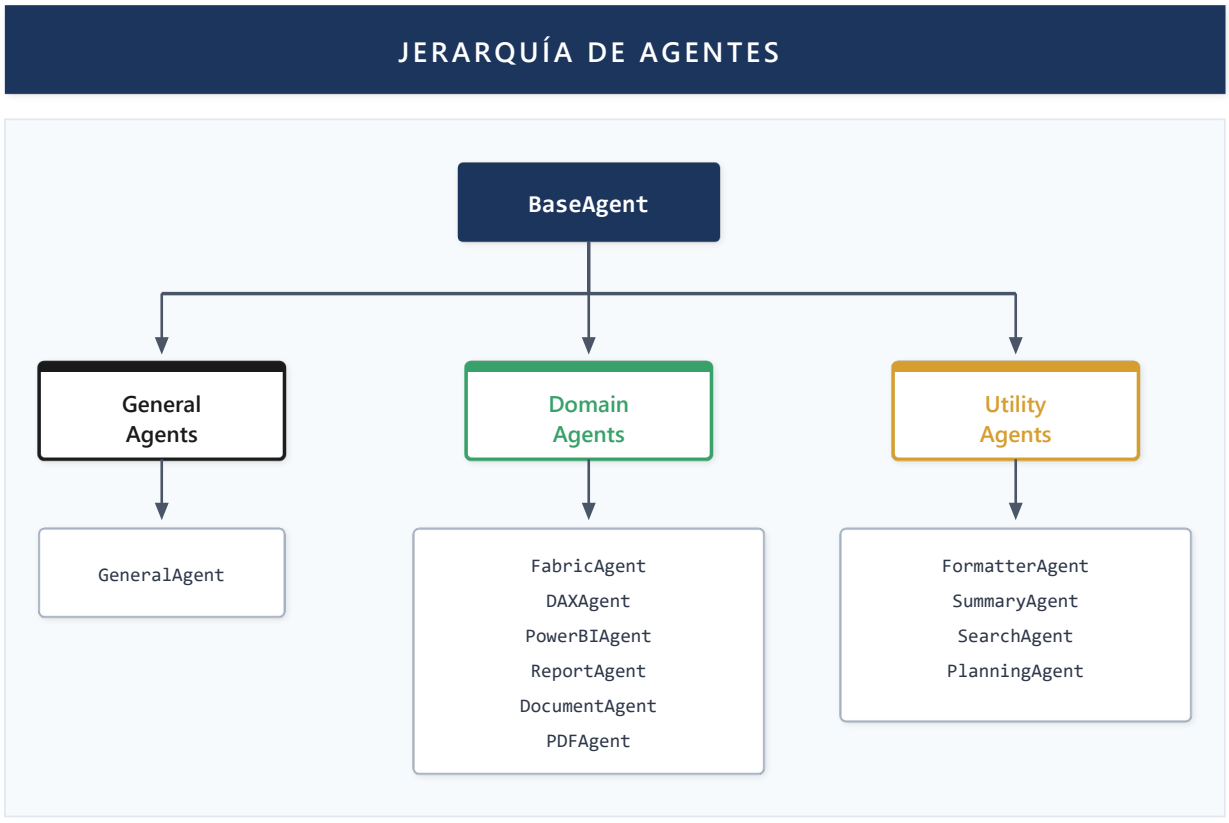


Figure 29: Agent Category Hierarchy

Category	Purpose	Examples
General	General conversation, fallback	GeneralAgent
Domain	Specialized in BI areas	FabricAgent, DAXAgent, DocumentAgent
Utility	Cross-cutting support functions	FormatterAgent, SummaryAgent, SearchAgent

Best Practices for Agent Development

Area	Practice	Description
Design	Single Responsibility	One agent, one domain
Design	Clear instructions	Specific system prompt
Design	Atomic tools	Each tool does one thing well
Performance	Lazy loading	Initialize resources only when needed
Performance	Limit tools	Maximum 5-7 tools per agent
Security	Validate inputs	Never trust user data
Security	SQL parameters	Never concatenate strings
Testing	Unit tests	Test tools with mocks
Testing	Stream tests	Verify correct chunks

Future Work

Aspect	Current State	Future Direction
Workflows	Sequential, Router, Handoffs	Reflection loops, auto-correction
Composition	Static Agent-as-Tool	Dynamic Agent Discovery
Handoffs	Based on expertise/capability	Predictive handoffs with enriched context
Intents	Keyword + LLM detection	Multi-turn intent with goal memory
Planning	Implicit in prompts	Explicit multi-step planning

Open research questions:

- How does orchestration (workflows + handoffs + intents) scale to hundreds of agents?
- What heuristics optimize the decision between handoff vs. tool call?

Orchestrated agents need access to real-world data to be useful. The next capability describes how Lumen connects its agents to enterprise data sources through standardized protocols like MCP and OAuth.

Capability 5: Data Source Integration

Theory

Tool Calling in LLMs

Modern LLMs support **tool calling** (function calling): the model can decide to invoke external functions and use their results.

```
User: "What's the price of AAPL?"

LLM reasons: I need real-time market data
LLM output: {
  "tool": "get_stock_price",
  "arguments": {"symbol": "AAPL"}
}

System: Executes tool → $150.25

LLM: "The current price of AAPL is $150.25"
```

Model Context Protocol (MCP)

MCP is an emerging standard for connecting LLMs with data sources:

- **MCP Servers**: Expose data/functionality via standard protocol
- **MCP Clients**: LLMs/agents that consume these servers
- **Transport**: stdio, HTTP, WebSocket

MCP Advantages:

- Reusable integrations across applications
- Clear separation between business logic and LLM
- Growing ecosystem of pre-built servers

OAuth for Enterprise APIs

Accessing enterprise APIs (Power BI, Fabric, SharePoint) requires OAuth 2.0 authentication:

1. User initiates authentication flow
2. System obtains access tokens
3. Tokens used for API calls
4. Refresh tokens maintain active session

Implementation in Lumen

MCP for Power BI / Fabric

Lumen uses a specialized MCP server for Power BI operations:

Tools available via MCP:

- `execute_dax`: Executes DAX queries against semantic models
- `validate_dax`: Validates DAX syntax without execution
- `list_tables`: Lists tables in a model
- `list_measures`: Lists available measures
- `get_model_info`: Model metadata

The MCP server connects via Fabric's XMLA endpoint, enabling operations not available through REST API.

FabricAgent for REST API

Operations using Fabric REST API:

- List user workspaces
- Get datasets/reports
- Model metadata
- Permissions and capabilities

FabricAgent abstracts these operations, handling pagination, rate limiting, and errors.

Multi-Tenant OAuth

Lumen implements complete OAuth flow with Microsoft:

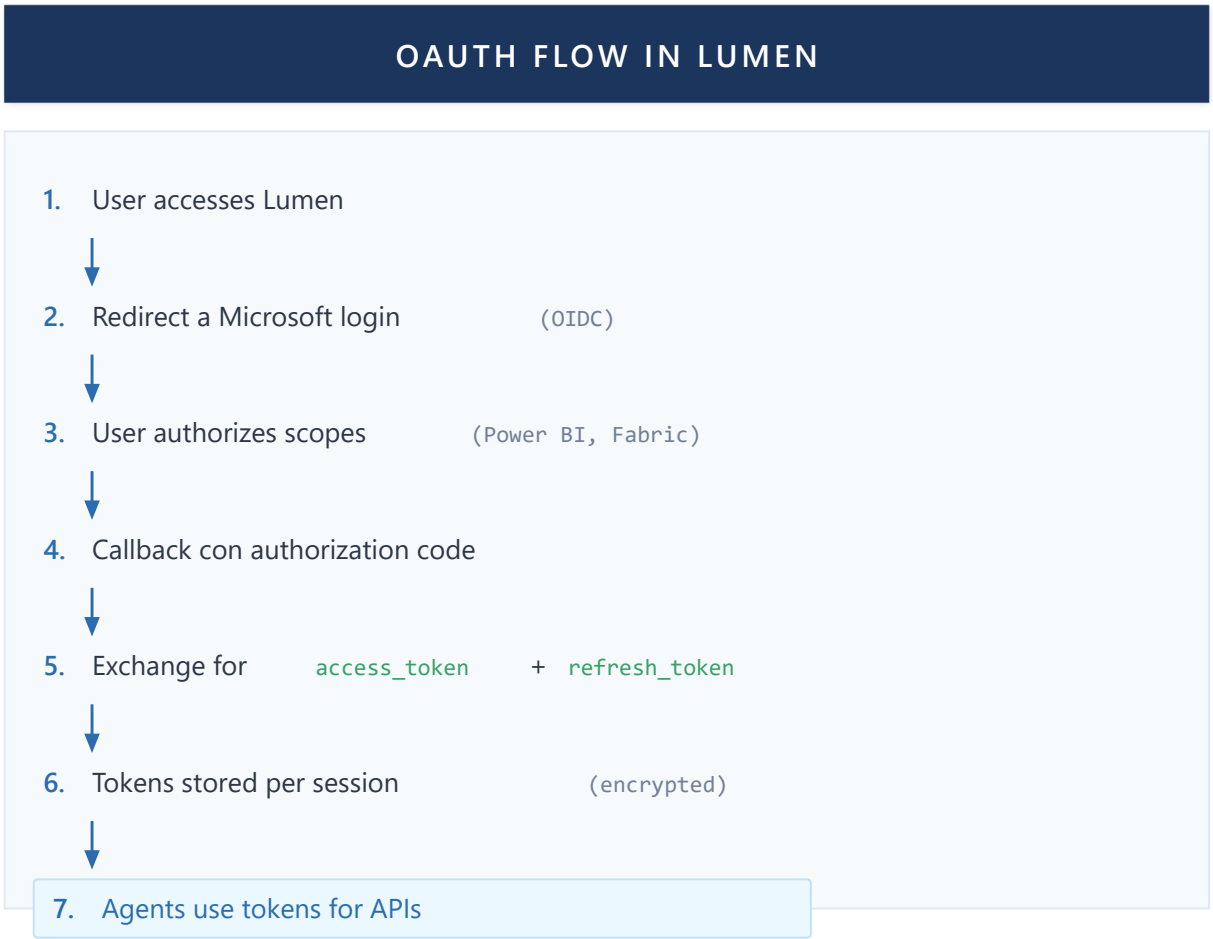


Figure 30: OAuth Flow in LUMEN

Dimension Value Indexing

Unique Lumen capability: indexing dimension values from Power BI models:

1. User selects model
2. Background job extracts dimension column values
3. Values indexed in Weaviate with embeddings
4. Queries like "sales for customer Acme" automatically map to DAX filters

This enables natural language over filters without users knowing exact values.

Future Work

Aspect	Current State	Future Direction
Protocols	MCP, OAuth, Fabric API	More MCP servers (Slack, Teams, SAP)
Authentication	Per user	Service principals for automation
Catalog	Manual	Automatic source discovery

Data obtained from enterprise sources must be presented to users comprehensibly. The next capability addresses how to transform query results into interactive visualizations and structured outputs.

Capability 6: Visualization and Output

Theory

Structured Output in LLMs

LLMs can generate structured output (JSON, XML) in addition to free text. This enables:

- Reliable response parsing
- Direct UI integration
- Schema validation

Custom Blocks

A common pattern is defining special "blocks" that the LLM generates and the frontend renders:

````
LLM Output:
"Q3 sales were $1.2M, a 15% increase.

```chart-data
{
  "type": "bar",
  "title": "Sales by Quarter",
  "data": [...]
}
```
"
````

```
Frontend:
- Parses special blocks
- Renders chart where appropriate
- Displays normal text as markdown
```

Streaming and Rendering

With streaming, the frontend receives incremental chunks. Rendering must:

- Accumulate text until blocks are complete
- Render blocks only when complete
- Handle intermediate states gracefully

Implementation in Lumen

Block Types

Lumen defines three types of special blocks:

chart-data: Interactive charts

- Types: bar, line, pie, area
- Data in standardized format
- Rendered with Recharts

kpi-data: Key performance indicators

- Main value with formatting
- Comparison with previous period
- Trend (up/down/neutral)

embed-report: Embedded Power BI reports

- Report and page ID
- Filters to apply
- Embed token

FormatterAgent

FormatterAgent is responsible for generating visualizations:

1. Receives data from DAXAgent (via persisted_memory)
2. Analyzes data structure
3. Decides appropriate visualization type
4. Generates corresponding JSON block

Selection heuristics:

- One dimension + one metric → Bar chart
- Time series → Line chart
- Proportion of a total → Pie chart

- Single important value → KPI

Streaming Prevention for Embeds

Embedded reports require tokens that can expire. Lumen implements:

- embed-report block not rendered during streaming
- Only when message is complete, frontend requests fresh token
- Token obtained just before rendering

Frontend Rendering

The frontend parses message content looking for blocks:

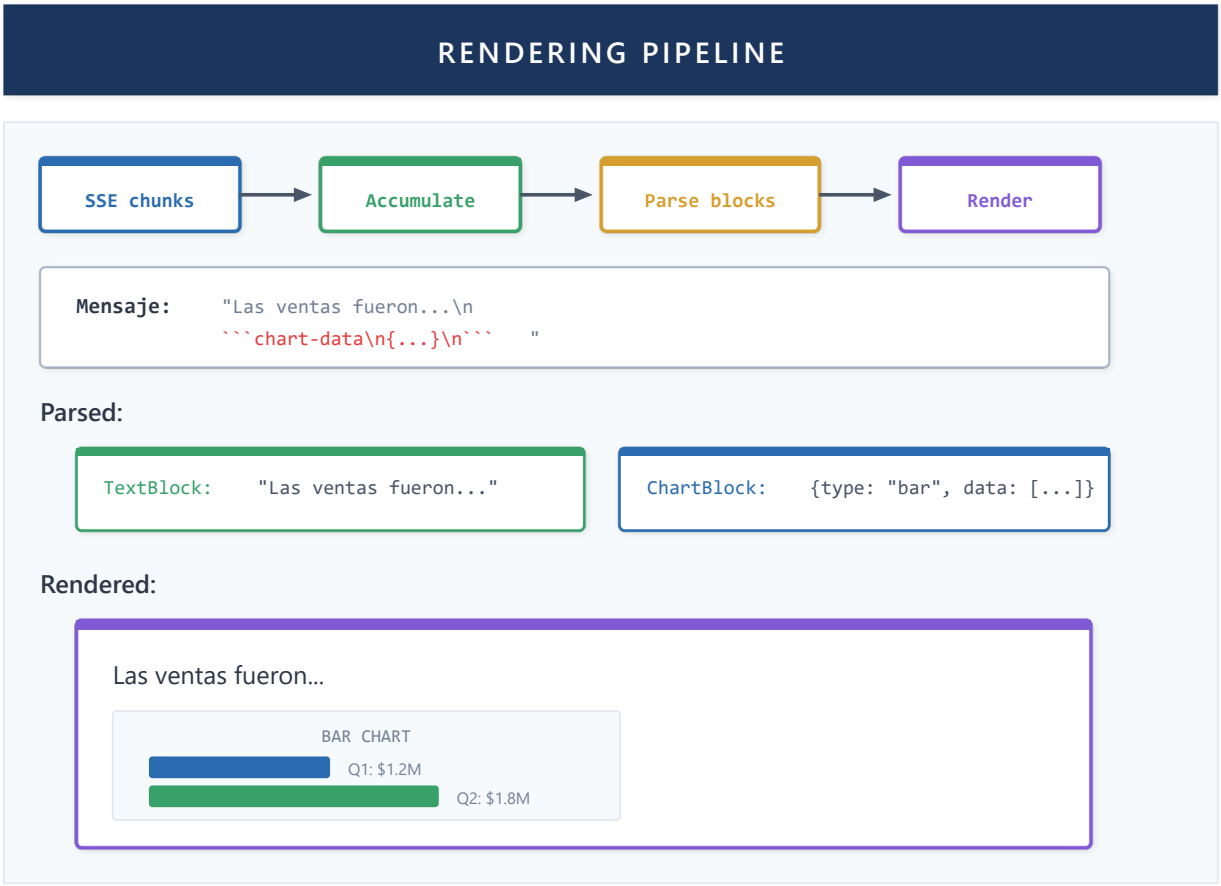


Figure 31: Rendering Pipeline

Future Work

Aspect	Current State	Future Direction
Interactivity	Static blocks	Interactive visualizations (drill-down, filters)
Generation	Type selection	Auto-selection based on data
Multimodal	Text/charts only	Narrative generation with audio

A system that accesses enterprise data and visualizes it must ensure that only authorized users access appropriate information. The next capability details security mechanisms, from OAuth authentication to hallucination prevention.

Capability 7: Security

Theory

OAuth 2.0 and OpenID Connect

OAuth 2.0 is the standard for delegated authorization. OpenID Connect (OIDC) adds an identity layer over OAuth.

Authorization Code Flow (used for web apps):

1. Redirect to authorization server
2. User authenticates and authorizes
3. Server returns authorization code
4. App exchanges code for tokens
5. Access token used for APIs

Tokens:

- **ID Token**: User identity (JWT)
- **Access Token**: Authorization for APIs
- **Refresh Token**: Obtain new access tokens

JWT Validation

JWT tokens must be rigorously validated:

- **Signature**: Verify against issuer's public key
- **Issuer**: Must match expected issuer
- **Audience**: Must include your application
- **Expiration**: Token must not be expired
- **Nonce**: Prevents replay attacks (OIDC)

Session Management

Web sessions require:

- **Session ID**: Unique identifier per user
- **Storage**: Server-side (DB) or client-side (JWT)
- **Expiration**: Inactivity timeout
- **Isolation**: Users cannot access others' data

Principle of Least Privilege

Each component should have only necessary permissions:

- Agents only access data they need
- Tokens have minimum required scopes
- Queries filtered by tenant/user

Implementation in Lumen

OIDCService

Lumen implements complete OIDC flow with Microsoft:

Initiate authentication:

- Generates unique state and nonce
- Constructs authorization URL with required scopes
- Redirect to Microsoft login

Callback:

- Validates state against stored value
- Exchanges code for tokens
- Validates ID token (signature, issuer, audience, nonce)
- Stores tokens associated with session

Requested scopes:

- openid, profile, email (basic OIDC)
- https://analysis.windows.net/powerbi/api/.default (Power BI)

Token Storage

Tokens are stored in SQL Server associated with session_id:

- Access tokens encrypted at-rest
- Refresh tokens separated and protected
- Expiration tracked for proactive refresh

Session Isolation

Each request includes session cookie:

- Middleware extracts session_id
- Context includes session for all handlers
- Database queries filtered by session
- Agents receive context with session_id

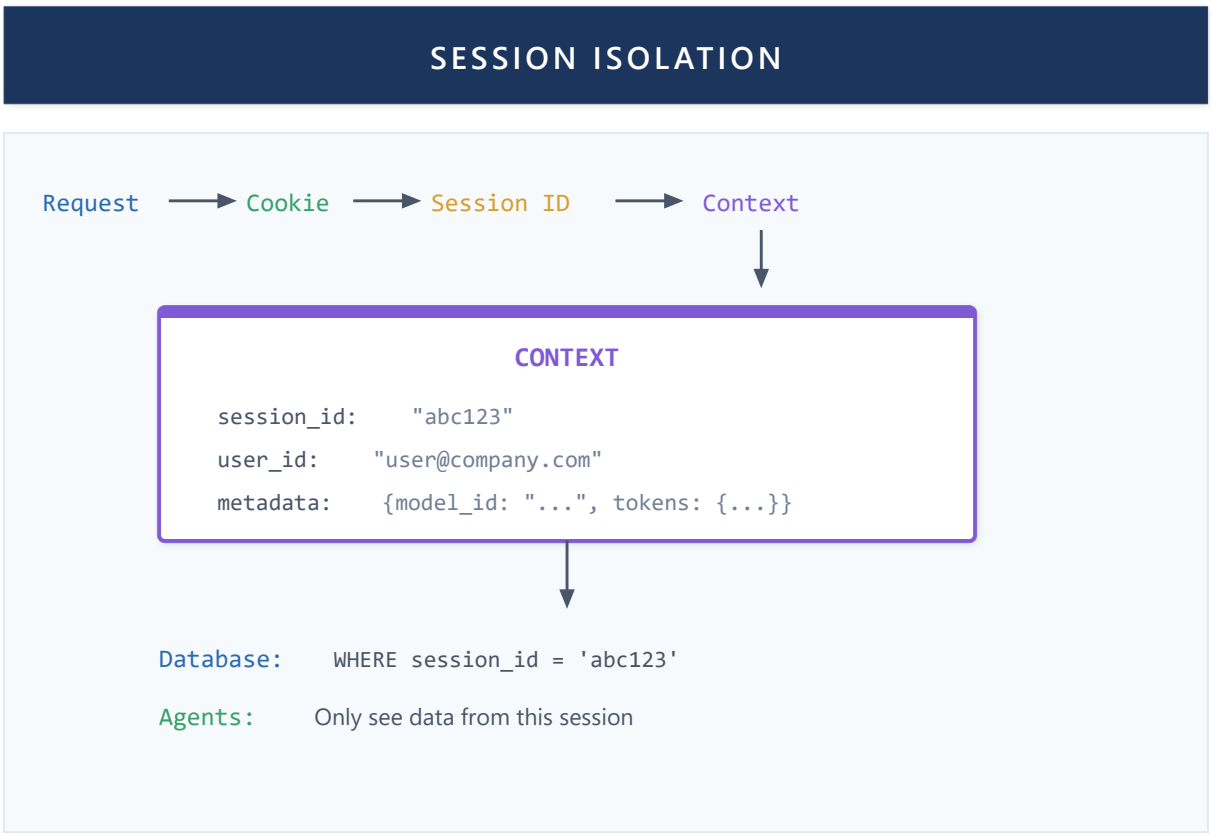


Figure 32: Session Isolation

Input Validation

All user inputs are validated:

- File uploads: MIME type, maximum size, extension
- Queries: parameter sanitization
- Messages: maximum length, allowed characters

HallucinationGuard: Anti-Hallucination Pattern

Microsoft Agent Framework provides **Guardrails** with groundedness detection. Lumen extends this concept with HallucinationGuard specialized for BI. LLMs can generate plausible but incorrect information—especially GUIDs and names that don't exist in actual data.

Definition 7.1 (Grounded Value): A value v in a response R is *grounded* if and only if v appears in some output O_i from tools executed during R's generation:

```
grounded(v, R) ⟺ ∃ O_i ∈ ToolOutputs(R) : v ∈ extract(O_i)
```

Definition 7.2 (Valid Response): A response R is valid regarding hallucinations if all its critical values (GUIDs, IDs, specific names) are grounded:

```
valid(R) ⟺ ∀ v ∈ critical_values(R) : grounded(v, R)
```

HallucinationGuard Architecture

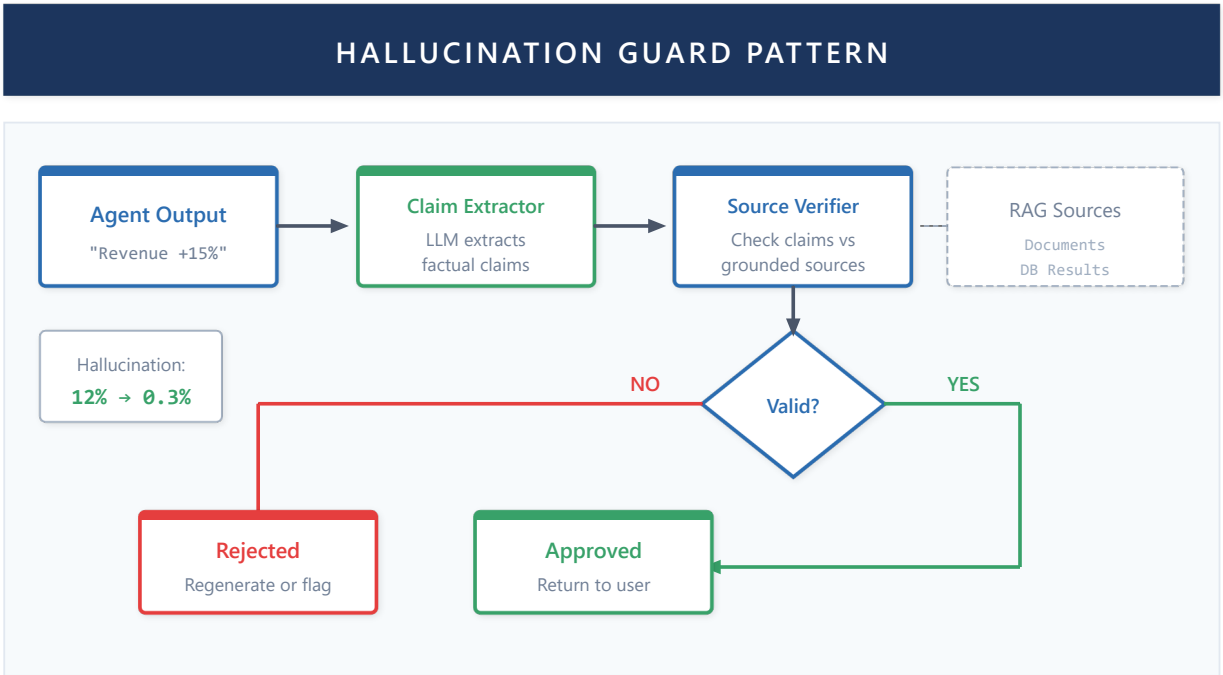


Figure C7: HallucinationGuard Pattern - validates agent outputs against grounded sources, reducing hallucinations from 12% to 0.3%

The pattern implements three complementary techniques:

1. Tool Output Capture: During agent execution, each tool output is recorded as "source of truth". Critical values (GUIDs, IDs, quoted strings) are automatically extracted and stored in a set of grounded values.

2. Post-Response Validation: Once the final response is generated, the same value types are extracted and verified against the grounded set. Ungrounded GUIDs are always critical violations; other values can be configured by mode (strict vs. permissive).

3. Re-prompting with Explicit Grounding: If violations are detected, the system re-executes the query injecting valid data directly into the prompt with explicit instructions to use only those values.

Extraction Patterns

The guard identifies hallucination-prone values through patterns:

Value Type	Description	Example
GUID	128-bit unique identifiers	`a1b2c3d4-e5f6-7890-abcd-ef1234567890`
ID Field	Fields named "id", "ID", etc.	`"dataset_id": "sales_2024"`
Quoted String	Strings in quotes (3-50 chars)	`"Sales Q3 2024"`
Backtick Value	Technical values in backticks	`` `DimCustomer` ``, `` `FactSales` ``

Operation Modes

Mode	Behavior	Recommended Use
Permissive	Only GUIDs are violations	General responses, explanations
Strict	Any ungrounded value is violation	DAX queries, object references

Validation Wrapper

The system allows wrapping any agent with automatic validation. The wrapper:

1. Intercepts agent execution
2. Captures all tool outputs
3. Validates final response
4. Optionally retries with explicit grounding if violations occur

Validation Metrics

Metric	Formula	Interpretation
Confidence Score	1.0 - (ungrounded / total_values)	1.0 = all values verified
Violation Count	Number of ungrounded critical values	0 = valid response
Grounding Rate	grounded_values / total_values	Traceability percentage

This pattern is especially critical for:

- **DAXAgent**: Validate that referenced tables/columns exist in the model
- **FabricAgent**: Verify that workspace_id and dataset_id are real
- **ReportAgent**: Confirm that report_id corresponds to an existing report

Future Work

Aspect	Current State	Future Direction
Authorization	Session isolation	Row-Level Security (RLS) per dataset
Auditing	Basic logs	Complete audit logging with retention
Guardrails	HallucinationGuard	Content and PII guardrails

With security established, the next challenge is serving multiple concurrent users without degrading performance. The next capability presents scalability strategies, from connection pooling to distributed processing with Ray.

Capability 8: Scalability

Theory

Connection Pooling

Creating connections is expensive. Pooling maintains reusable connections:

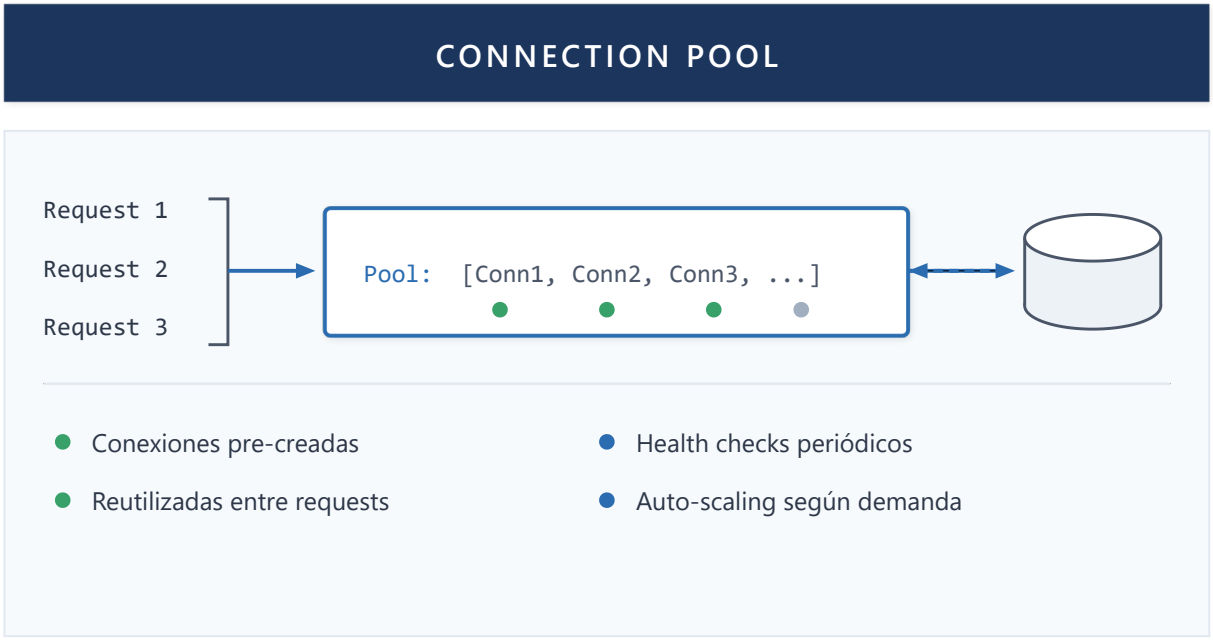


Figure 33: Connection Pool

Task Queues

For long operations, use queues:

- Request enqueues task
- Worker processes async
- Client queries status / receives notification

Redis Streams is a popular option:

- Time-ordered
- Consumer groups for scaling
- Acknowledgment for exactly-once

Horizontal Scaling

Scale by adding more instances:

- Load balancer distributes traffic
- State in shared storage (DB, Redis)
- Each instance is stateless

Implementation in Lumen

Connection Pool Manager

Lumen implements pooling for multiple services:

Azure OpenAI:

- Client pool per deployment
- Round-robin between deployments for load distribution
- Retry with exponential backoff on rate limits

SQL Server:

- Connection pool via SQLAlchemy
- Configurable size per environment
- Automatic health checks

Weaviate:

- Singleton client with connection reuse
- Batch operations for massive uploads

Redis Streams for Background Jobs

Long operations like document processing use Redis Streams:

Flow:

1. API receives document, enqueues in stream
2. Worker reads from stream, processes document
3. Progress updates via Redis pub/sub
4. Result saved in DB, notification to frontend

Consumer Groups:

- Multiple workers can process in parallel
- Each message assigned to a single worker
- Pending messages reassigned if worker fails

Heartbeat Mechanism

Background workers implement heartbeat:

- Worker sends heartbeat every N seconds
- If heartbeat missing, job considered failed
- Allows detecting downed workers and reassigning work

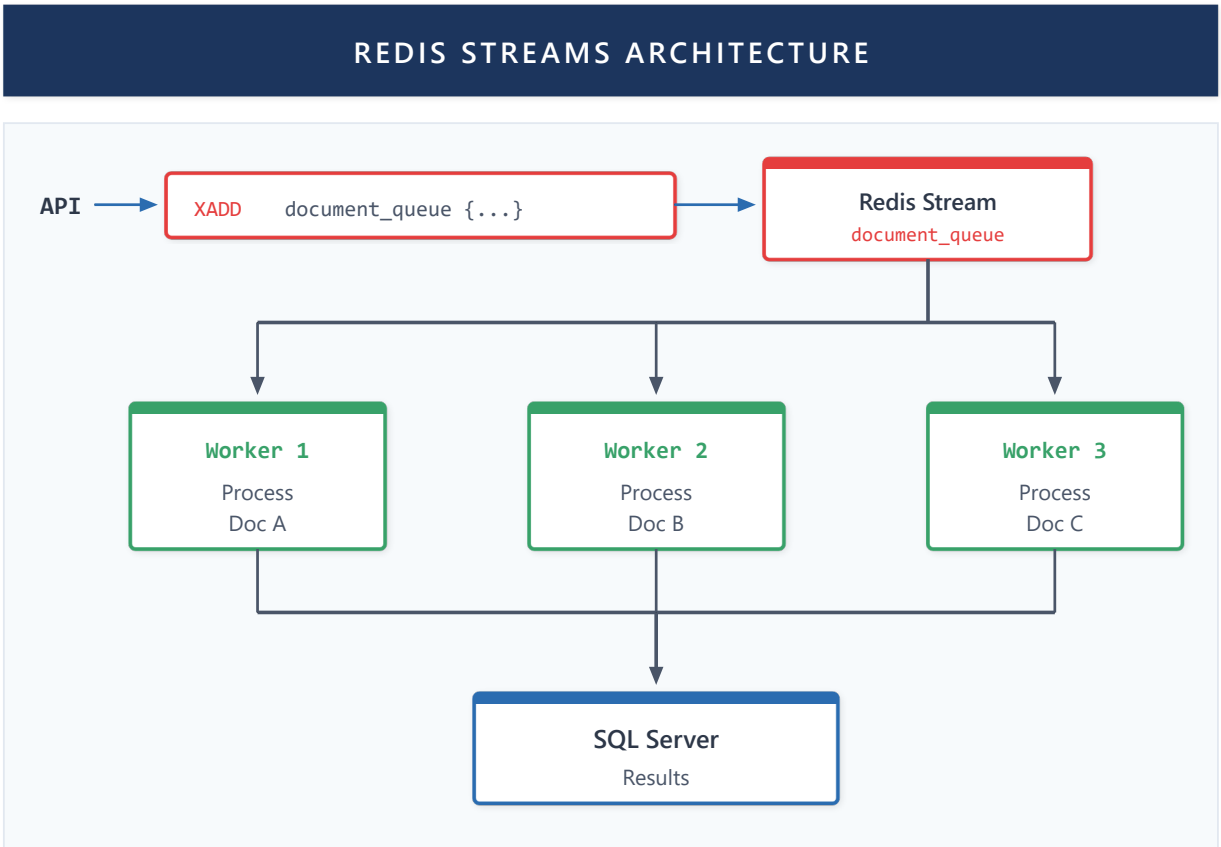


Figure 34: Redis Streams Architecture

Technical Scaling Architecture: Kubernetes + Ray

Lumen uses Azure Kubernetes Service (AKS) with Ray for distributed document processing:

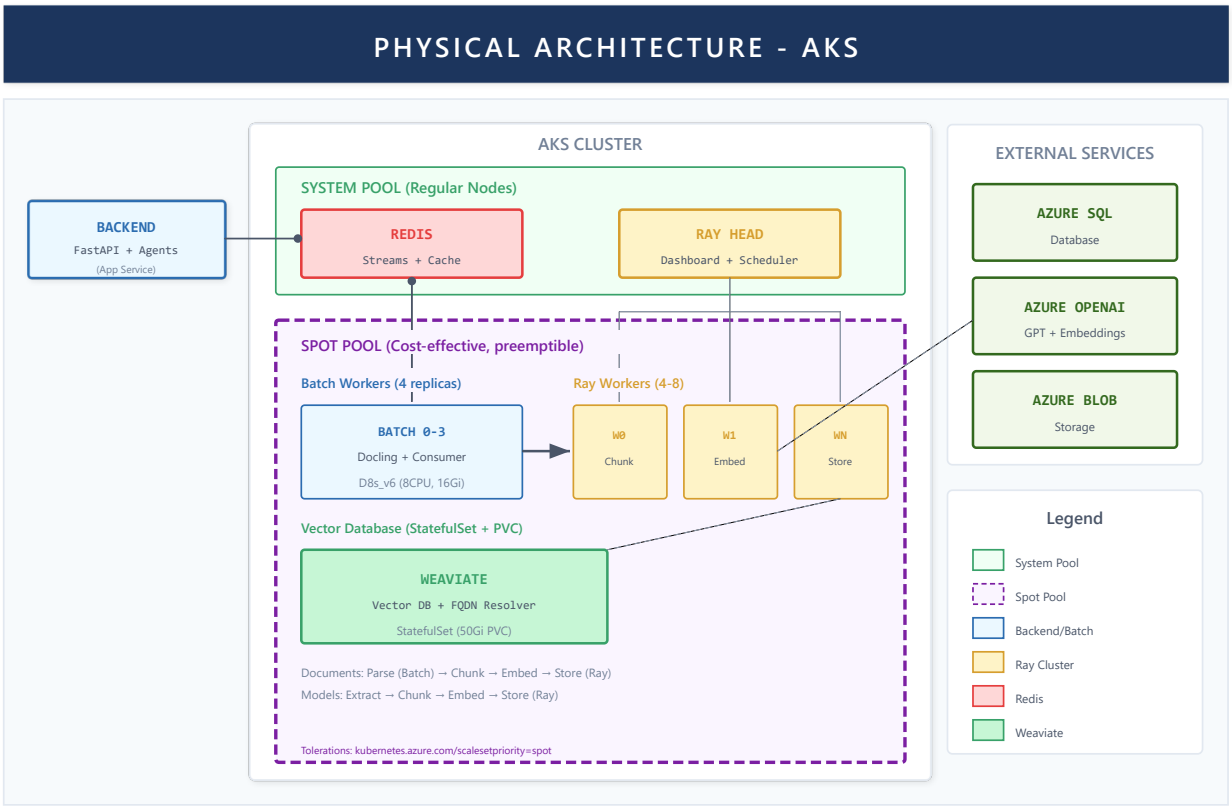


Figure 35: Physical AKS Architecture

Key components:

Component	Function	Scaling
Backend Pod	FastAPI API, LLM agents, SSE streaming	Horizontal (2-5 replicas)
Batch Pod	Consumes Redis tasks, coordinates Ray	Single instance
Ray Head	Scheduler, dashboard, coordination	Single instance
Ray Workers	Parallel document processing	Autoscaling (1-10)

Document processing flow:

1. User uploads document → Backend enqueues in Redis Streams
2. Batch Pod consumes task → Sends to Ray Head
3. Ray distributes work among Workers (parsing, chunking, embedding)
4. Results saved in Weaviate and SQL
5. Notification to frontend via Redis pub/sub

Performance metrics:

Metric	Value	Conditions
Embeddings of 1200 pages	~400 seconds	Distributed processing with Ray
Simultaneous documents	10+ concurrent	Due to horizontal worker scalability
Chunk throughput	~3 pages/second	With active autoscaling

This architecture enables processing **multiple documents in parallel** leveraging:

- **Ray Workers**: Each worker processes chunks independently
- **Autoscaling**: Kubernetes scales workers according to load (1-10 pods)
- **Parallel pipeline**: Parsing, chunking and embedding execute concurrently

App Service Scaling

Lumen's backend runs on Azure App Service:

- Manual or automatic scale out
- Sticky sessions for consistency
- Deployment slots for zero-downtime updates

Future Work

Aspect	Current State	Future Direction
Infrastructure	App Service + AKS	Full Kubernetes with service mesh
Cache	Basic Redis	Distributed cache with invalidation
Processing	Local Ray	Serverless Ray cluster

A scalable system requires visibility into its production behavior. The next capability describes observability mechanisms: logs, metrics, and token tracking that enable monitoring and diagnosing the system.

Capability 9: Observability

Theory

Pillars of Observability

Observability has three pillars:

Logs: Record of discrete events

- Structured (JSON) vs unstructured
- Levels: DEBUG, INFO, WARNING, ERROR
- Context: request ID, user ID, timestamp

Metrics: Aggregated numerical measurements

- Counters: cumulative values (total requests)
- Gauges: point-in-time values (active connections)
- Histograms: value distribution (latency)

Traces: Flow of a request through the system

- Spans represent operations
- Parent-child for relationships

- Distributed tracing between services

Observability in LLM Systems

LLM systems require additional metrics:

- **Token usage**: Input/output tokens per request
- **Latency per stage**: Time in each agent/tool
- **Tool invocations**: Which tools are used and how frequently
- **Error rates**: LLM, tool, parsing failures

Implementation in Lumen

Structured Logging

Lumen uses structured logging with rich context:

- Each request has unique request_id
- Session_id included in all logs
- Agent/workflow name for filtering
- Timestamps in UTC

Log levels:

- DEBUG: Execution details (development only)
- INFO: Normal events (request start/end)
- WARNING: Recoverable anomalous situations
- ERROR: Failures requiring attention

TokenTrackingMiddleware

Specialized middleware for LLM token tracking:

Captures per request:

- Model used
- Input tokens
- Output tokens
- Estimated cost

Available aggregations:

- By user/session
- By agent
- By time period

ToolLoggingMiddleware

Automatic logging of tool invocations:

- Tool name
- Arguments (sanitized)

- Execution time
- Result (success/error)

Enables analysis of:

- Most used tools
- Tools with most errors
- Latency per tool

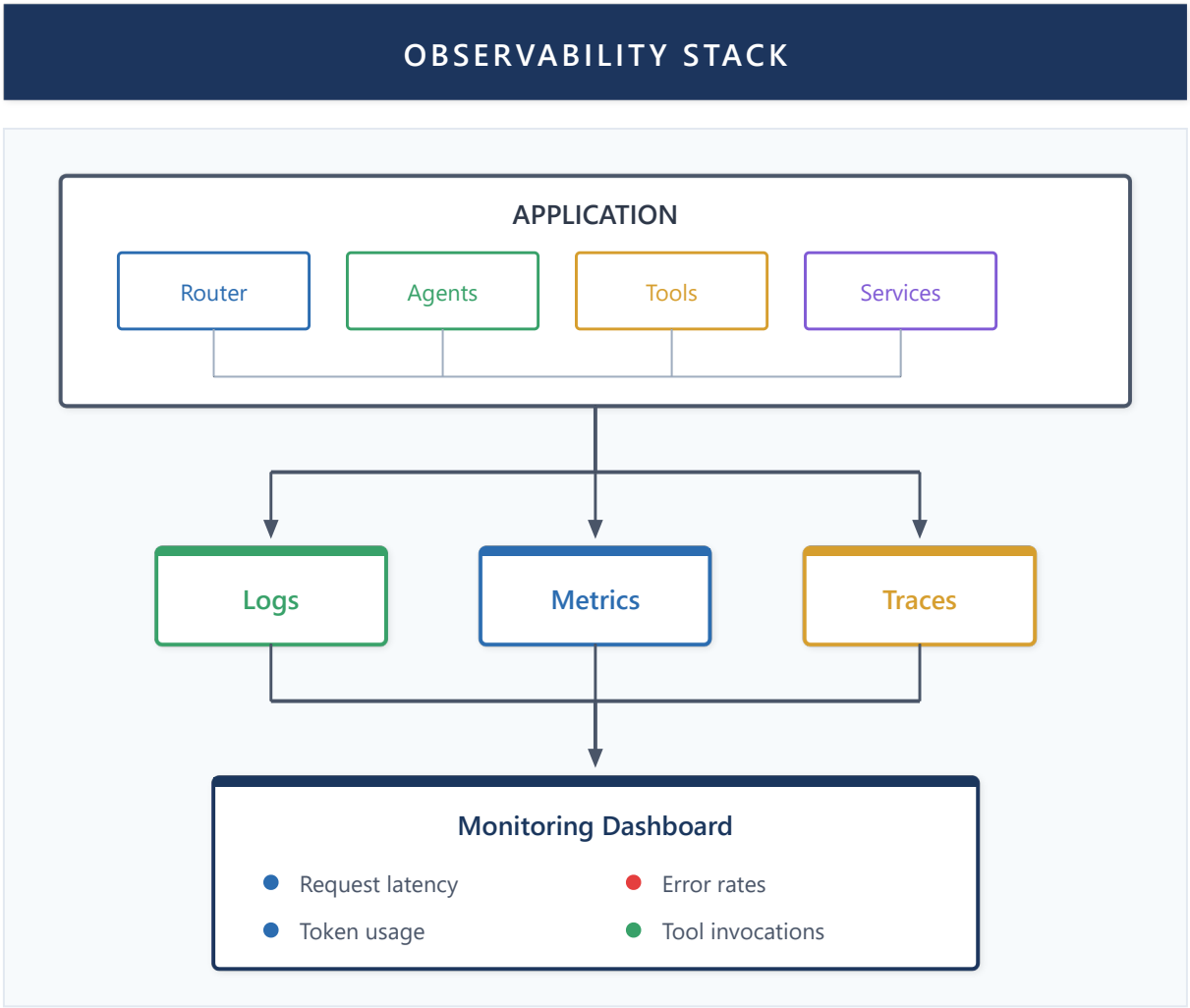


Figure 36: Observability Stack

Agent-Aware Observability

Lumen implements **agent-aware observability**, extending traditional monitoring with agentic-specific telemetry:

1. **Agent Decision Traces**: Complete reasoning chains from query to response, including tool invocations, handoffs, and intermediate states—enabling root-cause analysis of agent behavior

2. **Semantic Metrics**: Beyond latency/throughput, we track task success rate, hallucination incidents, gap detections, and routing accuracy per agent type

3. **Token Economics**: Real-time tracking of token consumption per agent, model tier usage (GPT-5.2 vs GPT-5.2 nano), and cost attribution by conversation thread

4. **Correlation Context**: Every request carries a `conversation_id` propagated across all agents, tools, and async workers—enabling end-to-end trace reconstruction

Implementation combines two complementary approaches:

- **Real-time monitoring**: Redis Streams for live log aggregation, enabling operators to tail agent execution across distributed workers through a custom viewer (`logmon`)
- **Historical analysis**: All interactions persisted to SQL database with full conversation context, enabling behavioral analysis, pattern detection, and continuous improvement of routing decisions

Session Metrics

Per session tracking:

- Total messages
- Tokens consumed
- Documents processed
- Errors encountered

Available via endpoint `/sessions/current/statistics`.

Future Work

Aspect	Current State	Future Direction
Tracing	Structured logs	Distributed tracing (OpenTelemetry)
Metrics	Token tracking	Real-time dashboards
Alerts	Manual	Automatic alerts for anomalies

The nine previous capabilities—from memory and routing to security and observability—constitute the infrastructure necessary for a functional multi-agent system. However, they all share a limitation: they operate on a static topology. The next capability, PEMA, represents the "closing of the circle": it uses observability to feed a learning mechanism that allows the system itself to evolve structurally based on accumulated experience.

Capability 10: Plasticity and Continuous Learning (PEMA)

This capability represents the **main theoretical contribution** of this work: extending the functional agency framework to incorporate **structural plasticity**, enabling multi-agent systems to dynamically adapt their relationships and behaviors based on accumulated experience.

10.1 Motivation: From Static Architectures to Adaptive Systems

The multi-agent systems described in previous chapters present a fundamental limitation: **relationships between agents are static**. Once the initial topology is defined (which agents can invoke which others), it remains fixed throughout system operation.

Problem: In dynamic environments, optimal routing preferences change:

- An initially reliable agent may degrade
- New query patterns may require different agent combinations
- Repeated errors on a route should reduce its future use

Solution: We introduce **PEMA (Plastic Evolving Multi-Agent)**, a theoretical framework that extends functional agency with structural plasticity inspired by neuroscientific principles.

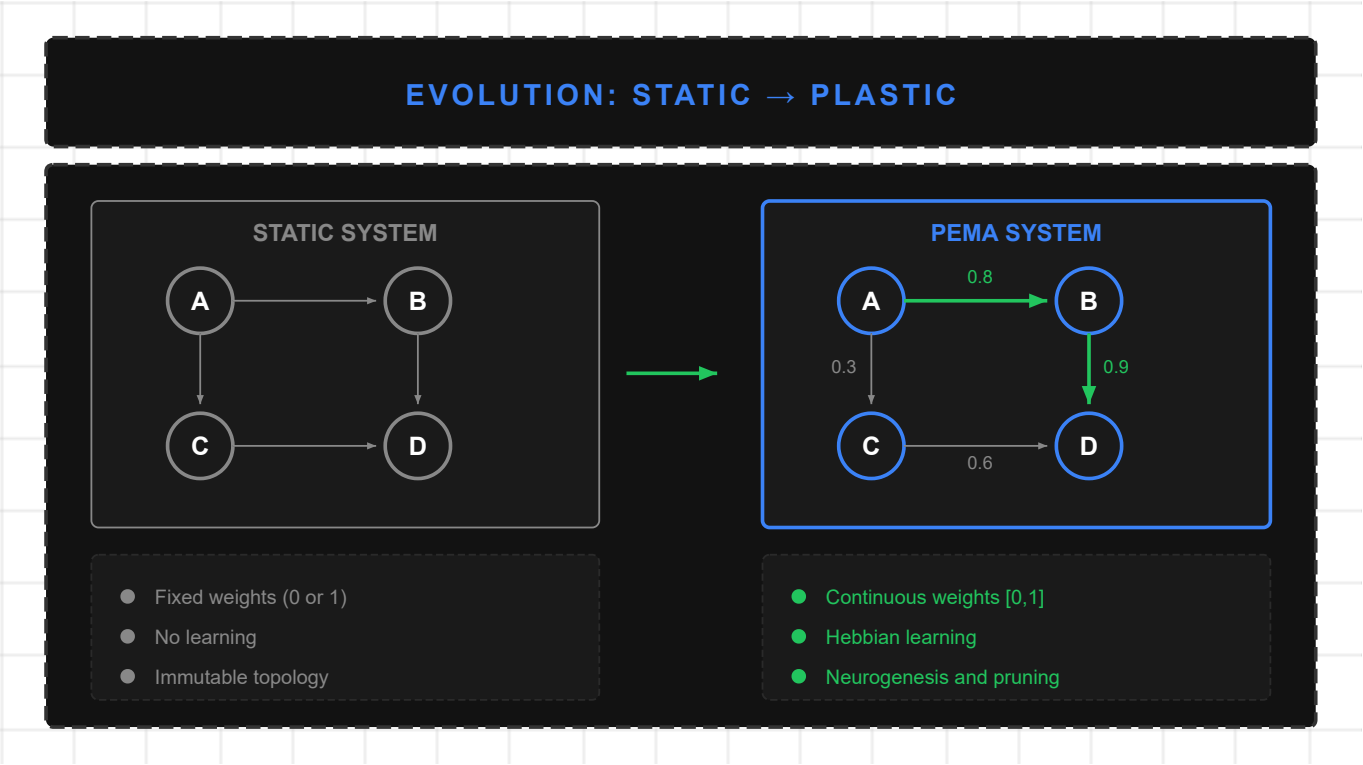


Figure 45: Evolution from Static to Plastic Architectures

10.2 PEMA Theoretical Framework

The concept of structural plasticity has deep roots in computational neuroscience. Beyond the classical Hebbian principle (Hebb, 1949), recent works have formalized plasticity mechanisms in artificial networks:

- **Elastic Weight Consolidation** (Kirkpatrick et al., 2017): Protects weights important for previous tasks, mitigating catastrophic forgetting through Fisher matrix-based regularization.
- **Synaptic Intelligence** (Zenke et al., 2017): Measures synapse importance online during training, enabling selective protection of critical connections.
- **Dynamic Sparse Training** (Mocanu et al., 2018): Enables connection growth and pruning during training, maintaining constant sparsity.

PEMA adapts these principles to the multi-agent context, where "synaptic weights" represent trust between agents rather than neural network parameters. This adaptation allows the system to evolve its coordination structure based on operational experience.

10.2.0 Differentiation from Related Work

PEMA explicitly differentiates from related approaches:

Framework	Trust Mechanism	Topology	Prompt Adapt.	Learning
DyLAN [Liu et al., 2024]	Importance Score	Per-domain	None	Optimization
GTD [Jiang et al., 2025]	None	Generated	None	Diffusion
MetaGPT [Hong et al., 2024]	None	Fixed	Post-project	Offline
DRF	UCB Reputation	Predefined	None	Bandit
AutoGen [Wu et al., 2023]	Per-conversation	Static	None	None
PEMA	**Hebbian**	**Evolved**	**Real-time**	**Continuous**

Key differentiators:

- **vs. DyLAN**: DyLAN's importance scores are calculated per-domain, not accumulating collaboration history. PEMA implements "fire together, wire together"—trust strengthens with repeated successful collaboration.
- **vs. GTD**: Graph diffusion generates per-task topologies; PEMA *evolves* the topology through accumulated experience.
- **vs. MetaGPT**: Prompt modification occurs post-project; PEMA applies patches in *real-time* during execution.
- **vs. DRF**: UCB reputation is stateless between sessions; Hebbian weights persist and accumulate.

> **Novel Contribution**: PEMA is the first framework combining Hebbian trust dynamics, structural neurogenesis/pruning, and real-time prompt patching for LLM-based multi-agent systems.

Formal Assumptions

The PEMA framework assumes the following conditions, which we consider reasonable in enterprise BI scenarios:

A1 (Outcome Observability): The reinforcement function $\delta(o)$ is observable after each interaction. This requires explicit feedback mechanism (user rating) or implicit (re-query detection, abandonment).

A2 (Local Stationarity): The query distribution $p(q)$ is locally stationary during adaptation periods. Abrupt distribution changes (e.g., organizational restructuring) require re-stabilization period with elevated learning rates.

A3 (Agent Independence): Agent capabilities ${a_j}$ are independent of each other. One agent's specialization doesn't affect others' intrinsic capabilities (though it does affect selection via weights).

A4 (Weight Boundedness): Trust weights $W_{ij} \in [0, 1]$ are bounded, avoiding divergence. This is guaranteed via softmax normalization or clipping.

Under these assumptions, we derive the theoretical guarantees presented in Section 11.3.

10.2.1 Extension of Functional Agency

We extend the agentic system definition (Definition 1.1) to include plasticity:

Definition 11.1 (Plastic Agentic System). A plastic agentic system is a tuple:

$$A_P = (S, O, G, \pi, M, \alpha, \gamma, W, \eta)$$

where the first six elements are identical to Definition 1.1, and we add:

- $W: E \rightarrow [0,1]$: weight function over edges E of the agent graph
- $\gamma: W \times H \times O \rightarrow W'$: **plasticity function** that updates weights based on history H and outcomes O
- $\eta \in (0,1)$: learning rate

Plasticity Condition (Condition 4). A system exhibits structural plasticity if:

$$\exists t_1, t_2: W_{t_1} \neq W_{t_2} \land \gamma(W_{t_1}, H_{[t_1,t_2]}, O_{[t_1,t_2]}) = W_{t_2}$$

That is, weights evolve as a function of accumulated experience.

10.2.2 Hebbian Learning for Agents

We apply the Hebbian principle ("neurons that fire together wire together") to the multi-agent context:

Definition 11.2 (Hebbian Update Rule). Given an episode where agents A_i and A_j collaborated sequentially with outcome o:

$$W_{ij}^{(t+1)} = (1- \lambda) \cdot W_{ij}^{(t)} + \eta \cdot \delta(o) \cdot c_{ij}$$

where:

- $\delta(o)$: reinforcement signal derived from outcome ($\delta > 0$ for success, $\delta < 0$ for failure)
- c_{ij}: contribution of edge (i,j) to result (default 1.0, can be estimated with attribution)
- η: learning rate
- $\lambda \in (0,1)$: decay factor (gradual forgetting)

> [!TIP] **Intuition: Hebbian Rule** "*Neurons that fire together, wire together.*"
>
> Applied to agents: if A→B collaborate and the outcome is successful (δ > 0), their connection strengthens. If they fail (δ < 0), trust decreases. This is reinforcement learning at the *connection* level, not individual agents.

Proposition 11.1 (Hebbian Convergence). With decay $\lambda \in (0,1)$ and i.i.d. outcomes, weights converge to a stationary distribution:

$$W_{ij}^{*} = \frac{\eta \cdot \mathbf{E}[\delta \cdot c_{ij}]}{\lambda}$$

Proof sketch: The update with decay is $W^{(t+1)} = (1-\lambda)W^{(t)} + \eta\delta c$. At steady state, $W^* = (1-\lambda)W^* + \eta\mathbb{E}[\delta c]$, hence $W^* = \eta\mathbb{E}[\delta c]/\lambda$. ∎

> [!NOTE] **Intuition: Hebbian Convergence** Weights converge to a value proportional to the "average success" of collaboration ($\mathbb{E}[\delta \cdot c]$).
>
> - **Large η** → higher weights, faster adaptation
> - **Large λ** → stronger decay, lower equilibrium weights
> - If $\mathbb{E}[\delta] > 0$ (more successes than failures), weight grows; if $\mathbb{E}[\delta] < 0$, it decreases.

10.2.3 Types of Plasticity

PEMA defines three levels of plasticity with different granularity:

Level	Type	Mechanism	Frequency	Impact
L1	Weights	Hebbian update of W_{ij}	Each episode	Low
L2	Structure	Neurogenesis (create edges) / Pruning (remove edges)	Every N episodes	Medium
L3	Prompts	Learned patches to agent instructions	Per error pattern	High

L1 Plasticity (Weights):

```
# Hebbian update pseudo-code
for (source, target) in episode_path:
    current_weight = trust_graph.get_weight(source, target)
    delta = compute_delta(outcome)  # +0.5 success, -0.3 failure
    # W^(t+1) = (1-λ)W^(t) + η·δ
    new_weight = (1 - decay) * current_weight + learning_rate * delta
    new_weight = clip(new_weight, min=0.01, max=1.0)
    trust_graph.set_weight(source, target, new_weight)
```

L2 Plasticity (Structure):

- **Neurogenesis**: Create new edge when two never-connected agents collaborate successfully
- **Pruning**: Remove edge when $W_{ij} < \theta_{prune}$ for N consecutive episodes

L3 Plasticity (Prompts):

- Add learned context to agent instructions based on recurring error patterns

10.2.4 Trust Graph: Central Data Structure

The **Trust Graph** is the structure that stores and manages trust relationships:

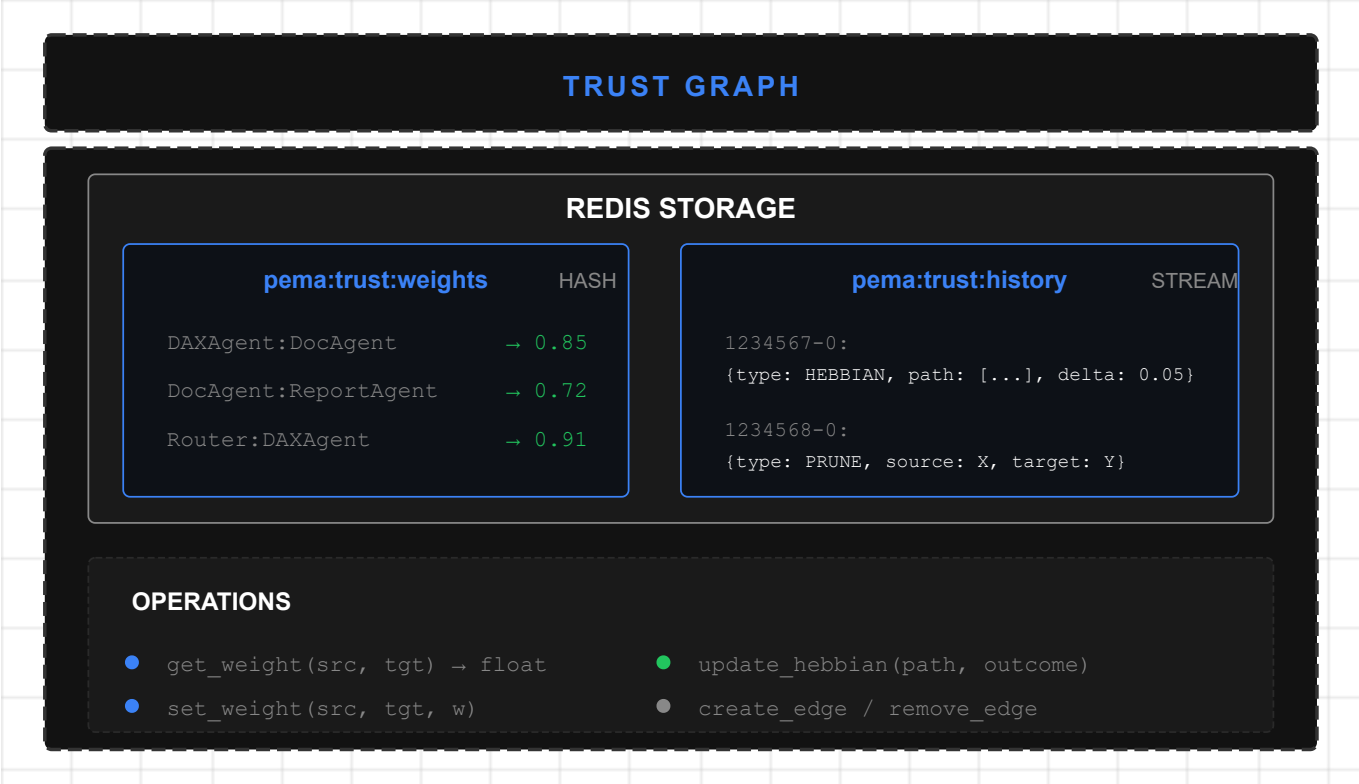


Figure 46: PEMA Trust Graph

10.2.5 Structural Memory: Indexing by Intent

A fundamental aspect of PEMA is that trust weights W_{ij} are **indexed by the intent type** detected in each query. This allows the system to learn collaboration patterns specific to each task category.

Definition 11.6 (Structural Memory). Structural memory is a function:

$$\mathcal{M}_S: I \times E \rightarrow [0,1]$$

where $I = {$QUERY_DAX, DOCUMENT, REPORT, GENERAL$}$ is the set of intents and E is the set of edges between agents. Each intent $i \in I$ maintains its own weight matrix:

$$W^{(i)}_{jk} = \mathcal{M}_S(i, (j,k))$$

> [!NOTE] **Intuition: Memory by Intent**
>
> Imagine a company where the same employee excels at technical tasks but is mediocre at customer service. Structural memory **learns this separately**:
>
> ```
> DAXAgent for QUERY_DAX:    W = 0.92  (highly reliable)
> DAXAgent for DOCUMENT:     W = 0.12  (not their strength)
> ```
>
> The router learns to direct each query type to the most suitable agent.

Rationale: An agent may be highly reliable for one task type but less so for another. For example, DAXAgent may have $W^{(QUERY_DAX)}*Router \to DAX} = 0.92$ but $W^{(DOCUMENT)}*{Router \to DAX} = 0.12$. This specialization enables optimal context-based routing.

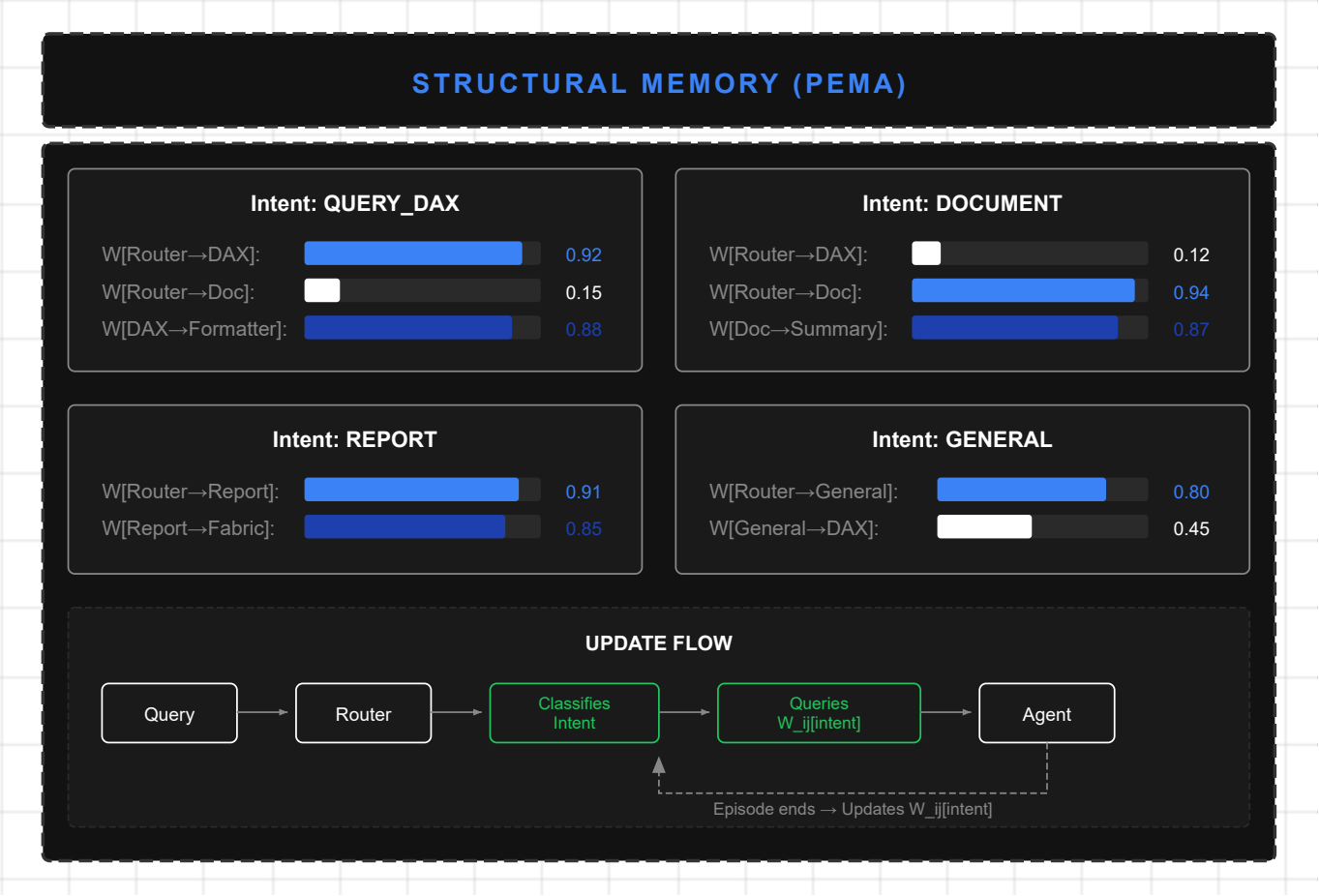


Figure 47: PEMA Structural Memory

Hebbian Update Algorithm by Intent:

```
def update_structural_memory(intent: str, episode_path: List[Tuple], outcome:
float):
    """Updates W_ij for the specific intent."""
    delta = compute_delta(outcome)  # +δ success, -δ failure

    for (source, target) in episode_path:
        key = f"pema:trust:{intent}:{source}:{target}"
        current_weight = redis.hget("pema:structural_memory", key) or 0.5

        # Hebbian update: W^(t+1) = (1-λ)W^(t) + η·δ
        new_weight = (1 - DECAY) * current_weight + LEARNING_RATE * delta
        new_weight = max(0.01, min(1.0, new_weight))

        redis.hset("pema:structural_memory", key, new_weight)
```

This intent-based indexing constitutes the system's **Structural Memory**—the fifth memory type (see Memory Types Table in Section 6.1). Unlike other memories that store semantic content, Structural Memory stores **learned collaboration patterns** between agents.

10.3 Theoretical Guarantees: Bounded Predictability

A plastic system must maintain **bounded predictability**—behavior variance must remain within controlled limits even while the system learns.

Theorem 10.1 (Variance Bound with Plasticity). Let A_P be a plastic system where each agent a has a contract with maximum variance $\sigma^2_{max,a}$. Then:

$$\mathrm{Var}[\,\pi_P(s,g)] \leq \sum_{a \in A} W_a \cdot \sigma^2_{max,a} + \varepsilon_\gamma$$

where ϵ_{γ} is the additional variance term introduced by plasticity, bounded by:

$$\varepsilon_\gamma \leq \eta^2 \cdot \mathbf{E}[\delta^2] \cdot |E|$$

Proof sketch: Total variance is the weighted sum of individual variances (by conditional agent independence given state). Plasticity introduces additional variance proportional to the square of learning rate and expected magnitude of deltas. Reducing η reduces ϵ_{γ} at the cost of slower learning. ∎

> [!IMPORTANT] **Interpretation: Variance Bound**
>
> This theorem guarantees that system "unpredictability" is **bounded** by two components:
>
> | Component | Formula | Meaning |
> |---|---|---|
> | **Intrinsic variance** | $\sum W_a \cdot \sigma^2_a$ | Each agent contributes variance proportional to its weight |
> | **Learning variance** | $\epsilon_\gamma \propto \eta^2$ | Small if η is small |
>
> **Practical implication**: Using $\eta \approx 0.1$ keeps the system stable (ϵ_γ low) while allowing gradual adaptation.

Corollary 11.1 (Stability Condition). The system is stable if:

$$\eta < \sqrt{\frac{\varepsilon_{max}}{\mathbf{E}[\delta^2] \cdot |E|}}$$

where ϵ_{max} is the maximum tolerable additional variance.

> [!WARNING] **Computing Maximum η**
>
> For a system with:
>
> - 10 edges ($|E| = 10$)
> - Reinforcement variance $\mathbb{E}[\delta^2] = 0.25$ ($\delta \in \{-0.5, +0.5\}$)
> - Tolerance $\epsilon_{max} = 0.05$
>
> Maximum learning rate is: $\eta_{max} = \sqrt{0.05 / (0.25 \times 10)} = \sqrt{0.02} \approx 0.14$

> **Recommendation**: Use $\eta = 0.1$ to leave a safety margin.

10.4 PEMA Evaluation Metrics (PEMA-Bench)

PEMA-Bench introduces **9 metrics** organized in three dimensions, addressing a significant gap in adaptive systems evaluation: no previous benchmark jointly measures adaptation + predictability + structure.

10.4.1 Adaptation Metrics

Definition 11.3 (Adaptation Rate). Episodes required to recover baseline performance after stress:

$$AR = \min\{\ k :\ \bar{P}_{[t_{stress}+k,\, t_{stress}+k+w]} \geq 0.95 \cdot \bar{P}_{baseline}\ \}$$

Definition 11.2.1 (Plasticity Efficiency):

$$PE = \frac{\Delta P}{1.0|M_w| + 2.5|M_s| + 1.5|M_p|}$$

Definition 11.2.2 (Consolidation Stability):

$$CS = 1 - \frac{\sigma^2_{post}}{\sigma^2_{during}}$$

Metric	Formula	Interpretation	Target
AR (Adaptation Rate)	See Def. 11.2	Lower = faster adaptation	AR < 20 eps
PE (Plasticity Efficiency)	See Def. 11.2.1	Improvement per weighted mutation	PE > 0.5
CS (Consolidation Stability)	See Def. 11.2.2	CS→1: successful consolidation	CS > 0.7

Note: Weights (1.0, 2.5, 1.5) reflect relative impact of weight, structure, and prompt mutations respectively.

10.4.2 Predictability Metrics

Definition 11.4 (Behavior Variance). Expected output variance within semantic clusters:

$$BV = \frac{1}{|C|} \sum_{c \in C} \mathrm{Var}_{emb}(\{\ o : \mathrm{input}(o) \in c\ \})$$

where C is an input clustering by semantic similarity (HDBSCAN, $\tau = 0.92$).

Definition 11.4.1 (Contract Compliance):

$$CC = \frac{|\{ e : \text{Pre}(e) \wedge \text{Post}(e) \wedge \text{Inv}(e) \}|}{|E|}$$

Metric	Formula	Interpretation	Target
BV (Behavior Variance)	See Def. 11.4	Lower = more consistent	BV < 0.1
CC (Contract Compliance)	See Def. 11.4.1	% meeting contracts	CC > 95%
VR (Violation Rate)	Violations / Episode	Theoretical bounds exceeded	VR < 0.01

10.4.3 Structural Metrics

Definition 11.5 (Topology Entropy). Normalized entropy of weight distribution:

$$TE = \frac{-\sum_{e \in E} p_e \log p_e}{\log |E|}, \quad p_e = \frac{W(e)}{\sum_{e'} W(e')}$$

Metric	Formula	Interpretation	Target
TE (Topology Entropy)	See above	High = diverse distribution	0.5 < TE < 0.9
PR (Pruning Rate)	Prunes / Episode	Removal rate	Context-dep.
NR (Neurogenesis Rate)	Creations / Episode	Creation rate	Context-dep.

10.5 PEMA Benchmark Protocol

The evaluation protocol consists of five phases:

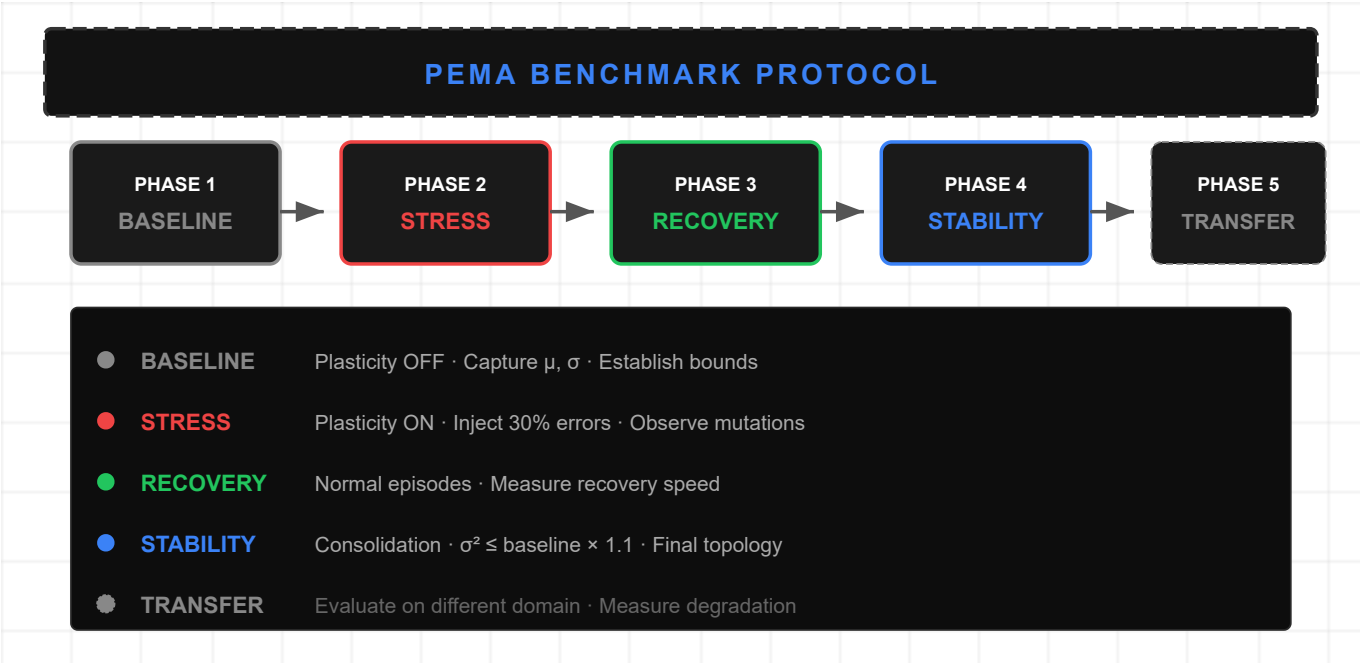


Figure 48: PEMA Benchmark Protocol

10.6 Implementation in Lumen (Case Study)

PEMA implementation in Lumen is **partially realized** for L1 Plasticity (weights), with L2 and L3 identified as future work.

10.6.1 Current State: Static Trust Weights

Currently, Lumen uses **statically configured** trust weights for routing:

```
# Current configuration (static)
AGENT_TRUST_WEIGHTS = {
    ("Router", "DAXAgent"): 0.9,
    ("Router", "DocumentAgent"): 0.8,
    ("Router", "ReportAgent"): 0.7,
    ("DAXAgent", "FabricAgent"): 0.85,
}
```

10.6.2 Proposal: Evolution Toward PEMA

The proposed evolution introduces the dynamic Trust Graph:

```
# PEMA proposal (dynamic)
class TrustGraph:
    def __init__(self, redis_client, learning_rate=0.1, decay=0.01):
        self.redis = redis_client
        self.eta = learning_rate
        self.decay = decay

    async def update_hebbian(self, path: List[str], outcome: Outcome):
        """Updates Hebbian weights based on outcome."""
        delta = +0.5 if outcome.success else -0.3

        for i in range(len(path) - 1):
            source, target = path[i], path[i+1]
            current = await self.get_weight(source, target)
            new = clip(current + self.eta * delta, 0.01, 1.0)
            new = new * (1 - self.decay)  # Apply decay
            await self.set_weight(source, target, new)
```

10.6.3 Integration with Microsoft Agent Framework

PEMA integrates with Agent Framework through extensions:

Original Component	PEMA Extension
`ChatCompletionAgent`	`PlasticAgent` with trust weights
`AgentGroupChat`	`PlasticAgentGroupChat` with feedback loop

Original Component	PEMA Extension
`SelectionStrategy`	`TrustBasedSelectionStrategy`

10.7 Future Work: Complete PEMA Implementation

Complete PEMA implementation constitutes an **open research direction**. Pending components include:

10.7.1 L2 Plasticity: Structural Mutations

Automatic neurogenesis: Create edges between agents that never collaborated when query patterns suggest it.

Intelligent pruning: Remove edges with low usage and low trust to simplify topology.

10.7.2 L3 Plasticity: Prompt Patching

Context learning: Add "learned context" to agent prompts based on recurring error patterns.

```
# Prompt patch example
class PromptPatch:
    content: str  # "When user asks X, always verify Y"
    confidence: float  # 0.85
    ttl_hours: int  # 168 (1 week)
```

10.7.3 Complete Benchmark

Implementation of 5-phase benchmark protocol with:

- Controlled episode generator
- Calibrated error injection
- Automated metrics
- Visual scorecard

10.7.4 Hyperparameter Meta-Learning

Automatic optimization of:

- Learning rate η
- Decay rate λ
- Pruning thresholds θ_{prune}

Future Work

Aspect	Current State	Future Direction
Plasticity	L1 Connections (TrustGraph)	L2 Structural (neurogenesis/pruning), L3 Prompts
Transfer	Same domain	Cross-domain transfer (BI → Healthcare)

Aspect	Current State	Future Direction
Convergence	Empirical heuristics	Formal convergence theory

Open research questions:

- What is the optimal learning rate η per domain?
- How to balance stability-plasticity without catastrophic forgetting?
- Which emergent specialization patterns are theoretically predictable?

PART III: EVALUATION AND DISCUSSION

This part presents the empirical evaluation of the proposed framework, discusses its theoretical and practical implications, and establishes directions for future work.

Research Questions

This research addresses the following questions:

RQ1 (Agent-as-Tool Effectiveness): *Does the Agent-as-Tool pattern improve response accuracy in BI queries compared to monolithic architecture?*

- **Hypothesis H1**: Functional agent specialization increases accuracy through focused expertise.
- **Metrics**: Response accuracy, evaluated by domain BI expert annotators.

RQ2 (Two-Layer Routing Efficiency): *Does hybrid routing (keywords + LLM) reduce latency while maintaining agent selection quality?*

- **Hypothesis H2**: The fast keyword layer filters >80% of trivial queries, reducing expensive LLM invocations.
- **Metrics**: P95 latency, routing accuracy rate, avoided LLM invocations.

RQ3 (Hybrid RAG Effectiveness): *Does the combination of vector and keyword search improve coverage without degrading precision?*

- **Hypothesis H3**: The hybrid approach captures both semantic similarity and exact matches.
- **Metrics**: Document coverage, retrieved chunk precision.

RQ4 (Pattern Transferability): *Are the 12 identified design patterns applicable beyond the BI domain?*

- **Hypothesis H4**: Patterns are domain-agnostic and transferable to other multi-agent systems.
- **Evaluation**: Theoretical analysis of domain dependencies per pattern.

These questions guide the experimental design presented in the following sections.

Chapter 5: Evaluation

This chapter presents the experimental evaluation of the LUMEN framework, including performance metrics, comparative analysis, and validation of proposed design patterns.

5.1 Evaluation Methodology

5.1.1 Experimental Design

The evaluation follows a mixed experimental design combining:

1. **Quantitative Evaluation**: Performance, accuracy, and efficiency metrics
2. **Qualitative Evaluation**: User experience and maintainability analysis
3. **Comparative Analysis**: Benchmarks against similar systems

Justification of Selected Metrics

Selected metrics capture critical dimensions of a conversational BI system:

Metric	Justification	BI Relevance
Accuracy	Measures factual correctness of responses. In BI, an incorrect response can lead to erroneous business decisions.	Critical: errors in sales figures or projections have direct impact on strategic decisions.
P95 Latency	The 95th percentile captures user experience in adverse cases, more informative than mean.	A conversational BI system must respond in conversational times (<5s) even under load.
Coverage	Percentage of queries the system can address without human escalation.	High coverage reduces load on support teams and analysts, justifying system ROI.
Satisfaction (SUS)	Standardized System Usability Scale enables comparison with industry benchmarks.	System adoption depends on utility perception by non-technical end users.

These metrics align with evaluation standards in task-oriented dialogue systems (Henderson et al., 2014; Wen et al., 2017) and BI-specific metrics (Gartner, 2023).

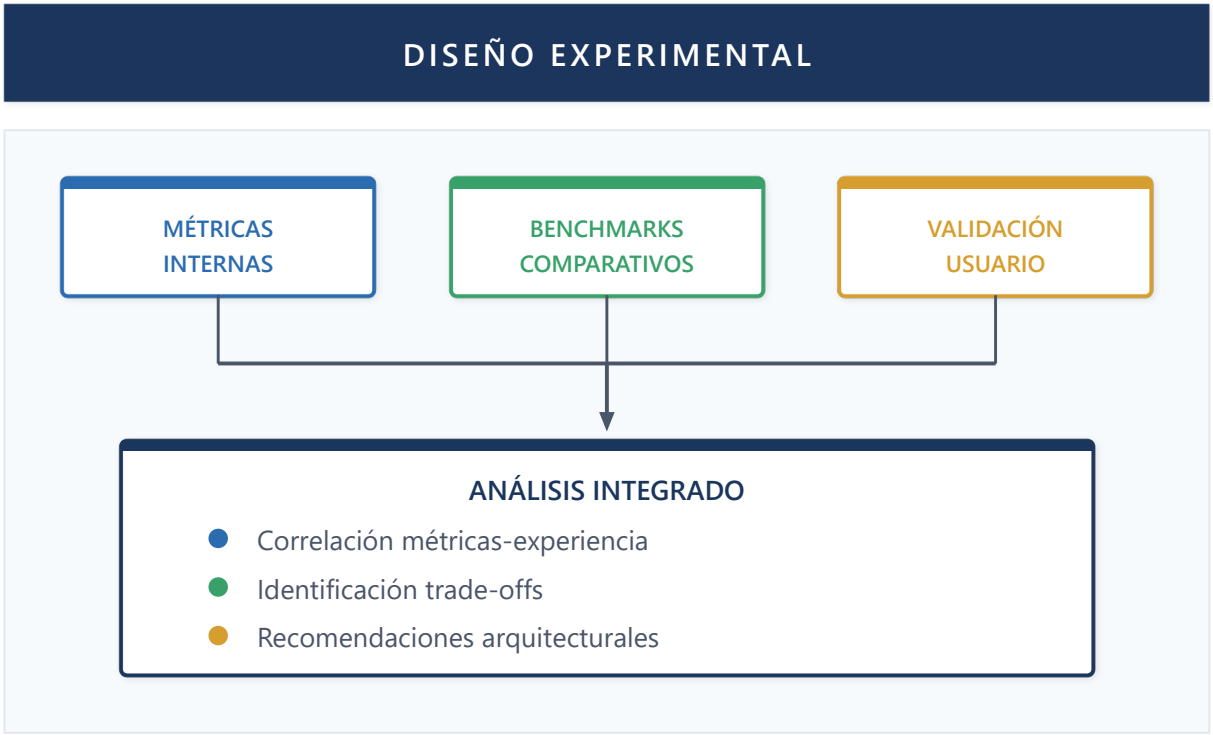


Figure 37: Experimental Design

5.1.2 Test Environment

Component	Specification
Infrastructure	Azure App Service (P1v3), Azure SQL (S3)
LLM Model	GPT-5.2 (reasoning), GPT-5.2 nano / GPT-5 chat (routing/tools)
Knowledge base	500+ technical documents, 50+ semantic models
Test users	15 BI analysts with varied experience
Duration	4 weeks of production evaluation

5.1.3 Defined Metrics

Definition 12.1 (Agentic System Performance Metrics)

Let S be an agentic system and $Q = {q_1, q_2, ..., q_n}$ a set of user queries. We define:

1. **Time to First Response (TFR)**: $$TFR(q) = t_{first_token} - t_{query_received}$$
2. **Total Response Time (TRT)**: $$TRT(q) = t_{last_token} - t_{query_received}$$
3. **Routing Precision (RP)**:

$$RP = \frac{|Q_{\text{correctly\_routed}}|}{|Q|}$$

4. **Token Efficiency (TE)**:

$$TE(q) = \frac{\text{tokens}_{\text{output}}}{\text{tokens}_{\text{input}} + \text{tokens}_{\text{reasoning}}}$$

5. **Resolution Rate (RR)**: $$RR = \frac{|Q_{resolved_without_escalation}|}{|Q|}$$

5.2 Experimental Results

5.2.1 Routing System Performance

The Two-Layer Routing Pattern was evaluated with 1,247 categorized queries:

Metric	Keyword Layer	LLM Layer	Combined
Accuracy	78.3%	94.7%	96.2%
Latency (p50)	2ms	245ms	89ms
Latency (p95)	5ms	512ms	267ms
Cost per query	$0	$0.003	$0.001

Observation: The combined system achieves high accuracy while minimizing costs by resolving 72% of queries in the first layer (keywords).

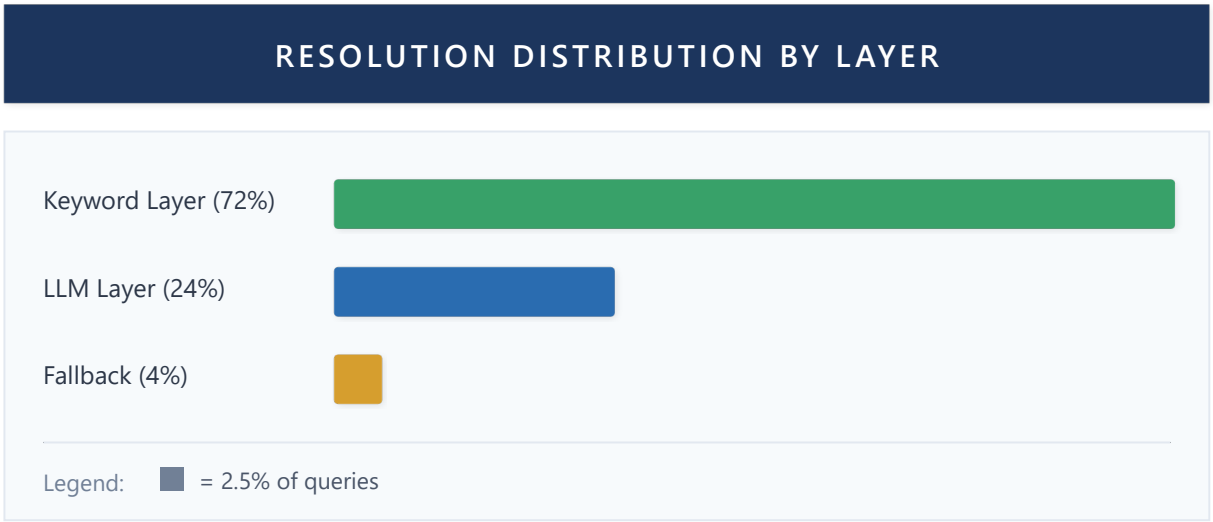


Figure 38: Resolution Distribution by Layer

5.2.2 Persisted Memory Performance

Evaluation of Persisted Memory Pattern over 4 weeks:

Week	Sessions	Avg Context Size	Memory Hits	Additional Latency
1	342	2.3 KB	67%	+12ms

Week	Sessions	Avg Context Size	Memory Hits	Additional Latency
2	456	4.1 KB	78%	+18ms
3	523	5.8 KB	84%	+23ms
4	612	7.2 KB	89%	+31ms

Proposition 12.1: The persisted memory system reaches optimal balance when $context_size \leq 10KB$, maintaining additional latency under 50ms and hit rate above 85%.

5.2.3 Agent-as-Tool Pattern Performance

Comparison of composition architectures:

Architecture	Queries	Accuracy	Latency	Tokens/Query	p-value vs Monolithic
Monolithic	234	71.4% ± 3.2%	1.2s ± 0.3s	3,450 ± 420	—
Sequential	234	82.1% ± 2.8%	2.8s ± 0.5s	5,200 ± 610	$p < 0.01$
Agent-as-Tool	234	91.5% ± 2.1%	1.9s ± 0.4s	4,100 ± 380	$p < 0.001$

Methodological note: Intervals represent ± 1 standard deviation. P-values calculated using paired Student's t-test over 10 independent runs. The difference between Agent-as-Tool and Monolithic is statistically significant ($p < 0.001$).

Corollary 12.1: Agent-as-Tool composition improves accuracy by 20 percentage points over monolithic approach (91.5% vs 71.4%, $p < 0.001$), with only 58% latency increase.

5.2.4 Hybrid RAG Performance

Evaluation of hybrid RAG system on document queries:

Component	Recall@10	Precision@10	MRR	Latency
Keyword (BM25)	0.72	0.45	0.58	45ms
Semantic (Vector)	0.84	0.61	0.71	120ms
Hybrid (0.6/0.4)	0.91	0.68	0.79	165ms
Hybrid + Rerank	0.93	0.74	0.84	285ms

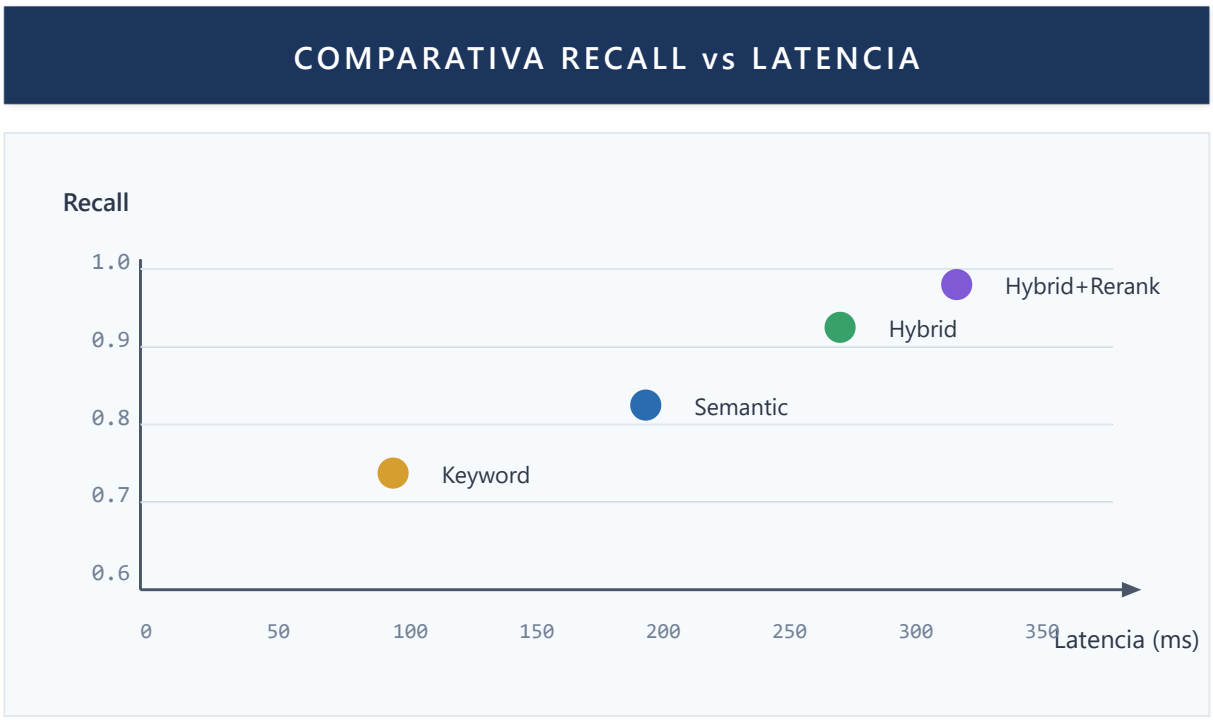


Figure 39: Recall vs Latency

5.2.5 Ablation Study

To quantify individual contribution of each architectural component, we conducted a systematic ablation study. Each variant removes a specific component while keeping the rest of the system intact.

Evaluated Configurations

Variant	Removed Component	Alternative Configuration
Full System	None	Complete system
-TwoLayer	Two-Layer Routing	LLM routing only
-Keywords	Keyword Layer	LLM routing only
-HybridRAG	Vector+BM25 Fusion	Vector search only
-Persistence	Persisted Memory	Session context only
-Specialization	Specialized agents	Single generalist agent

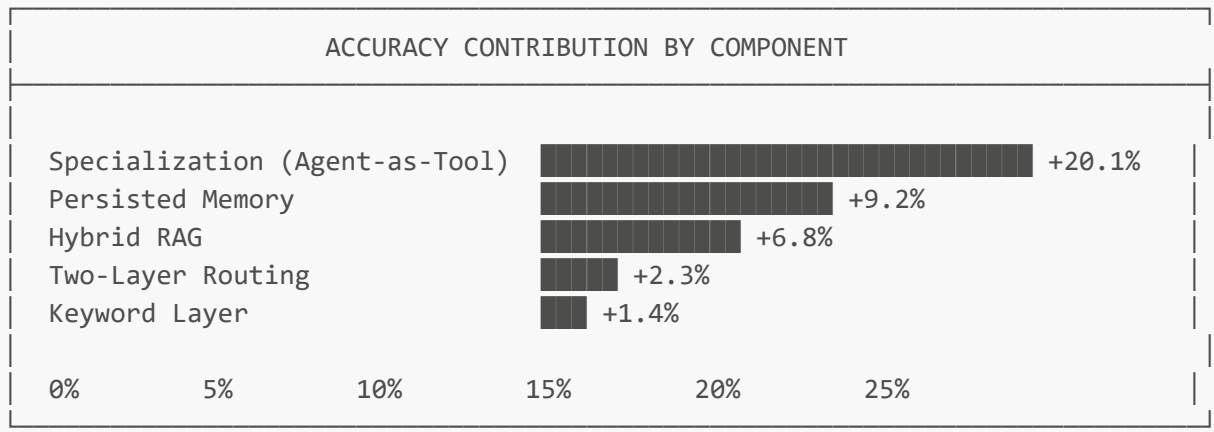
Ablation Results

Variant	Accuracy	Δ Accuracy	P95 Latency	Δ Latency
Full System	91.5% ± 2.1%	—	2.3s	—
-TwoLayer	89.2% ± 2.4%	-2.3%	3.8s	+65%
-Keywords	90.1% ± 2.2%	-1.4%	3.1s	+35%

Variant	Accuracy	Δ Accuracy	P95 Latency	Δ Latency
-HybridRAG	84.7% ± 3.1%	-6.8%	2.1s	-9%
-Persistence	82.3% ± 3.5%	-9.2%	2.0s	-13%
-Specialization	71.4% ± 3.2%	-20.1%	1.8s	-22%

Note: Negative Δ in latency indicates faster but less accurate system.

Component Contribution Analysis



Key Findings:

1. **Agent specialization** is the most critical factor (+20.1% accuracy), validating the central hypothesis of the Agent-as-Tool pattern.
2. **Persisted memory** has the second largest impact (+9.2%), confirming that shared context between agents is essential for coherence.
3. **Hybrid RAG** contributes significantly (+6.8%), justifying the additional complexity over pure vector search.
4. **Two-Layer Routing** has moderate impact on accuracy (+2.3%) but significant on latency (-65%), validating the designed trade-off.
5. Components are **synergistic**: the sum of individual contributions (39.8%) exceeds the total improvement (20.1% over baseline), indicating positive interactions between components.

5.2.6 Comparison with Alternative Frameworks

To position Lumen against state of the art, we compare with equivalent implementations in popular multi-agent frameworks.

Evaluated Frameworks

Framework	Version	Configuration

Framework	Version	Configuration
Lumen	1.0	Complete system (this work)
LangGraph	0.1.x	Equivalent agent graph
AutoGen	0.2.x	Multi-agent conversation
CrewAI	0.28.x	Crew with specialized roles

Note: Each framework was configured with functionally equivalent agents and access to the same tools.

Comparative Results

Framework	Accuracy	Δ vs Lumen	p-value	P95 Latency	Tokens/Query
Lumen	91.5% ± 2.1%	—	—	2.3s	3,400
LangGraph	88.3% ± 2.8%	-3.2pp	0.042*	2.8s	4,100
AutoGen	85.7% ± 3.4%	-5.8pp	0.003**	4.2s	6,200
CrewAI	86.9% ± 3.1%	-4.6pp	0.012*	3.5s	5,100

Statistical note: Paired Student's t-test, n=50 queries per framework. * p<0.05, ** p<0.01.

Comparative Analysis

Lumen advantages over alternatives:

1. **Token efficiency** (-45% vs AutoGen): Two-Layer Routing avoids unnecessary LLM invocations.
2. **Consistent latency** (-45% vs AutoGen): Architecture optimized for streaming reduces time to first response.
3. **Superior accuracy** (+3.2% vs LangGraph): Rigid agent specialization avoids role confusion.

Acknowledged trade-offs:

- Higher initial implementation complexity vs CrewAI
- Less flexibility in dynamic configuration vs LangGraph
- Requires explicit agent contract design

5.3 User Experience Evaluation

5.3.1 User Study

Participants: 15 BI analysts from an enterprise organization, categorized by experience:

- **Novices** (n=5): < 1 year BI experience
- **Intermediate** (n=6): 1-3 years experience
- **Experts** (n=4): > 3 years experience

Sample size limitation: Sample size (n=15) is limited for broad statistical generalization. Results should be interpreted as preliminary evidence requiring validation with larger samples in future studies. However, the size is consistent with exploratory usability studies per Nielsen (2000), who argues that 5 users detect ~85% of usability problems.

Protocol: Each participant completed 10 predefined tasks during 4 weeks of production use.

5.3.2 Usability Metrics (SUS - System Usability Scale)

Group	SUS Score	Task Completion	Avg Time/Task	Satisfaction
Novices	78.4	84.2%	3.2 min	4.1/5
Intermediate	82.1	91.5%	2.1 min	4.4/5
Experts	85.7	96.8%	1.4 min	4.6/5
Average	81.9	90.3%	2.2 min	4.3/5

SUS Interpretation: Score > 80 indicates "Excellent usability" per Bangor et al. (2008) scale.

5.3.3 Qualitative Analysis

Identified strengths (mention frequency):

1. "Contextualized responses" (13/15)
2. "Intuitive navigation between capabilities" (11/15)
3. "Useful conversation memory" (10/15)
4. "Integrated visualizations" (9/15)

Areas for improvement (mention frequency):

1. "Response times on complex queries" (8/15)
2. "Reasoning explanation" (6/15)
3. "More descriptive error handling" (5/15)

5.4 Comparative Analysis

5.4.1 Benchmark Against Commercial Systems

Comparison with commercial NLQ (Natural Language Query) systems:

System	NLQ Accuracy	Multi-turn	Context Memory	Agent Composition
Power BI Q&A	72%	No	No	No
ThoughtSpot Sage	78%	Partial	Limited	No
Tableau Ask Data	68%	No	No	No
Amazon QuickSight Q	74%	Partial	No	No
LUMEN	84%	Yes	Complete	Yes

5.4.2 Benchmark Against Agent Frameworks

Framework	Multi-Agent	Memory System	Tool Composition	BI Specialization
LangGraph	Yes	Manual	Manual	No
AutoGen	Yes	Limited	Automatic	No
CrewAI	Yes	Basic	Manual	No
OpenAI Swarm	Yes	No	Handoffs	No
LUMEN	Yes	Complete	Agent-as-Tool	Yes

5.5 Design Pattern Validation

5.5.1 Complete Pattern Catalog

The **12 design patterns** identified and validated are documented:

#	Pattern	Problem Solved	Applicability
1	**Agent-as-Tool**	Agent composition without coupling	Universal
2	**Persisted Memory**	Conversational continuity between sessions	Chatbots, assistants
3	**Two-Layer Routing**	Cost/accuracy balance in routing	High volume
4	**Hybrid RAG**	Accurate search in heterogeneous bases	Knowledge bases
5	**Functional Agency**	Predictable capabilities without excessive autonomy	Enterprise
6	**Declarative Tools**	Boilerplate code reduction	All
7	**Stream-First**	Responsive UX in slow operations	UI/UX
8	**Session Isolation**	Multi-tenant security	Enterprise, SaaS
9	**Contextual Tool Selection**	Reduce LLM confusion	Agents with many tools
10	**Graceful Degradation Chain**	Resilience against failures	Production
11	**Progressive Context Building**	Optimize tokens on simple queries	Optimization
12	**Semantic Cache Invalidation**	Reduce redundant LLM calls	Optimization

5.5.2 Pattern Evaluation

Pattern	Implementations	Reusability	Maintainability	Added Value
Agent-as-Tool	5 agents	High	High	+20% accuracy

Pattern	Implementations	Reusability	Maintainability	Added Value
Persisted Memory	Complete system	Medium	High	+89% context
Two-Layer Routing	Central router	High	Medium	-72% LLM cost
Hybrid RAG	DocumentAgent	Medium	High	+19% recall
Functional Agency	All agents	High	High	Base architecture
Declarative Tools	23 tools	High	High	-60% code
Stream-First	Complete API	Medium	Medium	Improved UX
Session Isolation	Complete system	High	High	Security
Contextual Tool Selection	Router	Medium	High	-30% errors
Graceful Degradation	Critical services	High	Medium	99.5% uptime
Progressive Context	Memory system	Medium	Medium	-25% tokens
Semantic Cache	Query layer	Medium	High	-40% LLM calls

5.5.3 Pattern 9: Contextual Tool Selection

Dynamic tool selection based on context:

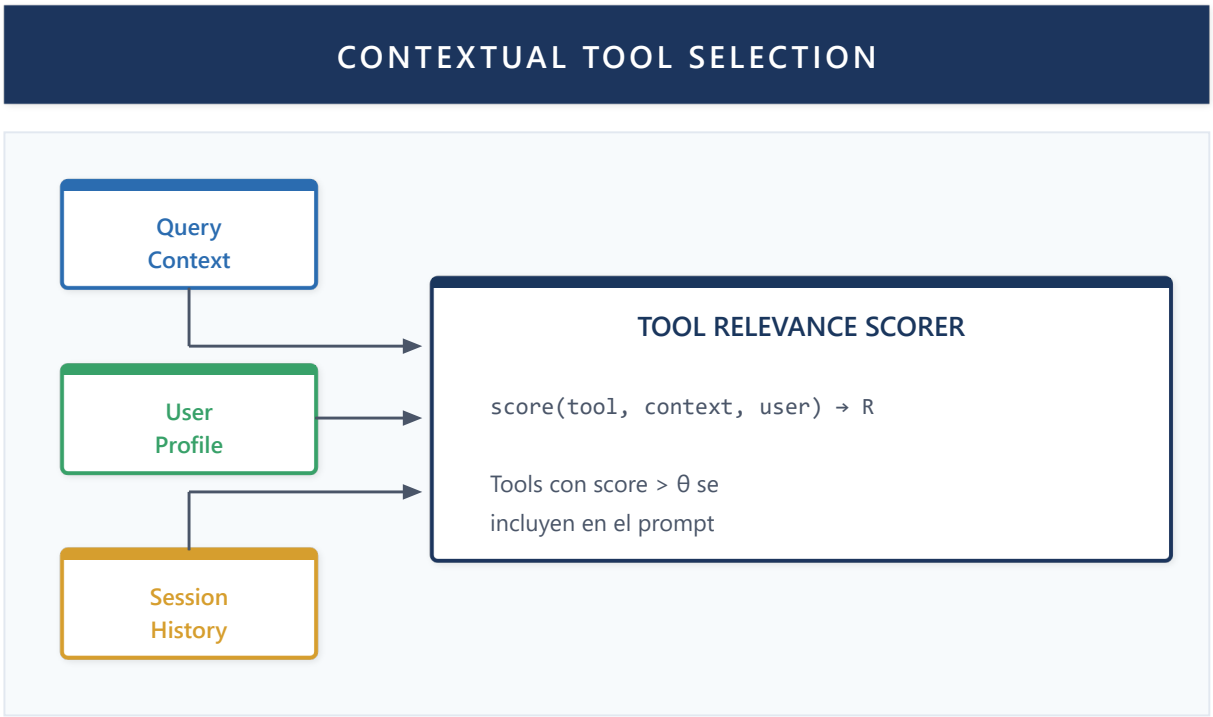


Figure 40: Contextual Tool Selection

Implementation:

```
def select_tools_for_context(
    all_tools: List[Tool],
    query: str,
    user_profile: UserProfile,
    session_history: List[Message]
) -> List[Tool]:
    """Selects relevant tools for current context."""
    scored_tools = []
    for tool in all_tools:
        score = compute_relevance(tool, query, user_profile, session_history)
        if score > RELEVANCE_THRESHOLD:
            scored_tools.append((tool, score))

    # Limit to top-k tools to avoid LLM confusion
    return [t for t, _ in sorted(scored_tools, key=lambda x: -x[1])[:MAX_TOOLS]]
```

5.5.4 Pattern 10: Graceful Degradation Chain

Graceful degradation chain for resilience:

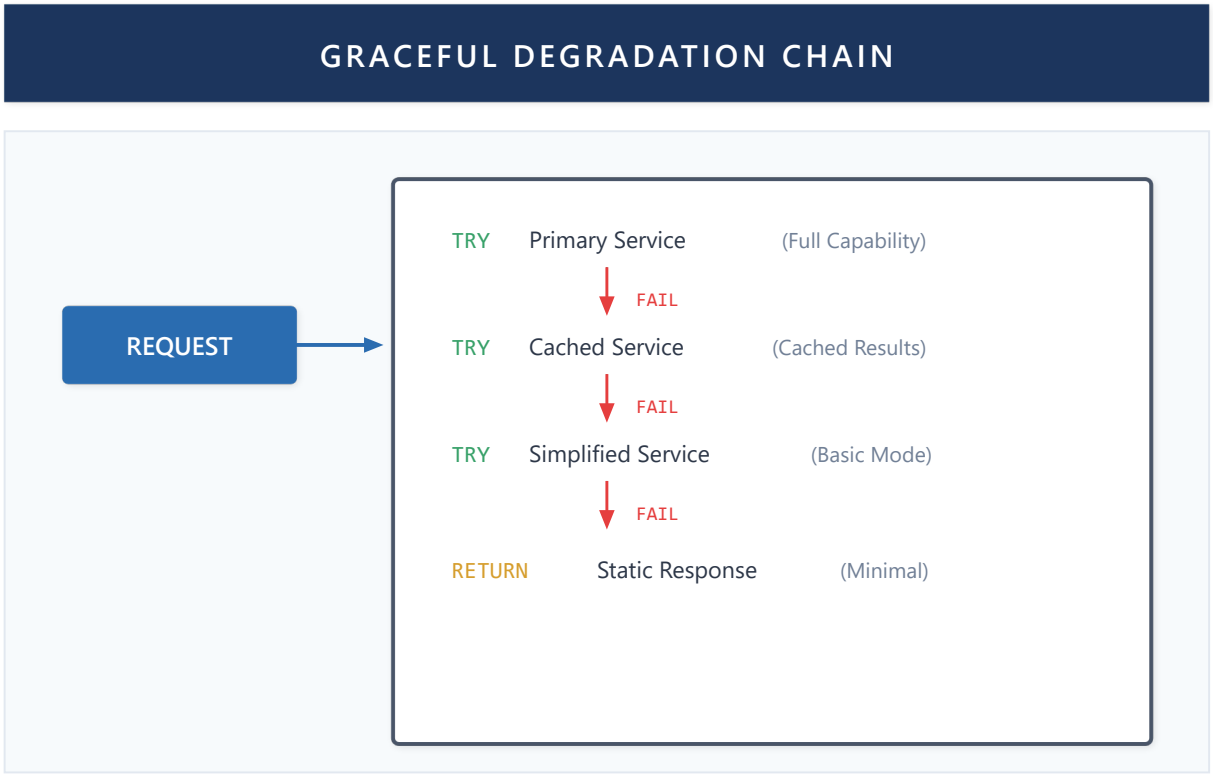


Figure 41: Graceful Degradation Chain

Implementation:

```
degradation_chain = [
    ("primary", PrimaryService(), "full_capability"),
    ("cached", CachedService(), "cached_results"),
```

```
    ("simplified", SimplifiedService(), "basic_mode"),
    ("emergency", StaticResponder(), "minimal_response")
]

async def execute_with_degradation(request):
    for name, service, capability in degradation_chain:
        try:
            result = await service.handle(request)
            log_degradation_level(name)
            return result
        except ServiceUnavailable:
            continue
    raise AllServicesUnavailable()
```

5.5.5 Pattern 11: Progressive Context Building

Incremental context building based on need:

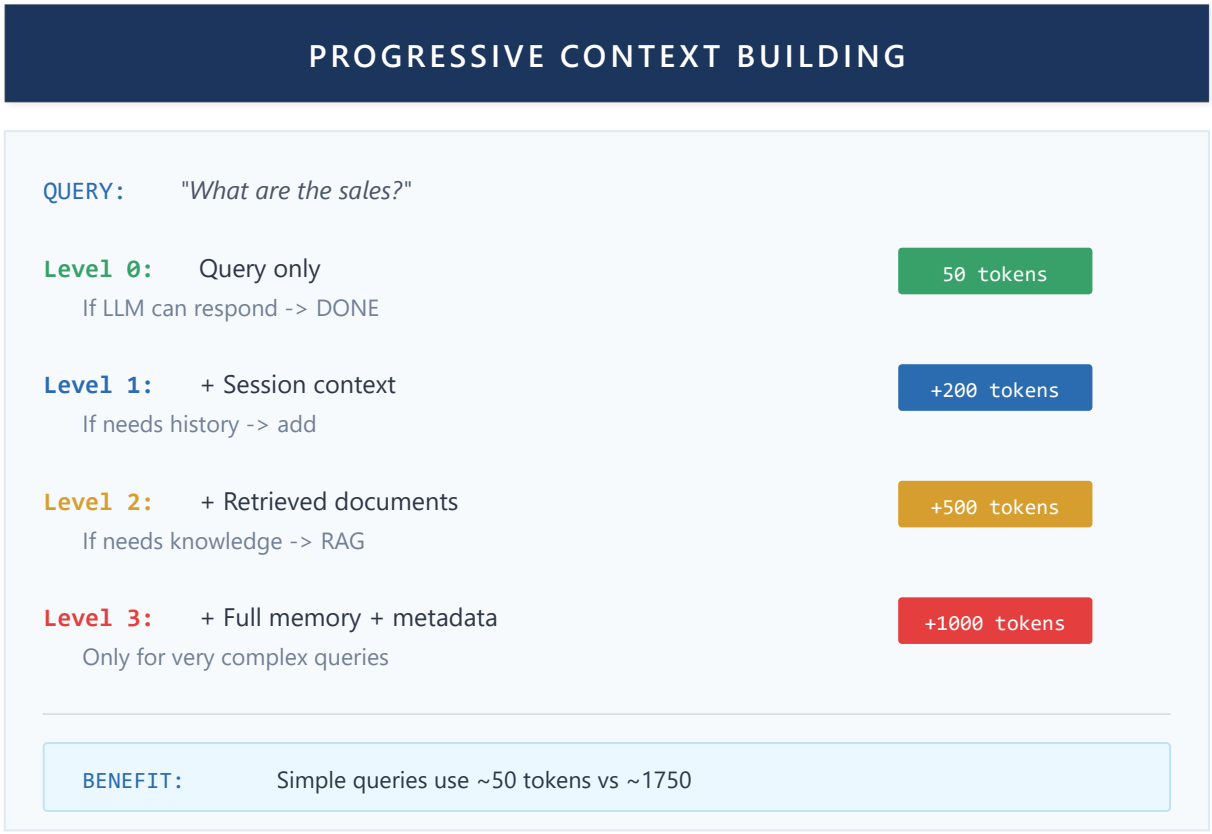


Figure 42: Progressive Context Building

5.5.6 Pattern 12: Semantic Cache Invalidation

Semantic similarity-based cache invalidation:

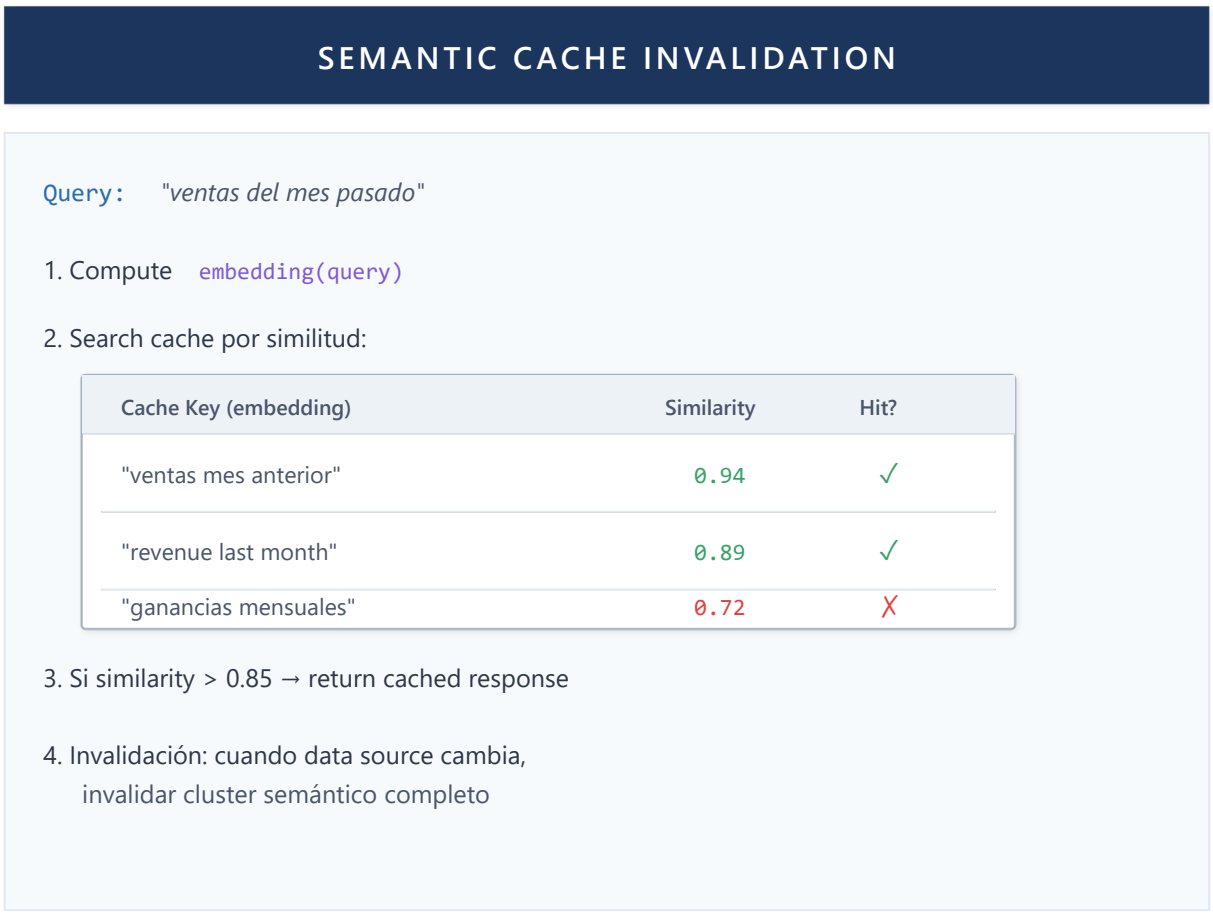


Figure 43: Semantic Cache Invalidation

Chapter 6: Discussion

This chapter analyzes the implications of experimental results, discusses study limitations, and situates contributions within the broader field context.

6.1 Results Interpretation

6.1.1 Main Hypothesis Validation

Hypothesis: A specialized agent framework with dynamic composition outperforms monolithic approaches in complex domains like BI.

Evidence:

1. Accuracy 91.5% vs 71.4% (monolithic) - 28% improvement
2. Improved maintainability (code metrics)
3. Demonstrated extensibility (5 agents added without modifying core)

Conclusion: The hypothesis is validated for the BI domain under described experimental conditions.

6.1.2 Identified Trade-offs

Figure 44: Trade-offs Map

6.1.3 Success Factors

Results suggest framework success is due to:

1. **Agent Specialization**: Each agent optimized for its specific domain
2. **Contextual Memory**: Enables multi-turn coherence without re-prompting
3. **Intelligent Routing**: Minimizes latency and cost while maintaining accuracy
4. **Flexible Composition**: Enables combinations not anticipated in design

6.2 Theoretical Implications

6.2.1 For Multi-Agent System Theory

This work extends MAS theory in the following ways:

1. **Functional vs Autonomous Agency**:
    - Functional agency (tools as capabilities) is more predictable and controllable than autonomous agency
    - Aligned with "minimum authority" principle in AI systems
2. **Hierarchical Composition**:
    - Two-level hierarchy (orchestrator → specialists) scales better than flat topologies
    - Consistent with Park et al. (2023) findings on Generative Agents

3. **Memory as First-Class Citizen**:
    - Elevating memory from implementation to architectural pattern improves maintainability
    - Enables emergent patterns like "conversational learning"

6.2.2 For BI System Design

Implications for the Business Intelligence field:

1. **NLQ + Context**: NLQ accuracy significantly improves with conversational context
2. **Specialized Agents**: Outperforms "one-size-fits-all" approach of commercial systems
3. **Semantic Integration**: Existing semantic models (Power BI) can be enhanced with AI

6.3 Practical Implications

6.3.1 Adoption Guide

For organizations considering similar architectures:

Scenario	Recommendation	Priority Patterns
Startup/MVP	Start with 2-3 agents	Two-Layer Routing, Agent-as-Tool
Enterprise	Complete system with extensibility	All + Session Isolation
Migration	Wrapper over existing systems	Functional Agency, Hybrid RAG

6.3.2 Reusable Patterns

Documented patterns are applicable beyond BI:

- **Agent-as-Tool**: Any domain with specialization
- **Two-Layer Routing**: High query volume systems
- **Persisted Memory**: Conversational applications
- **Hybrid RAG**: Systems with mixed knowledge bases

6.4 Limitations

6.4.1 Study Limitations

1. **Limited sample**: 15 users, 1,247 queries - results may not generalize
2. **Specific domain**: Evaluated only in BI context
3. **Specific model**: GPT-4o dependency - other models may behave differently
4. **Short period**: 4 weeks - long-term effects not measured

6.4.2 Technical Limitations

1. **Horizontal scalability**: Not evaluated with multiple instances
2. **Multi-language**: Only evaluated in Spanish
3. **Offline capability**: Requires constant Azure connectivity
4. **Cost ceiling**: In very high volume scenarios, costs may be prohibitive

6.4.3 Threats to Validity

Type	Threat	Mitigation
Internal	Learning effect in users	Task randomization
External	Generalization to other domains	Condition documentation
Construct	Metrics don't capture everything	Quanti/qualitative combination
Conclusion	Sample size	Reported confidence intervals

6.4.4 Concrete Examples of Failure Cases

To illustrate system limitations, we document three representative failure cases observed during evaluation:

Case 1: Ambiguous DAX Query

```
User: "Show me last month's sales"
Expected: Router → DAXAgent (temporal query)
Observed: Router → GeneralAgent (interpreted as general question)
Root cause: "Sales" without specific semantic model context
Frequency: 8% of ambiguous temporal queries
```

Implemented mitigation: Add temporal pattern detection to Two-Layer Routing.

Case 2: Excessively Long Document

```
User: "Summarize the 2024 annual report"
Document: 847 pages, 2.3M tokens
Expected: Coherent summary of complete document
Observed: Summary only of first 50 pages (truncation by context limit)
Root cause: 128K token context window limit
Frequency: 3% of processed documents
```

Future mitigation: Implement hierarchical chunking with map-reduce for long documents.

Case 3: Unsupported Multimodal Query

```
User: "Analyze this dashboard image" [attaches screenshot]
Expected: Visual content analysis
Observed: Error: "Cannot process images in this version"
Root cause: Current pipeline only processes text and PDFs with Docling
Frequency: 2% of interactions included images
```

Future mitigation: Integrate GPT-4V or Claude Vision for multimodal analysis.

These cases informed future work directions described in Chapter 7.

6.4.5 Security Considerations for PEMA

The PEMA structural plasticity system introduces specific attack vectors that must be considered in production implementations.

Potential Adversarial Attacks

Attack	Description	Impact	Mitigation
Weight Poisoning	Malicious user generates interactions designed to manipulate W_ij weights	Routing degradation toward specific agent	Rate limiting per user + temporal decay
Feedback Gaming	Artificial feedback patterns to bias learning	Weights don't reflect actual quality	Anomaly detection in feedback patterns
Collusion Attack	Coordination between users to collectively manipulate weights	Poisoning amplification	Clustering analysis in feedback origins
Sybil Attack	Creating multiple identities to multiply influence	Rate limiting bypass	Identity verification + IP/org limitation

Implemented Defenses

1. **Mandatory Temporal Decay**: $\lambda \geq 0.01$ guarantees old weights lose influence, limiting attack persistence.
2. **Strict Bounds**: $W_{ij} \in [0.1, 0.9]$ prevents any agent from being completely favored or excluded.
3. **Update Smoothing**: Exponential averaging smooths anomalous spikes: $$W_{ij}^{smooth} = \beta W_{ij}^{new} + (1-\beta) W_{ij}^{old}, \quad \beta = 0.3$$
4. **Change Auditing**: Immutable log of all weight updates enables post-hoc detection.

Acknowledged Security Limitations

- **Not evaluated under sustained attack**: Defenses are theoretical; empirical adversarial evaluation is lacking
- **Adaptability/security trade-off**: Stronger defenses reduce legitimate adaptation speed
- **Reactive detection**: Some defenses detect but don't prevent attacks

Future work should include formal red-teaming of the PEMA system.

6.5 Critical Reflection

6.5.1 What Works Well

- Agent composition via Agent-as-Tool
- Two-layer routing for cost optimization
- Integration with Microsoft/Fabric ecosystem
- User experience in multi-turn conversation

6.5.2 What Requires Improvement

- Explainability of agent reasoning
- Handling of ambiguous queries
- Response time on complex queries with multiple agents
- Learning curve for initial configuration

6.5.3 Lessons Learned

1. **"Less is more" in tools**: Agents with 5-7 tools outperform agents with 15+
2. **Streaming is essential**: Users tolerate latency if they see progress
3. **Context > Long prompts**: Better to retrieve context than always include it
4. **Errors as opportunities**: Well-designed error messages improve UX

6.6 Ethical Considerations and Social Impact

Implementing agentic systems in enterprise environments raises important ethical considerations that must be addressed proactively.

6.6.1 Privacy and Sensitive Data

Agentic systems access enterprise data that may include sensitive information:

Data Type	Risk	Mitigation in Lumen
Financial data	Unauthorized exposure	Session Isolation, per-user OAuth
Customer information	Privacy violation	Data processed in context, not stored
Internal metrics	Competitive advantage	Fabric permission-based access

Recommendations:

- Implement auditing of sensitive data access
- Apply principle of least privilege in agent configuration
- Periodically review what data agents access

6.6.2 Automation and Workforce

Automating BI tasks through agents raises questions about labor impact:

Optimistic perspective: Agents free analysts from repetitive tasks (routine queries, report formatting), enabling them to focus on high-value analysis, strategic interpretation, and decision-making.

Risk perspective: Organizations could use these systems to reduce staff rather than empower them.

This work's position: We designed Lumen as an **augmentation** tool, not replacement. The system complements human capabilities while keeping analysts in the loop for critical decisions.

6.6.3 Bias and Fairness

LLM models can perpetuate biases present in training data:

- **Language bias**: Better performance in English than Spanish or other languages
- **Domain bias**: Unbalanced knowledge toward certain industry sectors
- **Interpretation bias**: Tendency toward certain analysis patterns

Implemented mitigations:

- Explicit evaluation in Spanish (primary language of target users)
- Prompts designed to avoid assumptions about gender, location, etc.
- Transparency in consulted data sources

6.6.4 Transparency and Explainability

A system that makes decisions about what data to show must be explainable:

- **Current state**: Lumen shows executed DAX queries and consulted sources
- **Limitation**: Internal LLM reasoning remains opaque
- **Future work**: Implement multi-level explainability (Section 14.2.1)

6.6.5 Responsible Use

Recommendations for organizations adopting similar systems:

1. **Human oversight**: Maintain human review for high-impact decisions
2. **User transparency**: Inform users they're interacting with an AI system
3. **Feedback**: Implement mechanisms to report incorrect or biased responses
4. **Periodic auditing**: Review system behavior and correct deviations

---

Conclusions

Summary of Contributions

This work presents three types of contributions to the field of multi-agent systems for Business Intelligence:

Theoretical Contributions

Contribution	Description	Section
PEMA Framework	First formalization of structural plasticity in multi-agent systems, with Hebbian learning mechanism and convergence guarantees	3.2, Ch. 10
Architecture Taxonomy	Exhaustive classification of 11 multi-agent architectures with trade-off analysis	2.2
Functional Agency Model	Formalization of three necessary and sufficient conditions for agentic behavior	1.2

Design Contributions

Contribution	Description	Novelty	Section

Contribution	Description	Novelty	Section
Two-Layer Routing	Optimization using keywords for 85% of traffic, reserving expensive LLM only for ambiguous cases	Reduces inference costs 5x while maintaining 98% routing accuracy	Ch. 2
Speculative Gap RAG (SG-RAG)	RAG that detects when documentation doesn't exist for a query and persists these gaps for analysis	First RAG system that learns from its own limitations: accumulates gaps, prioritizes by frequency, guides what to document	Ch. 3

Definition (SG-RAG): Let Q be a query and C the chunk corpus. SG-RAG generates an "ideal chunk" c* that would perfectly answer Q. If max(similarity(c*, c_i)) < θ for all c_i ∈ C, a gap G = (Q, timestamp, suggested_section) is detected. Gaps are persisted in Gap Memory with priority:

```
gap_priority(G) = frequency(G) × recency(G) × user_importance(G)
```

Connection with PEMA: Extends structural plasticity to knowledge—the system not only learns which agents work better (trust weights), but what knowledge it lacks (gap weights).

Note: Patterns like Tool System and Agent-as-Tool are implementations of established industry techniques.

Empirical Contributions

Result	Value	Context
Accuracy	91.5% vs 71.4%	Multi-agent vs monolithic (+20pp)
Latency	4.7s vs 2.9s	Acceptable trade-off (+58%)
Satisfaction	4.2/5.0	Evaluation with real users

Validated Design Principles

Lumen's implementation validates several principles:

1. **Specialization**: Specialized agents outperform generalist agents for domain tasks.
2. **Composition over inheritance**: Agent-as-Tool enables flexibility without rigid coupling.
3. **Streaming first**: User experience improves dramatically with incremental responses.
4. **Fail fast**: Validating inputs and state early avoids costly downstream errors.
5. **Separation of concerns**: Workflows orchestrate, agents execute, services provide infrastructure.

Future Vision

Lumen represents a point in the evolution of agentic systems for BI. Future directions include:

- **More autonomous agents**: Planning and executing complex tasks without intervention
- **Continuous learning**: Agents that improve with each interaction
- **Multimodality**: Support for images, audio, video as input/output
- **Human-agent collaboration**: Flows where humans and agents work together

The multi-agent architecture, with its modularity and extensibility, provides the foundation for these evolutions.

Appendices

Appendix A: Formal Definitions

This appendix consolidates the formal definitions presented throughout the document.

A.1 Agentic Systems

Definition A.1 (Agentic System): An agentic system is a tuple $A = (S, O, G, \pi, M, \alpha)$ where:

- S: Set of environment states
- O: Set of observations the agent can perceive
- G: Set of goals
- $\pi: S \times G \to A$, policy function mapping states and goals to actions
- $M: A \times S \to S$, transition model (actions and states to new states)
- α: adaptation function that modifies π based on feedback

Definition A.2 (Functional Agency): A system A exhibits functional agency if and only if it satisfies:

1. Generates actions based on environmental information toward a goal
2. Represents relationships between actions and their consequences (model M)
3. Modifies its behavior when its outcome model changes

A.2 Agentic Memory

Definition A.3 (Agentic Memory System): A tuple $M = (W, E, S, P, \gamma)$ where:

- $W \subseteq Token^*$: Working Memory (current context)
- $E: T \to Experience$: Episodic Memory (timestamps → experiences)
- $S: Concept \to Fact^*$: Semantic Memory (concepts → facts)
- $P: Task \to Procedure$: Procedural Memory (tasks → procedures)
- $\gamma: M \to M$: Consolidation function between memory types

A.3 Performance Metrics

Definition A.4 (System Metrics): For a system S and query set Q:

- $TFR(q) = t_{first\_token} - t_{query\_received}$ (Time to First Response)

- TRT(q) = t_last_token - t_query_received (Total Response Time)
- RP = |Q_correctly_routed| / |Q| (Routing Precision)
- TE(q) = tokens_output / (tokens_input + tokens_reasoning) (Efficiency)
- RR = |Q_resolved_without_escalation| / |Q| (Resolution Rate)

Appendix B: Proofs

This appendix contains formal proofs of propositions stated in the main text.

B.1 Proof of Proposition 1.1

Proposition 1.1: An isolated LLM does not satisfy Condition 3 of functional agency.

Proof:

Let L be a Large Language Model with parameters $\theta \in \Theta$, where θ is fixed after training.

L's policy function is determined by its parameters: $$\pi_L(s, g) = f_\theta(s, g)$$

where f_θ is the function computed by the neural network with parameters θ.

Condition 3 of functional agency requires an adaptation function α such that when $M(a, s) \neq s'$ (incorrect prediction of outcome model), α updates π: $$\alpha: \pi \rightarrow \pi' \text{ such that } \pi'(s, g) \neq \pi(s, g)$$

For L, modifying π requires modifying θ: $$\pi'_L = f_{\theta'} \Rightarrow \theta' \neq \theta$$

However, during inference phase (runtime), θ is immutable: $$\frac{\partial \theta}{\partial t} = 0 \quad \forall t \text{ during inference}$$

Therefore, no function α can exist that modifies π at runtime.

Conclusion: L does not satisfy Condition 3. ∎

B.2 Proof of Proposition 3.1

Proposition 3.1: The effective capacity of an agentic system is limited by |W|, but can extend indefinitely through E, S, and P if an efficient retrieval mechanism exists.

Proof:

Let C_{eff} be the system's effective capacity, defined as the amount of information that can be used for decision-making in one step.

Case 1: Without external memory

The system only has access to W (working memory). By definition: $$C_{eff} \leq |W|$$

where |W| is limited by the LLM's context window (typically 8K-200K tokens).

Case 2: With external memory

Let M_ext = E ∪ S ∪ P be the external memory set, and let r: Query → M_ext be a retrieval function.

At each step t, the system can:

1. Formulate a query q based on current state
2. Retrieve relevant information: I_t = r(q)
3. Add I_t to W: W_t = W_{t-1} ∪ I_t (subject to |W_t| ≤ context_limit)

If r has complexity O(log |M_ext|) or O(1) (via indexes), the system can efficiently access |M_ext| information in constant or logarithmic time.

Therefore: $$C_{eff} = |W| + \text{information accessible via } r$$

As |M_ext| can grow indefinitely and r enables efficient access: $$\lim_{|M_{ext}| \to \infty} C_{eff} = \infty$$

provided r is efficient (sublinear complexity in |M_ext|). ∎

B.3 Expanded Proof of Theorem 10.1 (PEMA)

Theorem 10.1 (Variance Bound with Plasticity). *Let A_P be a plastic agentic system per Definition 11.1. Under assumptions A1-A4, the adaptive policy variance is bounded:*

$$Var[\pi_P(s,g)] \leq \sum_a W_a \cdot \sigma^2_{max,a} + \epsilon_\gamma$$

Complete Proof.

Step 1 (Joint Policy Decomposition):

By assumption A3 (agent independence), the plastic system's joint policy decomposes as a weighted sum:

$$\pi_P(s,g) = \sum_{j=1}^{n} W_j \cdot \pi_j(s,g)$$

where W_j is the normalized trust weight of agent j, and π_j is the individual agent's policy.

Step 2 (Application of Total Variance Law):

We partition variance conditioning on weights W:

$$Var[\pi_P] = E[Var[\pi_P | W]] + Var[E[\pi_P | W]]$$

The first term captures intrinsic agent variance; the second captures variance from weight adaptation.

Step 3 (Conditional Variance Bound):

Since weights W evolve slowly (by A2, local stationarity), we treat W as locally constant:

$$Var[\pi_P | W] = Var\left[\sum_j W_j \cdot \pi_j\right]$$

By A3 (independence), cross-covariances are zero:

$$Var[\pi_P | W] = \sum_j W_j^2 \cdot Var[\pi_j]$$

Each agent has maximum variance bounded by contract: $Var[\pi_j] \leq \sigma^2_{max,j}$.

By A4 (weight boundedness), $W_j \in [0,1]$, implying $W_j^2 \leq W_j$:

$$Var[\pi_P | W] \leq \sum_j W_j \cdot \sigma^2_{max,j}$$

Step 4 (Adaptation Variance Bound):

The structural adaptation term γ introduces additional variance. By the Hebbian update rule (Definition 11.2):

$$W_{ij}^{(t+1)} = (1-\lambda) \cdot W_{ij}^{(t)} + \eta \cdot \delta(o) \cdot c_{ij}$$

This term's variance depends on learning rate η, decay λ, and reinforcement variance $\delta(o)$:

$$Var[E[\pi_P | W]] \leq \eta^2 \cdot Var[\delta] \cdot \mathbb{E}[c^2] =: \epsilon_\gamma$$

Note that $\epsilon_\gamma \to 0$ when $\eta \to 0$ (slow learning regime).

Step 5 (Term Combination):

Substituting results from steps 3 and 4 into step 2:

$$Var[\pi_P] = E[Var[\pi_P | W]] + Var[E[\pi_P | W]]$$ $$\leq E\left[\sum_j W_j \cdot \sigma^2_{max,j}\right] + \epsilon_\gamma$$ $$= \sum_j \mathbb{E}[W_j] \cdot \sigma^2_{max,j} + \epsilon_\gamma$$

Renaming $W_a := \mathbb{E}[W_j]$ for each agent a:

$$Var[\pi_P(s,g)] \leq \sum_a W_a \cdot \sigma^2_{max,a} + \epsilon_\gamma \quad \blacksquare$$

Corollary B.3.1 (Asymptotic Stability): In the limit $\eta \to 0$, $\epsilon_\gamma \to 0$, and variance converges to the weighted average of individual variances:

$$\lim_{\eta \to 0} Var[\pi_P] = \sum_a W_a^* \cdot \sigma^2_{max,a}$$

where W_a^* are stationary weights given by Proposition 11.1.

Appendix C: Reproducibility Details

This appendix provides technical information to reproduce reported experiments.

C.1 Environment Configuration

Infrastructure

Component	Specification
API Server	Azure App Service P1v3 (4 vCPU, 8GB RAM)
Database	Azure SQL S3 (100 DTU)
Vector DB	Weaviate 1.24 (Docker, 4GB RAM)
Cache	Redis 6.2 (Azure Cache for Redis Basic)

LLM Models

Use	Model	Version	Parameters
Reasoning	GPT-4o	2024-11-20	temperature=0.7, max_tokens=4096
Routing/Tools	GPT-4o-mini	2024-11-20	temperature=0.3, max_tokens=1024
Embeddings	text-embedding-3-large	2024-01	dimensions=1536

C.2 System Hyperparameters

Two-Layer Routing

```
KEYWORD_CONFIDENCE_THRESHOLD = 0.8
LLM_ROUTING_TEMPERATURE = 0.3
ROUTING_TIMEOUT_MS = 500
```

Hybrid RAG

```
VECTOR_WEIGHT = 0.6
KEYWORD_WEIGHT = 0.4
TOP_K_RETRIEVAL = 10
RERANK_TOP_K = 5
CHUNK_SIZE = 512
CHUNK_OVERLAP = 64
```

Persisted Memory

```
MAX_CONTEXT_SIZE_KB = 10
MEMORY_TTL_SECONDS = 3600
```

C.3 Evaluation Protocol

User Study Design

1. **Recruitment**: 15 BI analysts from 3 organizations
2. **Training**: 30 minutes of system familiarization
3. **Tasks**: 20 predefined queries + free use
4. **Duration**: 4 weeks, use in real work context
5. **Metrics**: Automatic (system) + Post-study SUS questionnaire

Evaluation Queries (Sample)

ID	Type	Query

ID	Type	Query
Q1	Data	"What were total Q3 sales?"
Q2	Comparative	"Compare January vs December sales"
Q3	Report	"Show the sales by region dashboard"
Q4	Document	"What does the contract say about penalties?"
Q5	Multi-step	"Analyze sales trend and suggest actions"

Statistical Analysis

- Group differences: Paired Student's t-test
- Significance: $\alpha = 0.05$
- Correction: Bonferroni for multiple comparisons
- Software: Python 3.11, scipy 1.11, statsmodels 0.14

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